

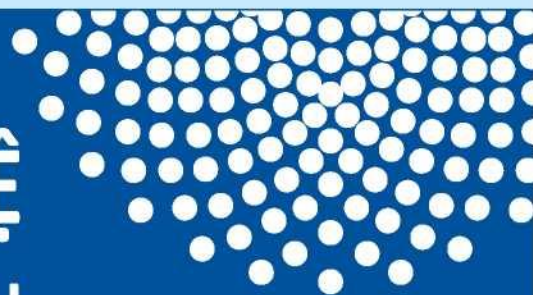
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học của cán bộ, giảng viên và sinh viên
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lượng đào tạo nguồn nhân lực và
chuyên gia công nghệ cho các đơn vị
sản xuất, doanh nghiệp có nhu cầu.

Đáp ứng nhu cầu phát triển nghiên cứu
khoa học và trao đổi học thuật ngày
càng tăng giữa các nhà khoa học, các
tổ chức nghiên cứu khoa học; khuyến
khích các nhà nghiên cứu trẻ và thu hút
các nhà khoa học có uy tín đăng tải kết
quả nghiên cứu.

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Validation of Quantification Process for Total Polyphenol and Flavonoid in Propolis Hydrogel by UV Spectrophotometry

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Abstract

Propolis is a resin rich in polyphenols and flavonoids, providing strong antibacterial, antioxidant, and wound-healing properties. Therefore, propolis hydrogels are therefore promising wound dressing materials that accelerate the healing of minor injuries. This study aimed to validate a Ultraviolet spectrophotometric method for the quantification of total polyphenol and total flavonoid content in propolis hydrogels. The quantifications were validated for their selectivity, linearity, accuracy and precision. The results demonstrated excellent linearity for polyphenols, and flavonoids over a concentration range of (10-120) $\mu\text{g/mL}$ at 754 nm and 427 nm, respectively. Furthermore, the method exhibited high specificity, accuracy (recovery rates between 95% and 102%), and precision (relative standard deviation $< 2\%$) for both analytes. The TPC and TFC in the propolis hydrogel were (4.2392 ± 0.0495) mgGAE/g and (2.1733 ± 0.0817) mgQE/g respectively. These validation results are crucial for establishing robust quality control criteria in the manufacturing of propolis hydrogels.

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Keywords

Propolis hydrogel; polyphenols; flavonoids; method validation; UV spectrophotometry.

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1 Introduction

A lesion is a damage or disruption of living tissue's cellular, anatomical, and functional integrity [1]. Wound healing is the body's natural response to restore the structure and function of the skin after injury. This process includes several complex steps such as hemostasis, anti-inflammation and tissue regeneration. Tissue repair time depends on many factors such as location, wound size, regeneration rate of damaged cells and the patient's health status. While minor wounds are naturally healed without medical interventions, large ones with a high risk of infection require external mediations to promote rapid recovery and ensure aesthetic.

Among numerous types of wound dressings, hydrogels are promising materials due to their rapid coagulation, antibacterial activity, biocompatibility and biodegradability [2]. In recent year, propolis has been loaded on hydrogels as wound dressing materials. Some polymers have been developed for the hydrogel film such as polyvinyl pyrrolidone (PVP), hydroxypropyl methylcellulose, polyacrylamide/methylcellulose, and chitosan/pectin [3-6]. The propolis hydrogels were developed to enhance wound healing, capitalizing on the 40% total polyphenol content (TPC) of propolis and the inherent swelling capability of hydrogels.

For determining TPC and the flavonoid content (TFC) of propolis hydrogel, UV spectrophotometry is a fast

and efficient analytical technique [3, 5, 6]. However, no articles demonstrate validation of TPC and TFC methods uniquely within hydrogels.

The present study validated quantification process of TPC and TFC of propolis hydrogel using UV spectrophotometry to provide a reliable assessment of the antioxidant properties of propolis. By establishing a standardized quantification process, this study contributed to make quality control criteria for propolis hydrogel in industrial manufacturing.

The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) guideline Q2(R2) on the validation of analytical procedures outlines several key elements for consideration, including specificity, working range, accuracy, and precision [7].

2 Materials and Methods

2.1 Materials and equipment

Propolis hydrogels were made in previous study [8]. Gallic acid (98% of purity), sodium carbonate, and aluminum chloride were purchased from Xilong, China. Quercetin (83.9% of purity) was purchased from National Institute of Drug Quality Control, Viet Nam. Folin-Ciocalteu reagent, and potassium acetate ($\geq 99\%$ of purity) were purchased from Merck (Merck Co., Darmstadt, Germany).

All equipment used in the study were located in the Faculty of Pharmacy at Nguyen Tat Thanh University, and routinely validated every 6 months.

2.2 Quantification procedure of total polyphenol content in propolis hydrogel

TPC in the propolis hydrogel was determined using a modified Folin–Ciocalteu method [9]. Four mL of 10% Folin-Ciocalteu reagent were added to 1 mL of the tested solution. The mixture was shaken well and left in the dark in 3 minutes. Then, 8 mL of 7.5% sodium carbonate solution was added, mixed and incubated in the dark for 30 minutes. Absorbance was recorded at a wavelength from (615-760) nm using a UV-Vis spectrophotometer [10]. Each sample was analyzed in triplicate. TPC was calculated using Equation (Eq.) 1 and expressed as a gallic acid equivalent.

$$TPC \left(\frac{mgGAE}{g} \right) = \frac{C \times V \times K}{m} \quad (Eq.1)$$

Whereas:

C: gallic acid concentration calculated (mg/mL)

V: test volume (mL)

K: dilution factor

m: propolis hydrogel weight (g)

2.3 Quantification procedure of flavonoid content in propolis hydrogel

TFC was determined using a modified aluminum chloride colorimetric method with slight modifications [11]. 1.5 mL of ethanol and 0.1 mL of 10% aluminum chloride solution were added to 0.5 mL of the sample solution. The mixture was stabilized for 6 minutes for proper reaction. Then, 0.1 mL of 1 M potassium acetate solution and 2.5 mL of distilled water were added. The mixture was shaken well and left at room temperature for 45 minutes. The absorbance was recorded at a wavelength of (410-430) nm [10], using a UV-Vis spectrophotometer. Each sample was analyzed in triplicate. TFC was calculated using Eq. 2 and expressed as quercetin equivalent.

$$TFC \left(\frac{mgQE}{g} \right) = \frac{C \times V \times K}{m} \quad (Eq. 2)$$

Whereas:

C: quercetin concentration calculated (mg/mL)

V: test volume (mL)

K: dilution factor

m: propolis hydrogel weight (g)

2.4 Validation of quantification procedure

2.4.1 Maximum absorbance wavelengths determination

The standard solutions of gallic acid (40 μ g/mL) and quercetin (20 μ g/mL) were prepared in 70% ethanol. These were then mixed with other reagents, following the same procedure used for quantifying TPC and TFC. The resulting mixtures were then analyzed with a UV-Vis spectrophotometer, which scanned their absorption spectra between 400 nm and 800 nm for gallic acid and between 200 nm to 800 nm for quercetin. The maximum absorbance wavelengths (λ_{max}) for both gallic acid and quercetin were determined from these scans and were used for further experiment. Blank was 70% ethanol.

2.4.2 Selectivity

The hydrogels spiked with gallic acid or quercetin and blank hydrogel were dissolved in 80 mL of 70% ethanol and sonicated for 15 minutes. The mixtures were left to cool and diluted to 100 mL. After filtration through a 0.45 μm membrane, they were mixed with reagents similar to quantification procedure of TPC and TFC. Subsequently, the resulting solutions were scanned for their spectra between 400 nm and 800 nm for gallic acid and between 200 nm to 800 nm for quercetin using a UV-Vis spectrophotometer. The requirement is that the spectra of blank hydrogels must not show any peak absorbance at the λ_{max} for gallic acid or the λ_{max} for quercetin, while the spectra of spiked hydrogels demonstrate absorbance peaks at corresponding wavelengths of gallic acid and quercetin [7].

2.4.3 Linearity and working range

Gallic acid (standard) or quercetin (standard) were dissolved in 70% ethanol at concentrations of (10, 20, 40, 60, 80, 100, and 120) $\mu\text{g/mL}$. The amount of gallic acid or quercetin was evaluated according to the quantification procedure for TPC or TFC. The response of the measured quantity (peak area) at each concentration was determined by establishing a regression equation and plotting a graph that shows the correlation between the response signal (peak area) and the concentration. Linearity was then assessed using statistical methods based on the regression equation $y = ax + b$. The significance of the coefficients

$$\text{Percent recovery} = \frac{\text{Accepted true value of added amount of analyte}}{\text{Known added amount of analyte}} \times 100 \quad (\text{Eq.5})$$

2.4.6 Precision

The precision test was performed by evaluating six determinations of a 40 $\mu\text{g/mL}$ gallic acid solution or a 20 $\mu\text{g/mL}$ quercetin solution. From the results, the relative standard deviation (RSD) was calculated. The procedure was considered precise if the RSD was less than or equal to 2% [7].

2.5 Determination of TPC and TFC in propolis hydrogels

The propolis hydrogel (approximate 1 g) was dissolved in 80 mL of 70% ethanol and sonicated for 15 minutes. The mixtures were left to cool down and diluted to 100 mL. After filtration through a 0.45 μm membrane, they were mixed with reagents similar to quantification procedure of TPC and TFC. The resulting solutions

in the regression equation was checked using a t -test, while the overall adequacy of the regression equation was checked using an F -test.

2.4.4 Detection limit and quantification limit

The detection limit (DL) and quantification limit (QL) are the lowest concentrations of an analyte that can be reliably detected and quantified, respectively. These limits are calculated using Eq. 3 and Eq. 4 [7]:

$$DL = \frac{3.3 \times SD}{a} \quad (\text{Eq. 3})$$

$$QL = \frac{10 \times SD}{a} \quad (\text{Eq. 4})$$

Whereas:

a: the slope of the calibration curve

SD: the standard deviation of the response

2.4.5 Accuracy

Sample solution with a concentration of 40 $\mu\text{g/mL}$ gallic acid or 20 $\mu\text{g/mL}$ quercetin was prepared. The solution was spiked with gallic acid standard solution or quercetin standard solution at 3 concentrations of (80, 100 and 120)% of initial concentration. The gallic acid or quercetin amount was determined following the quantification procedure for TPC and TFC. The experiment was repeated 3 times to obtain the average value of each concentration. Accuracy was reported as the mean percent recovery by known added amount of gallic acid or quercetin and the accepted true value as Equation 5. The procedure is accurate if the percent recovery is between (95 and 102)% [12].

were measured the absorbance at λ_{max} of gallic acid and quercetin. The experiment was repeated 3 times. The data were then used to calculate the total polyphenol content and total flavonoid content.

3 Results and Discussion

3.1 Maximum absorbance wavelengths of gallic acid and quercetin

The absorbance peaks for gallic acid and quercetin obtained in standard solutions from the scan spectrum are shown in Figure 1. The result exhibited that the λ_{max} for gallic acid and quercetin were 754 nm (Figure 1A) and 427 nm (Figure 1B), respectively (Figure 1A and Figure 1B)

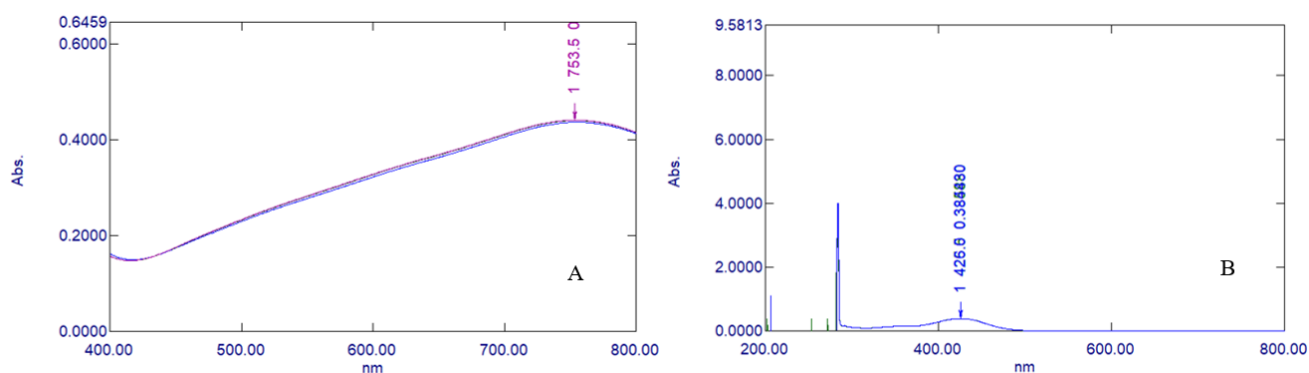


Figure 1 λ_{max} for gallic acid (A) and quercetin (B)

3.2 Selectivity

Figure 2 and Figure 3 showed the scan spectra of the blank hydrogel and the spiked hydrogel with gallic acid and quercetin, respectively. There are no absorption peaks at 754 nm and 427 nm on the scan spectra of the blank hydrogel (Figure 2A and Figure 3A,

respectively). Meanwhile, the spiked hydrogels' spectra display absorption bands maximum at 754 nm for gallic acid (Figure 2B) and 427 nm for quercetin (Figure 3B). This performance exhibited that the excipients contained in hydrogels did not interfere with the absorbance of gallic acid and quercetin.

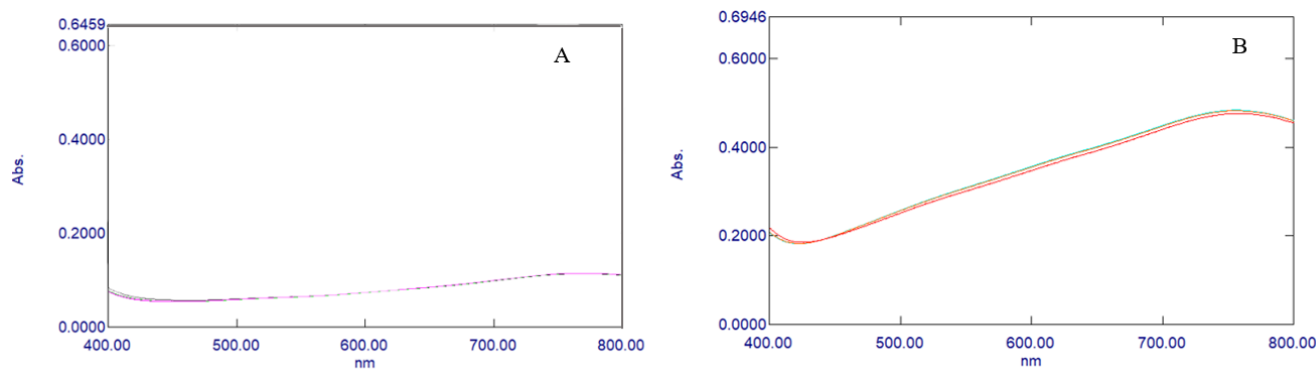


Figure 2 Scan spectrum of blank hydrogel (A) and spiked hydrogel with gallic acid (B)

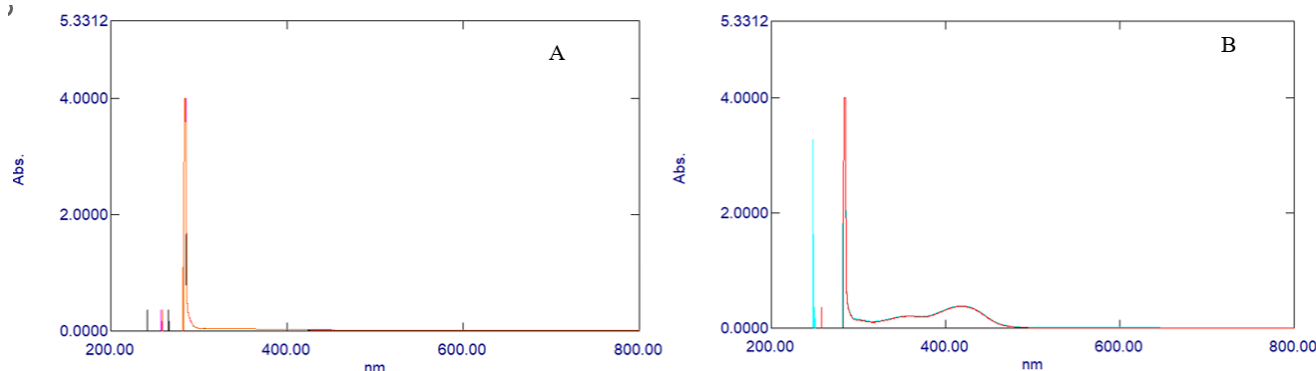


Figure 3 Scan spectra of blank hydrogel (A) and spiked hydrogel with quercetin (B)

The TPC and TFC quantification procedure in propolis hydrogel using UV spectrophotometry was selective.

3.3 Linearity and working range

Table 1 shows the absorbance of standard solutions of gallic acid and quercetin at concentrations from (10-120) $\mu\text{g/mL}$. The measurements were taken at 754 nm and 427 nm, respectively.

Table 1 Absorbance of gallic acid concentrations at 754 nm and of quercetin concentrations at 427 nm

Gallic acid concentration	Absorbance at 754 nm				
	No.1	No.2	No.3	Average	SD
10	0.0614	0.0590	0.0838	0.0681	0.0137
20	0.1492	0.1423	0.1456	0.1457	0.0035
40	0.3034	0.2984	0.3022	0.3013	0.0026
60	0.4509	0.4371	0.4469	0.4450	0.0071
80	0.6018	0.5857	0.5993	0.5956	0.0086
100	0.7604	0.7481	0.7488	0.7524	0.0069
120	0.9026	0.8854	0.9065	0.8982	0.0112
Quercetin concentration (µg/mL)	Absorbance at 427 nm				
	No.1	No.2	No.3	Average	SD
10	0.0757	0.0829	0.0809	0.0798	0.0037
20	0.1554	0.1495	0.1587	0.1545	0.0047
40	0.3556	0.3434	0.3383	0.3458	0.0089
60	0.5180	0.5074	0.5035	0.5096	0.0075
80	0.7017	0.6797	0.6860	0.6891	0.0113
100	0.8524	0.8626	0.8758	0.8636	0.0118
120	1.0417	1.0126	1.0553	1.0365	0.0218

As shown in Table 2, a statistical regression analysis revealed a high linear correlation between gallic acid concentration and its absorbance at 754 nm ($R^2 = 0.9999$), as well as between quercetin concentration and its absorbance at 427 nm ($R^2 = 0.9998$). The p -values for the coefficients in both cases are less than 0.05, indicating that the coefficients in the regression

equations are significant. The F -statistic for the significance of the gallic acid equation's y-intercept (b) is greater than 0.05, indicating it is not statistically significant. In contrast, the F -statistic for the quercetin equation's y-intercept is less than 0.05, meaning it is significant.

Table 2 Results of regression analysis

Value	TPC	TFC
Standard substance	Gallic acid	Quercetin
Regression equation $y = ax + b$	$y = 0.0075x - 0.0052$	$y = 0.0087x - 0.0115$
R square (R^2)	0.9999	0.9998
p -value of a	3.0099E-11	2.3066E-10
Significance F	$0.0796 > 0.05$: b is not significant	$0.0394 < 0.05$: b is significant
Standard error	0.0033	0.0058
Working range (µg/mL)	10-120	10-120
Linear equation	$y = 0.0075x$	$y = 0.0087x - 0.0115$

As a result, the procedure meets the linearity requirements, with a working range of (10-120) µg/mL for both experiments. The standard curves for TPC are $y = 0.0075x$ ($R^2 = 0.9999$) (Figure 4A), and for TFC it is $y = 0.0087x - 0.0115$ ($R^2 = 0.9998$) (Figure 4B).

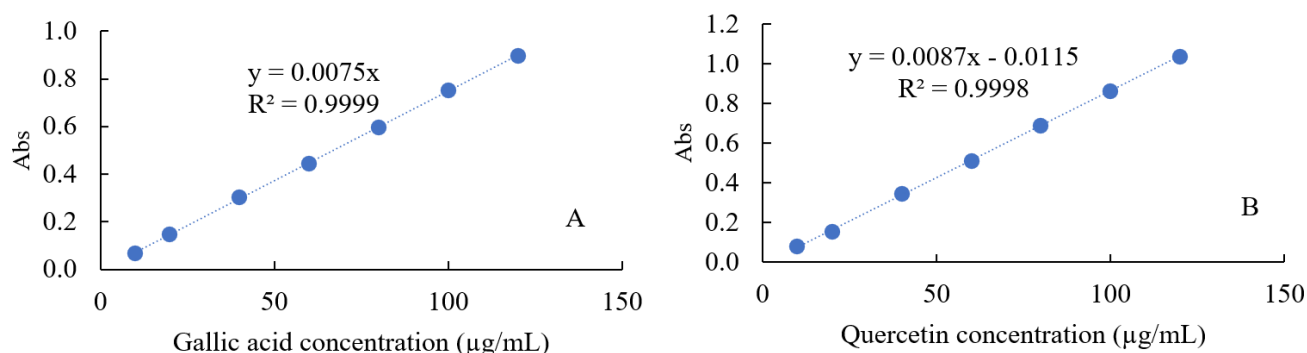


Figure 4 Standard curves of gallic acid (A) and quercetin (B)

3.4 Detection limit and quantification limit

The value of Standard Error and the slope of the calibration curve (a) shown in Table 2 was used to calculate the DL and QL. The results can be observed in Table 3.

Table 3 Detection limit (DL) and Quantification limit (QL)

Limit value	Unit	Total polyphenol content	Flavonoid content
Detection limit (DL)	µg/mL	1.4469	2.1744
Quantification limit (QL)	µg/mL	4.3845	6.5891

3.5 Accuracy

Table 4 Recovery rate of gallic acid in accuracy experiment

Sample	No.	Absorbance	Gallic acid concentration (µg/mL)	Average concentration (µg/mL)	SD	Found standard (µg/mL)	Added standard (µg/mL)	Percent recovery (%)
Sample (40 µg/mL)	1	0.3189	42.2923	38.9893	3.0212			
	2	0.2742	36.3656					
	3	0.2889	38.3100					
Sample + 80 % standard	1	0.5678	75.2927	70.5443	7.0784	31.5550	31.1914	101.2
	2	0.4706	62.4088					
	3	0.5575	73.9314					
Sample + 100 % standard	1	0.5816	77.1249	78.5884	3.1374	39.5992	38.9893	101.6
	2	0.6198	82.1901					
	3	0.5765	76.4503					
Sample + 120 % standard	1	0.6400	84.8590	86.0652	1.2090	47.0759	46.7871	100.6
	2	0.6582	87.2770					
	3	0.6490	86.0595					

Table 5 Recovery rate of quercetin in accuracy experiment

Sample	No.	Absorbance	Quercetin concentration (µg/mL)	Average concentration (µg/mL)	SD	Found standard (µg/mL)	Added standard (µg/mL)	Percent recovery (%)
Sample (20 µg/mL)	1	0.1787	21.7504	20.2348	1.6958			
	2	0.1682	20.5507					
	3	0.1494	18.4032					
Sample + 80 % standard	1	0.3129	37.1045	36.6657	0.6688	16.4310	16.1878	101.5
	2	0.3120	36.9967					
	3	0.30236	35.8960					

Sample + 100 % standard	1	0.3232	38.2767	40.5655	2.0512	20.3308	20.2348	100.5
	2	0.3486	41.1823					
	3	0.3578	42.2376					
Sample + 120 % standard	1	0.386	45.4951	44.4352	0.9470	24.2005	24.2817	99.7
	2	0.3703	43.6723					
	3	0.3744	44.1384					

To verify the accuracy of TPC and TFC quantification method, the concentration of the test sample and the samples spiked with (80, 100, and 120)% of the standard was determined, as shown in Table 4 and Table 5. The results for the gallic acid percent recovery at the (80, 100, and 120)% levels were (101.2, 100.6, and 100.6)%, respectively, which is within the required range of (95-102)%. Similarly, the results for the

quercetin percent recovery at the (80, 100, and 120)% levels were (101.5, 100.5, and 99.7)%, respectively, also falling within the required range of (95-102)%. Therefore, the total polyphenol and flavonoid content quantification procedure in propolis hydrogel using UV spectrophotometry was accurate.

3.6 Precision

Table 6 Absorbance of analyte concentrations in precision experiment

No.	Gallic acid		Quercetin	
	Absorbance	Concentration	Absorbance	Concentration
		($\mu\text{g/mL}$)		($\mu\text{g/mL}$)
1	0.3034	40.2270	0.1801	21.9128
2	0.2984	39.5633	0.1819	22.1175
3	0.2989	39.6394	0.1863	22.6220
4	0.3022	40.0782	0.1754	21.3747
5	0.3107	41.2047	0.1805	21.9677
6	0.3122	41.4042	0.1787	21.7473
Average		40.3528	-	21.9570
SD		0.7816		0.4130
RSD		1.9370		1.8812

The absorbance and concentrations of 6 determinations for gallic acid and quercetin were exhibited in Table 6. The average concentration for gallic acid and quercetin was $(40.3528 \pm 0.7816) \mu\text{g/mL}$ and $(21.9570 \pm 0.4130) \mu\text{g/mL}$, respectively. The RSD of 1.9370% and

1.8812% was less than 2%, meeting the requirements of the repeatability validation.

The TPC and TFC quantification method using UV spectrophotometry achieved repeatability.

3.7 Determination of total polyphenol and flavonoid content in propolis hydrogels

Table 7 Total polyphenol and flavonoid content in propolis hydrogels

Propolis hydrogel	Weight (g)	Moisture content (%)	Abs at 754 nm	Gallic acid concentration ($\mu\text{g/mL}$)	Total polyphenol content (mgGAE/g)	Abs at 427 nm	Quercetin concentration ($\mu\text{g/mL}$)	Flavonoid content (mgGAE/g)
1	1.0708	10.7300	0.3097	41.0681	4.2963	0.1646	20.1411	2.1070
2	1.0225	10.7300	0.2900	38.4606	4.2135	0.1692	20.6705	2.2646
3	1.0188	10.7300	0.2886	38.2684	4.2077	0.1593	19.5379	2.1482
Average				39.2657	4.2392	-	20.1165	2.1733
SD				1.5639	0.0495	-	0.5667	0.0817

The validated procedures for determining TPC and TFC in propolis hydrogels were applied to evaluate the total polyphenol and flavonoid content in the propolis hydrogels that were previously prepared [8]. The results are shown in Table 7. Accordingly, the TPC and TFC are (4.2392 ± 0.0495) mgGAE/g and (2.1733 ± 0.0817) mgQE/g, respectively.

4 Conclusion

The quantification procedure of total polyphenol and flavonoid content in propolis hydrogels using UV spectrophotometry was validated and achieved selectivity, linearity, accuracy, and precision. Accordingly, for total polyphenol, at 754 nm, the procedure demonstrated a linear equation $y = 0.0075x$ ($R^2 = 0.9999$) with a detection limit of $1.4469 \mu\text{g/mL}$ and a quantification limit of $4.3845 \mu\text{g/mL}$. For flavonoid, at 427 nm, the procedure demonstrated a

linear equation $y = 0.0087x - 0.0115$ ($R^2 = 0.9998$) with a detection limit of $2.1744 \mu\text{g/mL}$ and a quantification limit of $6.5891 \mu\text{g/mL}$. Both procedures verified a working range of $(10-120) \mu\text{g/mL}$, accuracy within $(95-102)\%$, and precision with an $\text{RSD} < 2\%$. The TPC and TFC are (4.2392 ± 0.0495) mgGAE/g and (2.1733 ± 0.0817) mgQE/g, respectively. This research is essential for establishing quality control criteria for propolis hydrogel in industrial manufacturing. Compared with more complex techniques such as HPLC, this method offers a faster, cost-effective and robust analytical approach that aligns with ICH Q2(R2) requirements.

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Thẩm định quy trình định lượng polyphenol tổng và flavonoid trong hydrogel keo ong bằng phương pháp quang phổ tử ngoại

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Tóm tắt Keo ong là một hỗn hợp nhựa giàu polyphenol và flavonoid, mang lại các đặc tính kháng khuẩn, chống oxy hóa và làm lành vết thương. Do đó, hydrogel keo ong là vật liệu băng vết thương đầy hứa hẹn, giúp đẩy nhanh quá trình làm lành các vết thương vừa và nhỏ. Nghiên cứu này nhằm mục đích thẩm định phương pháp định lượng polyphenol tổng (TPC) và định lượng flavonoid tổng (TFC) trong hydrogel keo ong. Các phép định lượng đã được thẩm định về tính đặc hiệu, độ tuyến tính, độ chính xác và độ đúng. Kết quả cho thấy độ tuyến tính cao đối với polyphenol và flavonoid trong khoảng nồng độ (10-120) $\mu\text{g/mL}$ lần lượt tại bước sóng 754 nm và 427 nm. Hơn nữa, phương pháp thể hiện tính đặc hiệu cao, độ chính xác (tỷ lệ thu hồi nằm trong khoảng 95 % đến 102 %) và độ lặp lại cao (độ lệch chuẩn tương đối dưới 2 %) cho cả hai chất phân tích. Hàm lượng TPC và TFC trong mẫu hydrogel keo ong lần lượt là $(4,2392 \pm 0,0495) \text{ mgGAE/g}$ và $(2,1733 \pm 0,0817) \text{ mgQE/g}$. Việc thẩm định này cần thiết trong xây dựng chỉ tiêu định lượng trong sản xuất hydrogel keo ong.

Từ khóa hydrogel keo ong, polyphenol, flavonoid, thẩm định phương pháp, quang phổ UV

Ultrasound-assisted Extraction of Bioactive Polyphenols from Cashew Testa (*Anacardium occidentale* L.) and Evaluation of Their Antioxidant and Antimicrobial Properties

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Abstract

Cashew testa (*Anacardium occidentale* L.) is an underutilized by-product rich in bioactive compounds. This study aims to evaluate the polyphenol content and bioactivities of cashew nut testa in Vietnam using ultrasound-assisted extraction with different solvents. Methanol and ethanol were identified as the most effective solvents, yielding extracts with high polyphenol content of (484.06 ± 4.26) mg GAE/g and (465.33 ± 4.69) mg GAE/g, respectively. The results from antioxidants assay demonstrated that at the concentration of 100 µg/mL, methanol and ethanol extracts exhibited > 87% DPPH and > 99% ABTS inhibition. In addition, both extracts showed pronounced antimicrobial activity against *Staphylococcus aureus* and *Candida albicans*, as indicated by inhibition zone diameters exceeding 11 mm and minimum inhibitory concentrations of 1.56 mg/mL. These results confirm the effectiveness of ultrasound-assisted extraction for producing polyphenol-rich extracts with antioxidant and antimicrobial potentials.

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ultrasound-assisted
extraction.

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1 Introduction

Viet Nam has remained the world's leading exporter of cashew nuts for 18 consecutive years, accounting for over 80% of global export volume. In 2024, the country exported more than 765,000 tons of cashew kernels and generated over 4.6 billion USD in revenue, which is a remarkable increase compared to previous year (WTO Center, Viet Nam) [1]. Binh Phuoc is often referred to as the “cashew capital” of Viet Nam, contributes nearly

50% of the national cultivation area and over 40% of the total raw cashew nut output [2].

Cashew nut testa is the thin reddish-brown skin surrounding the kernel that accounts for approximately (1-3)% of the total nut weight. It is generated in large quantities as a by-product of cashew nut processing with a rich source of polyphenols and flavonoids such as catechin, epicatechin, catechin gallate and procyanidins, along with other phytochemicals and nutrients which have demonstrated antioxidant and



antimicrobial properties [3]. Despite the rapid growth of cashew industry, this valuable by-product is often discarded or utilized only as low-value fuel or fertilizers [4, 5]. This highlights the cashew testa extract as a valuable but underutilized resource in fields such as pharmaceuticals, cosmetics, and food technology.

The efficient recovery of bioactive compounds from plant-derived materials strongly depends on the extraction process which directly influences yield, chemical composition, and biological activity [5, 6]. Conventional extraction techniques, including maceration, reflux extraction, Soxhlet extraction, and steam distillation, are still widely employed, yet are often associated with long extraction times, high solvent consumption, and potential degradation of thermolabile compounds [5]. Consequently, increasing attention has been directed toward advanced extraction technologies such as ultrasound-assisted extraction (UAE), microwave-assisted extraction (MAE), supercritical fluid extraction (SFE), and accelerated solvent extraction (ASE), which offer improved efficiency and reduced environmental impact [6]. Among these techniques, UAE has emerged as a promising approach for polyphenol recovery due to its ability to enhance mass transfer through acoustic cavitation, thereby improving extraction efficiency while reducing processing time and solvent usage. Furthermore, UAE is compatible with a wide range of solvents and is considered relatively safe and scalable for industrial applications. Research reported the optimization of UAE conditions for maximizing TPC from cashew testa, demonstrating the feasibility of this technique for phenolic recovery [7]. However, their study primarily focused on process optimization parameters and did not comprehensively evaluate the influence of solvent polarity on extraction efficiency, phytochemical profile, or biological activities of the resulting extracts. In addition, the antioxidant and antimicrobial properties of solvent-specific UAE extracts were not systematically compared.

Therefore, this study aims to evaluate the polyphenol content and bioactivities of cashew nut testa in Viet

Nam using UAE with different solvents. By systematically assessing extraction efficiency, antioxidant capacity, and antimicrobial activity, this work seeks to provide scientific evidence supporting the valorization of cashew testa as a high-value natural resource, thereby contributing to sustainable agricultural by-product utilization and the development of functional ingredients for food, cosmetic, and pharmaceutical applications.

2 Material and Method

2.1 Material

Cashew testa samples were collected from cashew processing factories in Binh Phuoc, Dong Nai.

Chemicals: ethanol, methanol, Na_2CO_3 , AlCl_3 , CH_3COONa , ascorbic acid, Folin - Ciocalteu reagent, quercetin, gallic acid, and 2,2-diphenyl-1-picrylhydrazyl (DPPH) were purchased from Sigma-Aldrich (St. Louis, MA, USA) and Merck (Germany). Brain Heart Infusion (BHI) media (Himedia) were also used.

Microbial strains: *S. aureus* ATCC 25923 and *C. albicans* ATCC 1023 were provided by Biotechnology Center of Ho Chi Minh City, Viet Nam.

2.2 Method

2.1.1 UAE procedure

Cashew testa was collected from Binh Phuoc, Dong Nai province. The raw material was cleaned and dried at 50 °C until the moisture content was below 12%, and ground into a fine powder. UAE was performed out using an ultrasonic bath (maximum power 600 W) with minor modifications [8]. Sample were extracted with different solvents (e.g. ethanol, methanol, and water) at a solid to solvent ratio of 1:10 for 180 min at 60 °C. The extracts were filtered, concentrated under reduced pressure at 40 °C using a rotary evaporator (Buchi, Switzerland), and freeze-dried to obtain powdered extracts. Extraction efficiency was calculated based on extraction yield.

2.2.2 Qualitative phytochemical analysis

Qualitative phytochemical characterization of cashew testa extracts was performed using established qualitative assays based on specific color reactions and precipitation

phenomena to detect major groups of secondary metabolites. Tannins were identified through gelatin precipitation reactions. Polyphenolic compounds were qualitatively detected using a 10% ferric chloride (FeCl_3) reagent, and flavonoids were identified by reaction with concentrated sulfuric acid (H_2SO_4) [9].

To further verify the occurrence of polyphenols and catechins, thin-layer chromatography (TLC) analysis was conducted [10]. Silicagel 60 F_{254} TLC plates (Merck, Germany) were activated at 110°C for 30 min prior to analysis. Separation was achieved using a mobile phase consisting of chloroform: methanol: acetic acid (9:3:3 v/v/v). Extracts and reference standards (gallic acid and catechin, 10 g/L) were spotted onto the plates and developed for (25-30) min. After development, the plates were dried, sprayed with 1% FeCl_3 solution, and heated at ($95\text{-}105^\circ\text{C}$) for (5-10) min. The presence of polyphenols and catechins was confirmed by comparing the color reactions and retention behavior of sample spots with those of the corresponding standards.

2.2.3 TPC determination

The TPC of cashew testa extracts was determined using the Folin-Ciocalteu method, with slight modifications [4]. Briefly, 1 mg of dried extract was dissolved in 1 mL of methanol to obtain a concentration of 1 mg/mL. An aliquot of 0.1 mL of this solution was mixed with 0.5 mL of 10% Folin-Ciocalteu reagent and allowed to react for 10 min. Subsequently, 0.4 mL of 7.5% Na_2CO_3 solution was added, and the mixture was incubated in the dark at 30°C for 1 h. The absorbance was measured at 765 nm using ELISA reader (VersaMAX). All experiments were performed in triplicate. A calibration curve was prepared using gallic acid as a standard under the same experimental conditions, at concentrations of (500, 250, 125, 62.5, 31.25, and 15.625) $\mu\text{g/mL}$. TPC values of the extracts were expressed as milligrams of gallic acid equivalents per gram of extract (mg GAE/g).

2.2.4 Antioxidant activity

2.2.4.1 DPPH scavenging assay

The DPPH scavenging activity was measured as previously described with slight modifications [11].

Briefly, 400 μL of 0.1 mM DPPH solution prepared in methanol was added to 400 μL of extract solution at different concentrations (1.95-62.5) $\mu\text{g/mL}$. The mixtures were incubated in the dark for 30 min, then the absorbance was recorded at 517 nm using a UV-VIS spectrometer (UV-11 model, Spectro, USA). Ascorbic acid at the concentrations of (1.95-62.5) $\mu\text{g/mL}$ was used as the positive control, while methanol was used as the negative control. All experiments were performed in triplicate. The percentage of DPPH radical scavenging activity was calculated using the following equation. Radical scavenging activity (%) = $[1 - (\text{Abs}_{\text{sample}}/\text{Abs}_{\text{control}})] \times 100$. The IC_{50} value, defined as the extract concentration required to inhibit 50% of DPPH radicals, was determined from the linear regression equation obtained by plotting inhibition percentage against sample concentration.

2.2.4.2 ABTS scavenging assay

The ABTS scavenging activity was determined as previously described with minor modifications [12]. The $\text{ABTS}^{+\bullet}$ radical cation was generated by mixing 7 mM $\text{ABTS}^{+\bullet}$ solution with 2.45 mM potassium persulfate ($\text{K}_2\text{S}_2\text{O}_8$) and allowing the mixture to stand in the dark at room temperature for (12-16) h. The solution was diluted with methanol to an absorbance of 0.70 ± 0.02 at 734 nm. Extracts solutions (1.95-62.5) $\mu\text{g/mL}$ were mixed with 900 μL of the $\text{ABTS}^{+\bullet}$ working solution and incubated in the dark for 6 min, and absorbance was measured at 734 nm using a UV-VIS spectrometer (UV-11 model, Spectro, USA). Ascorbic acid at the concentrations of (1.95-62.5) $\mu\text{g/mL}$ was used as positive control, methanol as negative control. All assays were performed in triplicate. ABTS radical scavenging activity was calculated as: Radical scavenging activity (%) = $[1 - (\text{Abs}_{\text{sample}}/\text{Abs}_{\text{control}})] \times 100$. IC_{50} values were determined from the linear regression of inhibition percentage versus concentration.

2.2.5 Antibacterial activity

2.2.5.1 Disc diffusion assay

The antibacterial activity of the cashew testa extract was evaluated using the disc diffusion method with minor

modifications [13]. *S. aureus* ATCC 25923 and *C. albicans* ATCC 1023 were cultured in Brain Heart Infusion (BHI) broth at 37 °C for 24 h, and microbial suspensions were adjusted to 10⁶ CFU/mL. Sterile Mueller-Hinton agar plates were inoculated, and wells (6 mm diameter) were filled with 50 µL of extract (400 µg/mL). Gentamicin (100 µg/mL) served as the positive control. After incubation at 37 °C for 24 h, diameters of inhibition zones were measured. Antibacterial activity was expressed as the inhibition zone diameter (r, mm), calculated as $r = D - d$, where D is the total diameter including the well and d is the well diameter (6 mm). All experiments were performed in triplicate.

2.2.5.2 Minimum inhibitory concentration (MIC)

The MICs were determined by broth microdilution in 96-well method [18]. The extract was poured into a sterile 96-well plate at various concentrations (0.19-100) µg/mL. Subsequently, 100 µL of bacterial suspension cultured BHI medium for (16-18) h at a concentration of (1.5 × 10⁶) CFU/mL was added. A bacterial suspension without extracts was used as a positive control, while negative control was made with only extracts and no bacterial suspension. The 96-well microplate was incubated at 37 °C for 24 h to 48 h. Afterward, 30 µL of 0.01% resazurin was added to each well and incubated further at 35 °C for 30 minutes. The MIC was determined as the lowest concentration of the extract at which the resazurin reagent did not turn pink, indicating the inhibition of microbial growth.

2.3 Data analysis

All experiments were performed in triplicate, and results are presented as mean ± standard deviation (S.D.). Statistical analyses were conducted using Statgraphics

and Microsoft Excel 2013. Differences among treatments were evaluated by analysis of variance (ANOVA), with statistical significance defined at $p < 0.05$.

3 Results and Discussion

3.1 Extraction yield of cashew testa extracts

UAE was performed using ethanol, methanol, and water to assess the effect of solvent type on the extraction efficiency of cashew testa. As shown in Figure 1, the freeze-dried methanol (ME) and ethanol (EE) extracts exhibited a darker and more homogeneous appearance than the water extract (WE), indicating a higher recovery of phenolic-rich components. Methanol achieved the highest extraction yield (36.07 ± 1.70)%, followed by ethanol (26.53 ± 2.87)% and water (15.47 ± 0.31)% (Table 1). These findings indicate that solvent polarity plays a crucial role in the extraction process, with methanol demonstrating the most effective recovery of cashew testa constituents. This enhanced performance is likely due to its suitable polarity, which improves the solubility of a broad range of phenolic compounds compared with ethanol and especially water [15]. Under UAE conditions, acoustic cavitation enhances solvent penetration and mass transfer, thereby facilitating the Additionally, under UAE conditions, acoustic cavitation enhances solvent penetration and mass transfer, facilitating the release of intracellular phenolic compounds [12]. Consistent with previous reports, organic solvents generally yield higher phenolic recovery than aqueous systems under both conventional and ultrasound-assisted extraction conditions [8, 15].

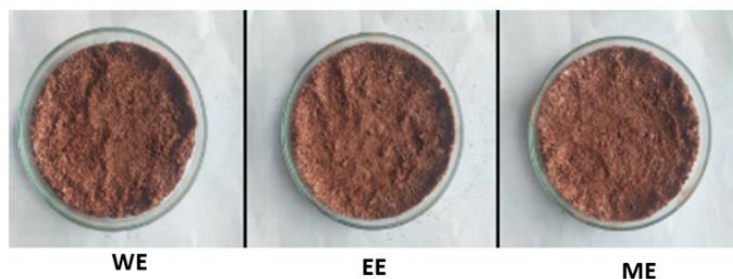


Figure 1 Cashew testa extract powder from different extraction solvent

WE: Water extract, EE: Ethanol extract, ME: Methanol extract.

Table 1 Extraction efficiency (%) of different extraction solvents

Sample	Extraction efficiency (%)
EE	26.53 ^b ± 2.87
ME	36.07 ^a ± 1.7
WE	15.47 ^c ± 0.31

Letter a, b, c: different letters indicate significant differences ($p < 0.05$).

The extraction efficiency of phenolic compounds from cashew testa was strongly influenced by solvent polarity. Methanol and ethanol yielded significantly higher extraction efficiencies than water, consistent with previous reports highlighting the superior solubilization capacity of organic solvents for phenolic compounds [15]. Although methanol showed the highest extraction efficiency, ethanol is generally preferred for food and pharmaceutical applications due to its lower toxicity and regulatory acceptance [16]. Under UAE conditions, acoustic cavitation further enhanced mass transfer and facilitated the release of intracellular phenolic compounds [8].

3.2 Phytochemical screening and TPC

Qualitative phytochemical screening of the cashew testa extracts demonstrated the presence of major bioactive

groups, including phenolics, tannins, and flavonoids, across all tested solvents (Table 2). The presence of phenolic compounds was further confirmed by thin-layer chromatography (TLC), which showed comparable migration behavior and color reactions among the extracts (Figure 2). These results confirm cashew testa as a rich source of natural bioactive compounds, particularly polyphenols. These metabolites are well known for their antioxidant and antimicrobial properties, contributing to the functional potential of the extracts. Moreover, UAE demonstrated effective recovery of phenolic constituents while preserving their native profiles, offering a practical advantage over conventional and enzyme-assisted extraction methods in terms of efficiency and process simplicity.

Table 2 Qualitative phytochemical results

Sample	tannin	flavonoid	polyphenol
EE	+	+	+
ME	+	+	+
WE	+	+	+

WE: Water extract, EE: Ethanol extract, ME: Methanol extract

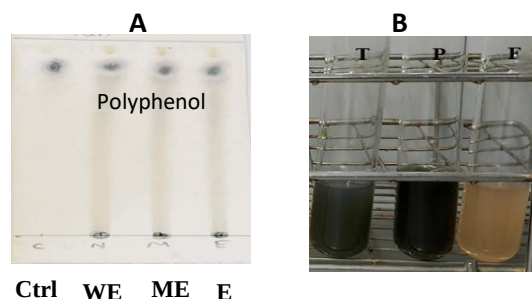


Figure 2 Qualitative phytochemical results (A. TLC method; B. classical biochemical reactions: T: Tanin, P: Polyphenol, F: Flavonoid)

3.3 TPC

The TPC of cashew testa extracts is shown in Table 3. The ME exhibited the highest TPC (484.06 ± 4.26) mg GAE/g, followed by EE, (465.33 ± 4.69) mg GAE/g and WE (453.13 ± 0.65) mg GAE/g. The superior performance of methanol may be attributed to its polarity and hydrogen-bonding capacity which enhance the solubility of phenolic compounds. These

results are consistent with previous studies in which lower TPC values were obtained from ethanol and methanol extracts of cashew testa under conventional and optimized UAE conditions [7]. The high TPC observed in this study confirms that methanol is an efficient solvent for phenolic recovery, while ethanol remains a more practical and safer option for large-scale food and pharmaceutical applications.

Table 3 TPC of cashew testa extract powder obtained from different extraction solvents.

Sample	EE	ME	WE
TPC (mg GAE/g extract)	$465.33^b \pm 4.69$	$484.06^a \pm 4.26$	$453.13^c \pm 0.65$

Notes: Different letters indicate significant differences ($p < 0.05$).

3.4 Antioxidant activity

3.4.1 DPPH radical scavenging activity

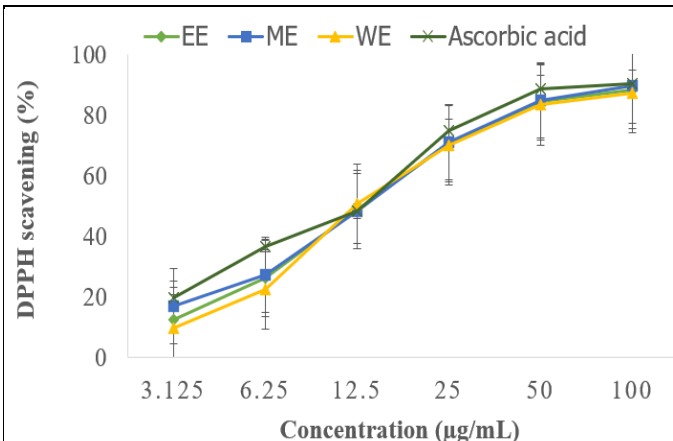


Figure 3 DPPH radical scavenging activity (%) of water extract (WE), ethanol extract (EE), and methanol extract (ME) at different concentrations (3.125-100) µg/mL.

The antioxidant capacity of the cashew testa extracts was first evaluated using the DPPH radical scavenging assay. As shown in Figure 3, all extracts exhibited a clear concentration-dependent increase in radical scavenging activity from 7.61% to 88.56%. Importantly, no significant difference was observed between the extracts and the positive control (ascorbic acid) at higher concentrations. The calculated IC_{50} values were (13.07 ± 0.35) µg/mL, (13.72 ± 0.50) µg/mL, and (14.68 ± 0.54) µg/mL for ME, EE and WE,

Table 4 IC_{50} value of cashew testa extract powder obtained from different extraction solvents

Sample	Equation	IC_{50} value (µg/mL)
EE	$y = 22.787\ln(x) - 9.674, R^2 = 0.9683$	13.72 ± 0.50
ME	$y = 22.451\ln(x) - 7.7085, R^2 = 0.9759$	13.07 ± 0.35
WE	$y = 23.913\ln(x) - 14.249, R^2 = 0.9607$	14.68 ± 0.54
Ascorbic acid	$y = 22.05\ln(x) - 3.6337, R^2 = 0.9603$	11.39 ± 0.96

respectively, compared with (11.39 ± 0.96) µg/mL for ascorbic acid (Table 4). These results indicate that all extracts possess strong radical scavenging potential, with methanol slightly outperforming ethanol and water. The differences in activity may be related to solvent dependent differences in polyphenol concentration, as higher polyphenol yields generally enhance antioxidant capacity.

3.4.2 ABTS radical scavenging activity

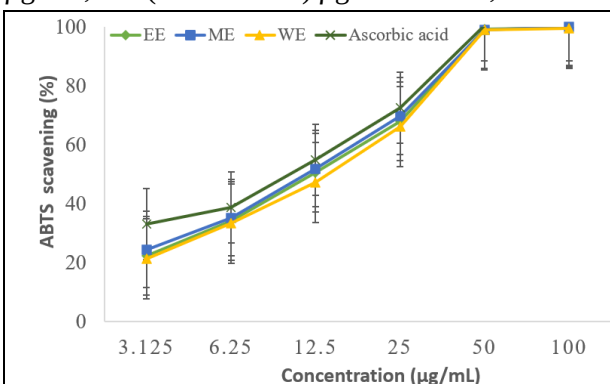


Figure 4 ABTS radical scavenging activity (%) of water extract (WE), ethanol extract (EE), and methanol extract (ME) at different concentrations (3.125-100) µg/mL.

The ABTS assay further confirmed the antioxidant potential of cashew testa extracts. As shown in Figure 4, radical scavenging activity increased in a clear dose-dependent manner from 21.00% to 99.79%. The IC_{50} values were (10.14 ± 0.31) µg/mL, (10.43 ± 0.27) µg/mL, and (11.36 ± 0.26) µg/mL for ME, EE and WE,

Table 5 IC_{50} value of Cashew testa extract powder obtained from different extraction solvents

Sample	Equation	IC_{50} value (µg/mL)
EE	$y = 24.282\ln(x) - 6.9244, R^2 = 0.9545$	10.43 ± 0.27
ME	$y = 23.646\ln(x) - 4.7629, R^2 = 0.9462$	10.14 ± 0.31
WE	$y = 24.956\ln(x) - 10.874, R^2 = 0.9526$	11.36 ± 0.26

respectively, compared with 8.32 µg/mL for ascorbic acid (Table 5). Although less potent than ascorbic acid, both methanol and ethanol extracts exhibited significantly higher ABTS scavenging activity than water extract. The antioxidant performance closely corresponded to the TPC of the extracts, supporting the

established role of phenolic compounds as primary contributors to ABTS scavenging activity. Comparable antioxidant activities of cashew testa extracts have been reported demonstrating strong DPPH $IC_{50} = (12.35 \pm 1.48) \mu\text{g/mL}$ and ABTS $IC_{50} = (33.77 \pm 1.04) \mu\text{g/mL}$ scavenging capacities [4]. Furthermore, comparison with our previous work employing conventional maceration revealed higher IC_{50} values, indicating lower antioxidant activity. This highlights the critical influence of extraction methodology on bioactivity, with UAE providing enhanced polyphenol recovery and preservation, resulting in stronger antioxidant performance.

Overall, both DPPH and ABTS assays demonstrated that all cashew testa extracts are capable of effectively scavenging free radicals. Flavan-3-ols such as catechin, epicatechin, and procyanidins act as effective antioxidants by donating electrons or hydrogen atoms to neutralize free radicals, while the resulting phenoxyl radicals are stabilized through resonance structures, thereby terminating oxidative chain reactions [17]. Phenolic-rich extracts have been reported to reduce intracellular reactive oxygen species (ROS), protect biomolecules from oxidative damage, and activate endogenous antioxidant defense pathways, including Nrf2-mediated signaling, in cellular models [18]. From an application standpoint, these properties support the use of cashew testa extracts as natural antioxidants in food, cosmetic, and pharmaceutical formulations. While ME and EE were more efficient in recovering bioactive phenolics, EE represents a safer and more sustainable solvent for industrial-scale applications.

Collectively, these findings highlight cashew testa as a valuable phenolic-rich by-product with strong antioxidant potential. The ME and EE extracts consistently exhibited higher activities than the water extract, reflecting their greater efficiency in extracting phenolic compounds. These results are consistent with previous studies in which cashew testa contained catechin, epicatechin, and procyanidins that exhibited strong free radical scavenging activities. These findings confirm cashew testa as a valuable source of natural antioxidants, with solvent selection (particularly methanol and ethanol) as a critical factor in maximizing antioxidant efficiency.

3.5 Antibacterial activity

The antimicrobial assays demonstrated that all cashew testa extracts exhibited inhibitory effects against *S. aureus* and *C. albicans* (Figure 5). Among them, EE and ME showed the strongest activity, as indicated by inhibition zones exceeding 11 mm (Table 6). Specifically, the ME produced the largest inhibition zone against *S. aureus* (13.5 ± 0.5) mm, followed by EE (12.5 ± 0.5) mm and WE (11.17 ± 0.29) mm. For *C. albicans*, both EE and ME demonstrated inhibition zones greater than 11 mm, whereas WE displayed the weakest effect (10.5 ± 0.5) mm. The MIC assay further supported these observations (Figure 6). Both EE and ME exhibited strong antimicrobial effects, with MIC values of 1.56 mg/mL and 0.78 mg/mL, respectively. These findings suggested that ME and EE are more effective than water in recovering antimicrobial compounds from cashew testa, likely due to their favorable polarity.

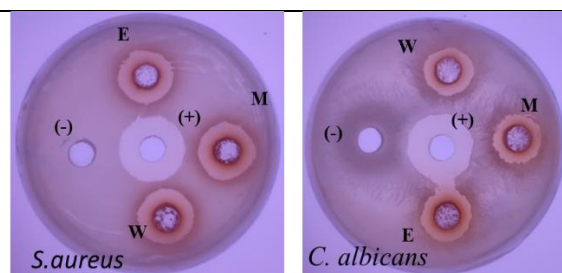


Figure 5 Antibacterial activity of WE, EE and ME against *S. aureus* and *C. albicans* by disk diffusion

Table 6 Measurement of inhibition zone diameters (mm) of WE, EE and ME against *S. aureus* and *C. albicans*

Sample	Diameter of inhibition zone of (mm)	
	<i>S.aureus</i>	<i>C.albicans</i>
EE	$12.5^a \pm 0.5$	$11.17^a \pm 0.29$
ME	$13.5^a \pm 0.5$	$11.33^a \pm 0.58$
WE	$11.17^b \pm 0.29$	$10.5^a \pm 0.5$

Different letters indicate significant differences ($p < 0.05$).

The antimicrobial activity observed for cashew testa extracts is consistent with previous studies reporting strong antibacterial and antifungal properties of polyphenol-rich extracts. In the present study, EE and ME extracts produced significantly larger inhibition zones (> 11 mm) against *S. aureus* and *C. albicans* and exhibited lower MIC values (0.78 - 1.56) mg/mL than WE, indicating superior antimicrobial potency.

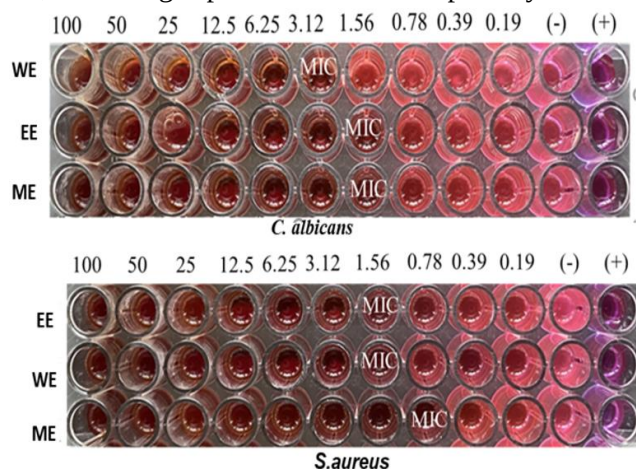


Figure 6 MIC results of WE, EE and ME.

This enhanced activity is closely associated with their higher polyphenol content and more efficient recovery of bioactive compounds. Polyphenols, particularly catechins, flavonoids, and coumarin derivatives, are known to disrupt microbial cell membranes, interfere with essential metabolic enzymes, and inhibit cell

division. Moreover, synergistic interactions among these phenolic constituents may further amplify antimicrobial efficacy. The solvent-dependent differences in inhibition zones and MIC values underscore the critical role of extraction solvent polarity in maximizing the recovery and biological effectiveness of antimicrobial compounds from cashew testa.

4 Conclusion

This study demonstrated that UAE is an effective method for recovering polyphenols from cashew testa, yielding extracts with strong antioxidants and antimicrobial activities. Among the tested solvents, the ME and EE exhibited the highest bioactivity, as evidenced by its superior radical scavenging capacity and inhibitory effects against *S. aureus* and *C. albicans*. These findings highlighted the influence of extraction methods on the biological efficacy of cashew testa extracts and underscore their potential as sustainable value added products derived from agricultural by-products.

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Ứng dụng phương pháp siêu âm trong chiết xuất polyphenol từ vỏ lụa hạt điều (*Anacardium occidentale* L.) và đánh giá hoạt tính chống oxy hóa, kháng khuẩn

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Tóm tắt Vỏ lụa hạt điều (*Anacardium occidentale* L.) là phụ phẩm của ngành công nghiệp chế biến hạt điều, chứa nhiều hợp chất có hoạt tính sinh học. Nghiên cứu này nhằm đánh giá hàm lượng polyphenol và hoạt tính sinh học của cao chiết từ vỏ lụa hạt điều tại Việt Nam bằng phương pháp chiết xuất hỗ trợ siêu âm với các dung môi khác nhau. Kết quả nghiên cứu cho thấy dung môi methanol và ethanol tỏ ra hiệu quả nhất, cho hàm lượng polyphenol tổng số cao nhất lần lượt là 465,33 mg GAE/g và 484,06 mg GAE/g, xác định theo phương pháp Folin - Ciocalteu. Hoạt tính chống oxy hóa của các chiết xuất được đánh giá bằng phương pháp DPPH và ABTS. Ở nồng độ 100 µg/mL, cao chiết methanol và ethanol thể hiện khả năng bắt gốc tự do mạnh, với % ức chế > 87 % (DPPH) và > 99 % (ABTS). Giá trị IC₅₀ tương ứng là 13,07 và 10,14 µg/mL đối với cao methanol, và (13,72 và 10,43) µg/mL đối với cao ethanol. Hoạt tính kháng khuẩn của chúng được đánh giá đối với *Staphylococcus aureus* và *Candida albicans*. bằng phương pháp khuếch tán đĩa thạch và phương pháp Nồng độ ức chế tối thiểu. Kết quả cho thấy cao chiết methanol và ethanol đều cho hiệu quả ức chế rõ rệt, với đường kính vòng vô khuẩn > 11 mm và nồng độ ức chế tối thiểu đạt 1,56 mg/mL. Nhìn chung, kết quả cho thấy chiết xuất hỗ trợ siêu âm kết hợp với lựa chọn dung môi phù hợp là giải pháp hiệu quả để thu nhận cao chiết giàu polyphenol từ vỏ lụa hạt điều, góp phần định hướng ứng dụng phụ phẩm này như nguồn chất chống oxy hóa và kháng khuẩn tự nhiên cho các lĩnh vực thực phẩm, mỹ phẩm và dược phẩm.

Từ khóa Chống oxy hóa; kháng khuẩn; vỏ lụa hạt điều; polyphenols; chiết xuất hỗ trợ siêu âm.

Study on Antioxidant and Cytotoxicity Activities of Fractions from *Coptosapelta Flavescens* Korth. Rubiaceae in Human Breast Cancer MDA-MB-231 Cell Line

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Abstract

Coptosapelta flavescens has been reported to possess various biological activities, particularly antioxidant and cytotoxicity effects. This study evaluated the antioxidant and cytotoxic properties of fractionated extracts against the MDA-MB-231 breast cancer cell line. The fractions were obtained by solvent partitioning with increasing polarity, namely n-hexane, chloroform, dichloromethane, ethyl acetate, and methanol. Total polyphenol content was determined using the Folin–Ciocalteu method, antioxidant activity by the 2,2-diphenyl-1-picrylhydrazyl radical scavenging assay (DPPH), and cytotoxicity by the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide assay (MTT). Polyphenols were primarily distributed in the ethylacetate, chloroform, and dichloromethane fractions, with their concentrations of (186.65, 211.95, and 219.66) μg GAE/mg, respectively. These fractions also exhibited antioxidant capacity with IC_{50} values of (15.24, 12.27, and 11.59) $\mu\text{g/mL}$, respectively. Furthermore, they demonstrated cytotoxic activities against MDA-MB-231 cells, reducing cell viability by 50% after 72 h at concentrations of (88.12, 106.67, and 104.72) $\mu\text{g/mL}$, respectively. These findings indicate that these fractions are promising sources of bioactive compounds with antioxidant and cytotoxic potential.

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Keywords

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Coptosapelta flavescens;
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Polyphenol.

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1 Introduction

Cancer is currently recognized as one of the most pressing global health concerns worldwide. According to the International Agency for Research on Cancer (IARC), an estimated 19.2 million new cancer cases were reported worldwide in 2020 [1]. Among cancer types, breast cancer is the second most prevalent after lung cancer and remains one of the top five causes of cancer-related mortality [2]. In Viet Nam, the incidence of breast cancer

has been steadily rising, with an increasing trend toward younger age at onset. The disease occurs across nearly all post-pubertal age groups, social strata, and occupational backgrounds, underscoring its broad public health impact. Breast cancer is conventionally treated with surgery, radiotherapy, chemotherapy, and hormone therapy. While these modalities improve survival, they are limited by high treatment costs – especially in Viet Nam, where most drugs are imported – and by adverse effects such as



hematological toxicity and alopecia, which impair quality of life [3]. These issues underscore the need for affordable, safer, and more effective alternatives, including plant-derived agents with anticancer potential. Polyphenols, widely present in herbal sources, act as natural antioxidants that scavenge free radicals, modulate nitric oxide activity, and exert antiproliferative and phytoestrogenic effects, offering considerable potential for cancer prevention and therapy [4-6].

Coptosapelta flavescens Korth., Rubiaceae (CF) is a promising medicinal plant in Viet Nam, particularly abundant in the south, with notable pharmacological potential. For a long time, ethnic minorities in Ninh Thuan province (now part of Khanh Hoa province) have used this herb to create a traditional remedy called "Cao Khai" – an aqueous extract, which was utilised for wound cleansing, antimicrobial, and anti-inflammatory purposes. Recent studies have identified key constituents from the roots of *C. flavescens*, including 3 β -O- β -D-quinovopyranosid pyrocincholic acid, acid pyroquinoic

[7] and several anthraquinones [9,67] 1-hydroxyl-2-ethoxycarbonyl anthraquinon, 1-hydroxyl-2- methyl - 4methoxy-anthraquinon, 1-hydroxyl-2-hydroxymethyl-anthraquinon [8]. In 2015, the cytotoxicity of anthraquinones and root extracts against the RD-A cell line was documented, and 1-hydroxyl-2-ethoxycarbonyl anthraquinone showed an IC₅₀ of 9.37 μ M. [8]. These findings highlight *C. flavescens* as a promising source of bioactive compounds with potential anticancer activity, including applications in breast cancer. Therefore, this research was conducted to screen the antioxidant and cytotoxic activities of the CF's fractions with three specific objectives: (1) to determine the total polyphenol content of CF's fractions using the Folin-Ciocalteu method, (2) to evaluate CF's fractions antioxidant activity by the DPPH assay, (3) to assess CF's fractions cytotoxic effects on the MDA-MB-231 breast cancer cell line.

2 Materials and Methods

2.1 Materials

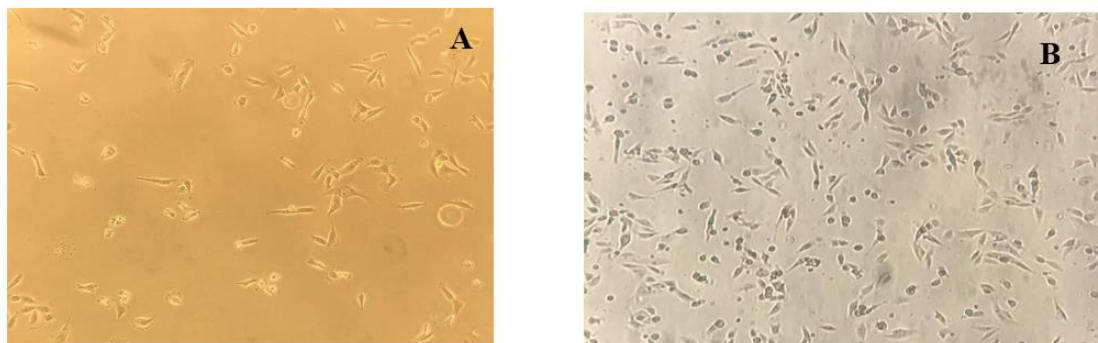


Figure 1 Morphology of MDA-MB-231 cells under inverted microscopy (20X)

A: 2 days after subculture in a 75 cm² flask; B: 1 day after seeding in a 96-well plate at (1.5 × 10⁴) cells/cm²

Samples: six fraction extracts of *C. flavescens* – total (CK), n-hexane (n-hex), chloroform (CHCl₃), dichloromethane (CH₂Cl₂), ethyl acetate (EA), and methanol (MeOH) – were obtained through solvent partitioning with increasing polarity as previously reported [9].

Cell lines: the human breast cancer cell line MDA-MB-231 was sourced from the American Type Culture Collection (ATCC, USA). Cell preservation and culture procedures were performed at the Faculty of Pharmacy, University of Medicine and Pharmacy, Ho Chi Minh City. MDA-MB-231 is a human breast

adenocarcinoma cell line derived from a 51-year-old Caucasian female. With a 38-hour doubling time, it expresses EGF and TGF- α receptors, carries the WNT7B oncogene, and is widely used in *in vitro* models to predict *in vivo* toxicity.

Chemicals and reagents: Dulbecco's Modified Eagle Medium (DMEM), fetal calf serum (FCS), L-glutamine, penicillin-streptomycin, trypsin-EDTA, MTT, dimethyl sulfoxide (DMSO), doxorubicin, 2,2-diphenyl-1-picrylhydrazyl (DPPH), Folin-Ciocalteu reagent, gallic acid monohydrate, quercetin, methanol,

isopropanol, hydrochloric acid (HCl), phosphate-buffered saline (PBS).

Equipment: Analytical Balance (Mettler Toledo), 96-Well Culture Plates, Inverted microscope, UV-Vis Spectrophotometer, Elisa Microplate Reader, Vortex Mixer (VX-200), micropipettes (1 μ L to 10 μ L and 100 μ L to 1000 μ L), Autoclave, CO₂ Incubator, Biosafety Cabinet (AC2-4E1).

2.2 Methods

2.2.1 Determination of total phenol content of CF's fraction extracts

The total phenol content (TPC) was determined using the Folin-Ciocalteu reagent assay [10]. A volume of 50 μ L of each fraction was mixed with 250 μ L of 10% Folin-Ciocalteu reagent in 1.5 mL tubes and reacted for 5 min at room temperature. Subsequently, a volume of 200 μ L of 7.5% Na₂CO₃ was added, and the mixture was incubated in the dark for 1 h. After incubation, the reaction mixture was transferred to a 96-well plate, and absorbance was measured at 765 nm. Each experiment was performed in triplicate, and the mean values were calculated.

Gallic acid served as the standard, and the total polyphenol content was expressed as μ g gallic acid equivalents per mg extract (μ g GAE/mg) according to the following equation:

$$\text{TPC} = (C \times V \times K \times \text{purity})/m \quad (1)$$

Where:

TPC: total polyphenol content (μ g GAE/mg extract)

C: x value obtained from the gallic acid calibration curve (μ g/mL)

V: volume of the sample extract (mL)

K: dilution factor

m: mass of the extract present in the tested volume (mg)

The purity of gallic acid used in the experiment was 0.98.

2.2.2 Determination of total antioxidant capacity (TAC)

Antioxidant activity was initially screened by thin-layer chromatography (TLC) on silica gel 60 F₂₅₄ plates. Six CF fractions and quercetin (1,000 μ g/mL) were spotted (10 μ L), allowed to dry, and briefly immersed in 1 mM DPPH solution for 10 s. Fractions exhibiting yellow spots on a purple background were considered active and selected for subsequent IC₅₀ determination.

Determination of IC₅₀ value of potential fractions using the DPPH radical scavenging assay [11]: Potential fractions were dissolved in appropriate solvents at 1000 μ g/mL and further diluted to concentrations ranging from (0-20) μ g/mL. A volume of 40 μ L of each potential fraction was mixed with 160 μ L of 0.1 mM DPPH in a 96-well plate and incubated in the dark for 30 min. Absorbance was measured at 517 nm. Experiments were performed in triplicate, and mean values were calculated. Quercetin in methanol was used as a positive control. The percentage of DPPH radical scavenging activity was calculated using the following formula (2). Based on the results, a dose-response curve was constructed, from which the IC₅₀ values were determined.

$$\% \text{HTCO} = \frac{A_{\text{control}} - A_{\text{sample}}}{A_{\text{control}}} \times 100 \quad (2)$$

Where

A sample: absorbance of the CF's fractions or the quercetin mixed with the DPPH solution

A control: absorbance of the DPPH solution

2.2.3 *In vitro* Evaluation of cytotoxic activity of CF's fractions on MDA-MB-231 Breast Cancer Cells

Cell viability of MDA-MB-231 cells was evaluated using the MTT assay [12], which relies on mitochondrial succinate dehydrogenase activity. CF's fractions were prepared in 1% DMSO and diluted to (25-200) μ g/mL in culture medium. After 72 h treatment, cells were incubated with 0.5 mg/mL MTT in serum-free medium at 37 °C, 5% CO₂ for (3-4) h. Formazan crystals were dissolved in acidified isopropanol, and absorbance at 570 nm was measured. Cell viability inhibition was calculated as described in formula (3). Based on the results, IC₅₀ values were determined from dose-response curves of percentage inhibition versus extract concentration. Each experiment was performed in triplicate, and the mean values were calculated.

$$\% \text{inhibition} = \frac{A_{\text{negative control}} - A_{\text{sample}}}{A_{\text{negative control}}} \times 100 \quad (3)$$

Where

A_{sample}: absorbance of the CF's fractions or the doxorubicin-treated group



$A_{\text{negative control}}$: absorbance of the negative control group treated with the sample solvent diluted in culture medium

2.2.4 Statistical analysis

The results are shown as mean SD (standard deviation of mean). Pearson correlation analysis was performed using SPSS 22.0 to evaluate the relationship between antioxidant activity and total polyphenol content in these samples.

3 Results and Discussion

3.1 Contents of total polyphenol compounds

Absorbance values at 765 nm of gallic acid standards at concentrations ranging from (1 to 10) $\mu\text{g/mL}$ are presented in Table 1.

Based on these data, a correlation between absorbance and gallic acid concentration was constructed, with the

regression equation $y = 0.0381x + 0.0088$ and a correlation coefficient of $R^2 = 0.9994$, where y represents absorbance and x represents gallic acid concentration ($\mu\text{g/mL}$).

Table 1 Concentration and absorbance of gallic acid standards (Mean $\pm$ SD)

Concentration of GAE ($\mu\text{g/mL}$)	Absorbance (MEAN $\pm$ SD)	Concentration of GAE ($\mu\text{g/mL}$)	Absorbance (MEAN $\pm$ SD)
1	0.044 $\pm$ 0.002	6	0.241 $\pm$ 0.004
2	0.083 $\pm$ 0.002	7	0.273 $\pm$ 0.003
3	0.126 $\pm$ 0.003	8	0.310 $\pm$ 0.002
4	0.164 $\pm$ 0.002	9	0.351 $\pm$ 0.002
5	0.201 $\pm$ 0.004	10	0.390 $\pm$ 0.006

Absorbance values at 765 nm of six fractions are presented in Table 2.

Table 2 Concentration and absorbance of six fractions (Mean $\pm$ SD)

Samples	Concentration ($\mu\text{g/mL}$)	Absorbance (MEAN $\pm$ SD)	Samples	Concentration ($\mu\text{g/mL}$)	Absorbance (MEAN $\pm$ SD)
Total (CK)	100	0.177 $\pm$ 0.003	CH_2Cl_2	30	0.265 $\pm$ 0.003
n-hex		0.328 $\pm$ 0.009	CHCl_3		0.256 $\pm$ 0.004
MeOH		0.213 $\pm$ 0.008	EA		0.2265 $\pm$ 0.0005

Using this regression equation, the x values were determined and then substituted into the formula described in section 2.2.1; TPC of each fraction ($\mu\text{g GAE/mg}$) was calculated, as presented in Table 3.

Table 3 TPC of the total extract and all six fractions

Samples	TPC ($\mu\text{g GAE/mg}$)	Samples	TPC ($\mu\text{g GAE/mg}$)
Total (CK)	43.26	CH_2Cl_2	219.66
n-hex	82.10	CHCl_3	211.95
MeOH	52.52	EA	186.65

TPC of the extracts ranged from (43.26 to 219.66) $\mu\text{g GAE/mg}$, with all fractionated extracts higher than the total extract. The order of TPC from lowest to highest was $\text{CK} < \text{MeOH} < \text{n-hex} < \text{EA} < \text{CHCl}_3 < \text{CH}_2\text{Cl}_2$. Notably, three fractions - EA, CHCl_3 , and CH_2Cl_2 – showed comparable high polyphenol content (186.65, 211.95 and 219.66) $\mu\text{g GAE/mg}$, respectively). Polyphenols are plant compounds with hydroxyl groups on benzene rings, which confer antioxidant properties. Their content is positively correlated with radical-scavenging activity, helping protect the body from oxidative stress [10]. These findings indicated that the CHCl_3 , CH_2Cl_2 , and EA fractions had high antioxidant potential and were therefore selected for further evaluation using *in vitro* DPPH radical scavenging assay.

3.2 Antioxidant potential

The preliminary antioxidant screening of the fractions on silica gel 60 F_{254} TLC plates is shown in Figure 2. Four yellow spots on a purple background (spots 1, 4, 5, and 6) after dipping the silica gel 60 F_{254} TLC plate in 1 mM DPPH, corresponding to quercetin, CH_2Cl_2 , CHCl_3 , and EA fractions, indicating their stronger *in vitro* antioxidant activity compared to the other fractions. In contrast, n-hex, MeOH, CK, and the MeOH solvent control showed no visible activity at 1,000 $\mu\text{g/mL}$. Based on these findings, the CHCl_3 , CH_2Cl_2 , and EA fractions were selected for IC_{50} determination using the DPPH radical scavenging assay.

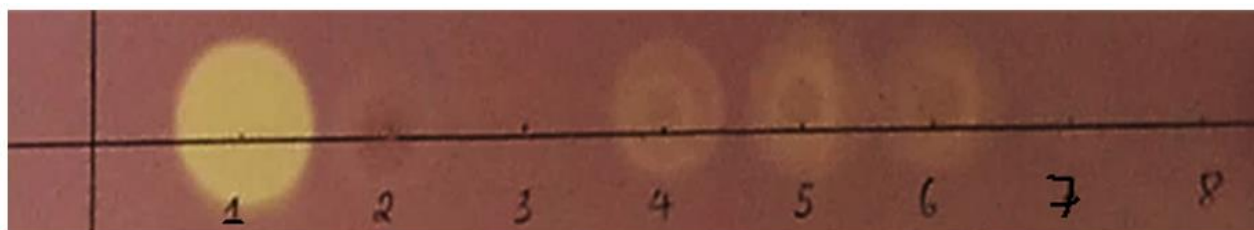


Figure 2 The preliminary antioxidant screening of the fractions

Note: (1) Quercetin; (2) CK; (3) n-hex; (4) CHCl_3 ; (5) CH_2Cl_2 ; (6) EA; (7) MeOH; (8) MeOH solvent

The *in vitro* antioxidant activity of the three potential fractions at concentrations of (2-20) $\mu\text{g/mL}$ and the reference compound quercetin at (0.8-4) $\mu\text{g/mL}$, assessed using the DPPH assay, is presented in Table 4.

Table 4 *In vitro* antioxidant activity of quercetin and CF's potential fractions (Mean $\pm$ SD)

Samples	Concentration $\mu\text{g/mL}$	HTCO (%)	Samples	Concentration $\mu\text{g/mL}$	HTCO (%)
CH_2Cl_2	2	28.75 ± 1.75	EA	2	29.30 ± 0.09
	4	31.41 ± 0.96		4	31.17 ± 1.09
	8	40.94 ± 1.95		8	37.28 ± 0.79
	12	49.83 ± 3.01		12	44.46 ± 0.85
	16	60.39 ± 1.73		16	51.85 ± 1.21
	20	70.79 ± 4.65		20	57.79 ± 0.96
CHCl_3	2	30.14 ± 0.69	Quercetin	0.8	34.80 ± 0.89
	4	31.06 ± 1.87		1.6	42.85 ± 1.75
	8	43.82 ± 3.21		2.4	50.14 ± 3.04
	12	51.85 ± 0.17		3.2	60.10 ± 3.40
	16	55.14 ± 2.08		4	66.65 ± 3.13
	20	65.13 ± 0.17			

The antioxidant activity of all fractions increased proportionally with extract concentration. Specifically, quercetin showed DPPH radical scavenging activity rising from 34.80% at 0.8 $\mu\text{g/mL}$ to 66.65% at 4 $\mu\text{g/mL}$. The CHCl_3 fraction exhibited an increase from 30.14% at 2 $\mu\text{g/mL}$ to 65.13% at 20 $\mu\text{g/mL}$, CH_2Cl_2

from 28.75% to 70.79%, and EA from 29.30% to 57.79% over the same concentration range.

Based on the obtained results in Table 4, dose-response curves were plotted in Microsoft Excel to establish the linear regression equations ($y = ax + b$), from which IC_{50} values were calculated by substituting $y = 50$, as presented in Table 5.

Table 5 IC_{50} values of quercetin and CF's potential fractions

Sample	Linear Equation	R^2	IC_{50} ($\mu\text{g/mL}$)
Quercetin	$y = 10.119x + 26.624$	$R^2 = 0.9969$	2.31
CH_2Cl_2 fraction	$y = 2.365x + 22.579$	$R^2 = 0.9964$	11.59
CHCl_3 fraction	$y = 1.9656x + 25.878$	$R^2 = 0.9764$	12.27
EA fraction	$y = 1.6356x + 25.073$	$R^2 = 0.9965$	15.24

Compared to quercetin, the CHCl_3 and CH_2Cl_2 fractions of CF showed notable antioxidant activity ($\text{IC}_{50} = 12.27$ $\mu\text{g/mL}$ and 11.59 $\mu\text{g/mL}$), about six times higher than

quercetin ($\text{IC}_{50} = 2.31$ $\mu\text{g/mL}$), while the EA fraction was weaker ($\text{IC}_{50} = 15.24$ $\mu\text{g/mL}$), with potency ranking $\text{EA} < \text{CHCl}_3 < \text{CH}_2\text{Cl}_2$. Although less potent than quercetin,

these fractions still demonstrate significant potential as natural antioxidants, which may help mitigate oxidative stress – a key factor in DNA damage, chronic inflammation, and cancer development.

Previous studies reported a positive correlation between antioxidant activity and anticancer effects [13]. Chronic oxidative stress damages biomolecules, lipid peroxides, proteins, enzymes, and DNA, disrupting redox homeostasis and contributing to cancer development. Preventive strategies include minimizing free radical generation, enhancing antioxidant intake, maintaining a healthy lifestyle (avoiding smoking and alcohol, reducing chemical exposure, exercising regularly), and proper nutrition [13]. Some environmental carcinogens can directly generate free radicals and activate inflammatory cells to produce ROS and RNS *in vivo* [13]. Thus, the potent antioxidant activity of *C. flavescentis* fractions suggests potential for preventing oxidative stress-related diseases and supporting cancer therapy. Beyond scavenging ROS and protecting DNA, promoting apoptosis in cancer cells is a current therapeutic strategy. Therefore, this study further evaluated the cytotoxic effects of the fractions *in vitro* using the MTT assay on MDA-MB-231 breast cancer cells.

3.3 Correlation between polyphenol content and antioxidant activity

Pearson's correlation coefficient (r) was used to evaluate the relationship between two quantitative variables. Values of r approaching 1 or -1 indicate a strong linear correlation: positive when close to 1,

negative when close to -1 . Statistical significance was assessed, with $p < 0.05$ indicating a significant linear correlation.

The correlation between polyphenol content and IC_{50} values of the $CHCl_3$, CH_2Cl_2 , and EA fractions was determined using SPSS and is shown in Table 6.

Table 6 Pearson correlation between polyphenol content and IC_{50} of CF's potential fractions

Pearson correlation (r)	Polyphenol	IC_{50}
Polyphenol	1	-0.999^{**}
IC_{50}	-0.999^{**}	1

** : $p < 0.01$, statistically significant

As shown in Table 6, $r = -0.999$ with $p < 0.01$ indicates a strong, statistically significant negative correlation. This means that higher TPC corresponds to lower IC_{50} values and stronger DPPH radical scavenging activity, suggesting that polyphenol content largely determines the antioxidant potential of the tested CF fractions.

3.4 Assessment of the cytotoxic activity of CF's fractions on the MDA-MB-231 human breast cancer cell line

Evaluation of the growth-inhibitory effect of CF's fractions on MDA-MB-231 cells at 200 μ g/mL.

The results of the MTT assay evaluating cell viability after 72 h of treatment with these fractions are presented in Table 7.

Table 7 Cytotoxic activity of these fractions against human breast cancer MDA-MB-231 cells at a concentration of 200 μ g/mL after 72 h of treatment.

Fractions	CK	MeOH	$CHCl_3$	CH_2Cl_2	EA	n-hex
% inhibition	24.30 ± 0.66	19.87 ± 0.87	82.34 ± 0.92	87.48 ± 0.48	85.27 ± 0.61	75.65 ± 0.69

As shown in Table 7, at the concentration of 200 μ g/mL, the dichloromethane (CH_2Cl_2), ethyl acetate (EA), chloroform ($CHCl_3$), and n-hexane (n-hex) fractions exhibited strong cytotoxic activity against MDA-MB-231 cells, with inhibition rates of 87.48%, 85.27%, 82.34%, and 75.65%, respectively. In contrast, the crude extract (CK) and methanol (MeOH) fractions showed relatively weak inhibitory effects, with inhibition rates of 24.30% and 19.80%, respectively, compared with the negative control (DMSO). Accordingly, the cytotoxicity assay was performed

further on the four most promising fractions ($CHCl_3$, CH_2Cl_2 , EA, n-hex fractions) to determine their IC_{50} values, defined as the concentration at which 50% cell growth is inhibited.

3.4.1 Determination of the IC_{50} values of CF's potential fractions against human breast cancer MDA-MB-231 cells

The growth-inhibitory effects of the four most promising fractions and reference of doxorubicin relative to the negative control are summarized in Table 8 and Table 9.

Table 8 Cytotoxic activity and IC₅₀ values of the four most promising fractions against MDA-MB-231 breast cancer cells after 72 h of treatment

Fractions	Inhibition rate compared with negative control (%)				Regression equation	R ²	IC ₅₀ (μg/mL)
	200 μg/mL	100 μg/mL	50 μg/mL	25 μg/mL			
CHCl ₃	82.34 ± 0.92	46.20 ± 0.94	21.24 ± 0.71	4.65 ± 0.57	$y = -0.0011x^2 + 0.6938x - 11.5$	0.9997	106.67
CH ₂ Cl ₂	87.48 ± 0.48	48.09 ± 0.71	26.00 ± 1.18	12.71 ± 1.26	$y = -0.0005x^2 + 0.5287x + 0.1176$	0.9998	104.72
EA	85.27 ± 0.61	52.84 ± 0.72	39.83 ± 0.59	26.11 ± 0.61	$y = -0.0003x^2 + 0.3866x + 18.261$	0.9931	88.12
n-hex	75.65 ± 0.69	31.99 ± 0.45	14.07 ± 1.15	9.91 ± 0.19	$y = 0.0008x^2 + 0.1895x + 3.849$	0.9989	149.36

Based on the dose-response data, the IC₅₀ values of the four fractions and doxorubicin were calculated (Table 8 and Table 9). Among them, the EA fraction exhibited the strongest cytotoxic effect with an IC₅₀ of 88.12 μg/mL, followed by CH₂Cl₂ (104.72 μg/mL) and

CHCl₃ (106.67 μg/mL). In contrast, the n-hexane fraction showed weaker activity with an IC₅₀ of 149.36 μg/mL. In comparison, the standard chemotherapeutic agent doxorubicin showed markedly stronger activity, with an IC₅₀ value of 1.2 μM.

Table 9 Cytotoxic activity and IC₅₀ values of the reference doxorubicin against MDA-MB-231 breast cancer cells after 72 hours of treatment

Fractions	Inhibition rate compared with negative control (%)				Regression equation	R ²	IC ₅₀ μM
	5 μM	2.5 μM	1 μM	0.5 μM			
Doxorubicin	75.49 ± 0.32	71.32 ± 0.46	46.72 ± 1.04	31.91 ± 0.49	$y = -4.0224x^2 + 31.611x + 17.917$	0.9981	1.2

Overall, the CH₂Cl₂, CHCl₃, and EA fractions consistently demonstrated high polyphenol contents, strong antioxidant activity, and moderate cytotoxicity against MDA-MB-231 cells. However, these findings should be considered preliminary because the study was limited to a single cancer cell line and did not include a non-cancerous control cell line. When compared with previous studies, these IC₅₀ values indicate a moderate cytotoxic effect. For example, in 2014, isolated anthraquinone derivatives from *C. flavescens* were reported to exhibit lower IC₅₀ values against Vero cells, ranging from (7.1 to 48.9) μg/mL [14]. Similarly, 1-hydroxyethoxycarbonyl-anthraquinone 1-hydroxy-2-hydroxymethyl-anthraquinone exhibited cytotoxic effects against RD-A cells with IC₅₀ values of 9.371 μM and 92.66 μM, respectively.

The observed cytotoxicity may be associated with the presence of polyphenols, triterpenoids, anthraquinones,

and saponins previously reported in *C. flavescens* [7,15]. In particular, the CH₂Cl₂, CHCl₃, and EA fractions contained high levels of polyphenols and showed strong antioxidant activity, suggesting that these fractions may contribute to cancer cell inhibition partly through the reduction of oxidative stress. Excessive production of reactive oxygen species (ROS) can disrupt the structure of proteins, lipids, lipoproteins, and DNA, leading to a cascade of pre-cancerous events such as cell cycle promotion, gene mutation, abnormal gene expression, cellular senescence, and necrosis [4]. ROS may directly damage DNA or activate aberrant signaling pathways that enhance cancer cell survival [5-6]. Therefore, inhibiting or scavenging ROS is considered an important therapeutic strategy for cancer. These findings suggest that the CH₂Cl₂, CHCl₃, and EA fractions are promising sources of polyphenol-rich compounds with strong antioxidant activity and

potential anticancer effects, providing an initial basis for further screening and mechanistic studies.

4 Conclusions

This research initially screened that polyphenols were enriched in the ethylacetate, chloroform, and dichloromethane fractions of *C. flavescens* Korth (ranging from (186.65 to 219.66) μg GAE/mg), which showed strong antioxidant activity ($\text{IC}_{50} = (11.59\text{-}15.24)$ $\mu\text{g/mL}$) and notable cytotoxicity against MDA-MB-

231 cells ($\text{IC}_{50} = (88.12\text{-}106.67)$ $\mu\text{g/mL}$). These results suggest that these fractions are promising sources of bioactive compounds with dual antioxidant and anticancer potential, warranting further phytochemical and mechanistic studies.

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Đánh giá hoạt tính kháng oxy hóa và độc tế bào trên dòng tế bào ung thư vú MDA-MB-231 của các phân đoạn Dây khai

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Tóm tắt Dây khai (*Coptosapelta flavescens*) được báo cáo có nhiều hoạt tính sinh học, đặc biệt là khả năng chống oxy hóa và độc tế bào. Nghiên cứu này nhằm đánh giá hoạt tính chống oxy hóa và độc tế bào của các cao phân đoạn từ dây khai trên dòng tế bào ung thư vú MDA-MB-231. Các phân đoạn được thu bằng phương pháp lắc phân bố với các dung môi có độ phân cực tăng dần, bao gồm n-hexan, chloroform, dichloromethan, ethylacetat và methanol. Hàm lượng polyphenol được xác định theo phương pháp Folin-Ciocalteu, khả năng chống oxy hóa bằng phương pháp DPPH, và hoạt tính độc tế bào bằng phương pháp MTT. Polyphenol tập trung chủ yếu ở các phân đoạn ethylacetat, chloroform và dichloromethan với hàm lượng tương ứng (186,65; 211,95 và 219,66) $\mu\text{g GAE/mg}$. Các phân đoạn này thể hiện khả năng chống oxy hóa với giá trị IC_{50} lần lượt là (15,24; 12,27 và 11,59) $\mu\text{g/mL}$ và thể hiện hoạt tính độc tế bào trên dòng MDA-MB-231, làm giảm 50 % tỷ lệ sống của tế bào sau 72 giờ ở các nồng độ tương ứng là (88,12; 106,67 và 104,72) $\mu\text{g/mL}$. Kết quả cho thấy các phân đoạn ethylacetat, chloroform và dichloromethan là nguồn hợp chất sinh học tiềm năng với hoạt tính chống oxy hóa và độc tế bào, làm cơ sở cho các nghiên cứu phân lập và xác định cấu trúc hợp chất trong tương lai.

Từ khóa Chống oxy hóa; *Coptosapelta flavescens*; độc tế bào; MDA-MB-231; polyphenol.

Cloning and Expression of Colicin N Exhibiting Activity Against Select Pathogenic *Vibrio* Species in Aquaculture

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Abstract

Antimicrobial resistance in aquaculture pathogenic bacteria is a prevalent threat that contributes to the increasingly uncontrolled misuse of antibiotics. Currently, bacteriocins have been selected as a viable alternative to antibiotics in aquaculture due to their high safety profile. In this study, Colicin N from *Escherichia coli* was cloned, expressed, and purified. Among seven tested *Vibrio* strains, recombinant Colicin at a concentration of 25 µg/mL exhibited inhibitory activity against *Vibrio parahaemolyticus* and *Vibrio owensii*. Furthermore, recombinant Colicin N showed a significant inhibitory effect against *Vibrio vulnificus* and *Vibrio harveyi* at a concentration of 12.5 µg/mL. The results of this study demonstrate the in vitro inhibitory activity of Colicin N against several pathogenic *Vibrio* species, highlighting its potential for further development as an antimicrobial agent in shrimp aquaculture.

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1 Introduction

The sector of export-oriented aquaculture, particularly shrimp cultivation, is expanding annually on a larger scale. Nonetheless, aquaculture farming enterprises are constantly challenged by a high incidence of shrimp diseases primarily attributed to bacterial pathogens [1]. The transition from extensive to intensive farming practices, coupled with the frequent or uncontrolled use of antibiotics, has led to the widespread dissemination of antibiotic-resistant bacteria in shrimp farms [1-3]. The major pathogen groups consistently documented within the aquaculture environment and farming facilities include *Vibrio* spp., *Aeromonas*, *Enterobacterales* (such as *Escherichia coli* and *Klebsiella*), and *Enterococcus* [1-3]. Aquaculture dependent on antibiotics is the main

reason for the rapid dissemination of these organisms, frequently harboring Antibiotic Resistance Genes (ARGs), which are typically located on plasmids and associated with mobile genetic elements (MGEs) [1]. This is concerning because the prevalent pathogens in shrimp ponds (*Vibrio* and *Aeromonas* species) are indigenous to the surrounding marine and estuarine environments, which enables drug-resistant bacterium strains to become dominant [1]. One strategy to mitigate antibiotic use is to identify and develop of novel antimicrobial agents.

Bacteriocins are ribosomally synthesized peptides or proteins secreted by bacteria into the environment, primarily functioning to eliminate competing bacterial strains [4]. The potential of bacteriocins to effectively

inhibit antibiotic resistant bacteria represents a promising research direction for treating drug resistant bacterial diseases [4]. Colicins are a specific class of bacteriocins produced by *E. coli* that exhibit antimicrobial activity by piercing the cell membrane of target bacteria, leading to death by cytoplasm leakage [5]. Colicins use discrete structural modules that are designated to distinct functional steps in recognition, import, and killing. Each module's sequence and fold tune receptor specificity and the mechanism of target inactivation. T-domain mediates translocation across the outer membrane and interacts with import machinery or pores on the target cell surface. R-domain binds specific outer-membrane receptors to dock the Colicin. C-domain executable toxic activity results in bactericidal activity [6].

Colicin activity depends on inter-domain mechanics, surface electrostatics, catalytic motifs, and the immunity-binding architecture. The opening of a hinge or extension of the translocation segment is required for some Colicins to enter or plug an outer-membrane channel, thereby enabling import or channel occlusion [6]. Additionally, amphipathic helices and membrane-inserting C-terminal segments specifically form conductive pores in pore-forming Colicins. Full length architecture and surface charge distribution can modulate potency, as increasing the positive surface charge on colicin N enhanced cytotoxicity in experimental mutagenesis [7].

Among the various types of Colicins, Colicin N is noted for having the smallest molecular size, which potentially enhances its ability to penetrate target cells [8]. Furthermore, Colicin N has been demonstrated to possess the capability to inhibit lung cancer cells [9]. In this study, we hypothesized that the structural compactness and positive surface charge of Colicin N enabled it to overcome barriers specific to a genus and target conserved outer membrane porins in *Vibrio* species. Because colicin N is the smallest pore-forming colicin that has been reported [10]. To test this, we cloned and expressed recombinant Colicin N to evaluate its efficacy against diverse *Vibrio* isolates from Vietnamese shrimp farms. The potency of Colicin N was

systematically assessed based on MIC thresholds and its comparative activity across different pathogenic strains.

2 Materials and Methods

2.1 Bacterial strains and reagents

Vibrio parahaemolyticus, *Vibrio alginolyticus*, *Vibrio cholerae*, *Vibrio vulnificus*, *Vibrio harveyi*, *Vibrio owensii*, and *Vibrio mimicus* strains were kindly provided by Dr. Vu Van Van from NTT Hitech Institute, Nguyen Tat Thanh University, Viet Nam. *Escherichia coli* BL21 (DE3) and *E. coli* DH5 α were kindly provided by Dr. Nguyen Hoang Dung from the Institute of Life Sciences, Viet Nam Academy of Science and Technology, Viet Nam.

NdeI (R0111L), XhoI (R0146L), *Taq* 2X Master Mix (M0270L), T4 DNA Ligase (M0202L), Monarch[®] DNA Gel Extraction Kit (T1020L), 1 kb DNA Ladder (N3232L), Color Prestained Protein Standard (P7719L), and Monarch[®] Plasmid Miniprep Kit (T1010L) were purchased from New England Biolabs, USA. Ni Sepharose[™] 6 Fast Flow (GE17-5318-02), Imidazole (I202-500G), sodium phosphate (S9763-500G), isopropyl β -D-1-thiogalactopyranoside (IPTG) (I6758-1G), and Sodium chloride (S9888-10KG) were purchased from Merck, Germany. PD-10 desalting columns packed with Sephadex[™] G-25 resin were purchased from Cytiva, USA. Bacterial culture media were purchased from BD, Canada.

Primers for PCR reaction were purchased from Integrated DNA Technologies, USA.

Colicin N DNA sequence (GenBank: CP027259) was synthesized by the gene synthesis service of GenScript, China.

2.2 Primer design

Based on the sequences of the colicin N gene (GenBank: CP027259), the primers NdeI-colicin (5'-TATATACATATGGGTAGTAATGGCGCAGATA-3') and XhoI-colicin (5'-TATATACTCGAGTCATCGAATAACACTAGATA-3') were designed for the purpose of amplifying the Colicin gene, introducing a restriction site for the enzyme NdeI at the 5' end and a restriction site for XhoI at the 3' end. The size of the amplified gene fragment

is approximately 1185 bp. The primers Pet28bF (5'-CCTGGTGGCGCGCGGCAGCC-3') and Pet28bR (5'-TCAGTGGTGGTGGTGGTGGT-3'), which amplify the cloning site on the pET-28b(+) vector, were designed for use in a Colony PCR reaction. The size of the amplified gene fragment is approximately 1213 bp.

2.3 Construction of the recombinant plasmid pET28b(+)-colicin

pET-28b(+) plasmid and PCR product amplifying the colicin gene were double-digested using NdeI and XhoI, followed by ligation using T4 DNA Ligase. The recombinant plasmids were transformed into *E. coli* DH5 α , and the positive clones were screened by kanamycin resistance. Colony PCR was utilized to screen bacterial colonies that carry the Colicin gene inserted into the pET-28b(+) plasmid. The recombinant plasmid pET-28b(+)-colicin was recovered from *E. coli* DH5 α colonies via plasmid extraction and subsequently transformed into *E. coli* BL21 (DE3) for protein expression.

2.4 Expression of Colicin protein

To express Colicin protein, a colony of *E. coli* BL21 (DE3) carrying the pET-28b(+)-colicin plasmid was inoculated into 10 mL of Luria Broth (LB) medium supplemented with kanamycin (50 μ g/mL) and incubated at 37 °C while shaking at 250 rpm for 3 h. Subsequently, the entire 10 mL culture was transferred into 500 mL of LB medium supplemented with kanamycin (50 μ g/mL) and cultivated at 37 °C with shaking at 250 rpm until the bacterial density reached an OD₆₀₀ of 0.5 (approximately 5×10^8 CFU/mL). Afterward, we evaluated the impact of (0.1, 0.2, 0.4, 0.8 and 1) mM IPTG (final concentration) on protein expression at 20 °C. After 18 h of induction, *E. coli* cells were harvested by centrifugation, and the resulting pellet was collected. The recombinant Colicin protein was then lysed using a Sonicator (operating at 20 kHz and 150 W). Colicin protein was purified using a Ni-NTA agarose column, and subsequently, the purity of the protein was verified by SDS-PAGE and Coomassie Brilliant Blue staining. Following Ni-NTA purification, the recombinant Colicin N was

concentrated by Vivaspin column and stored in Tris pH = 7.0 with 15% glycerol at -20 °C. Before performing the MIC assays, the protein underwent a secondary purification step using PD-10 desalting columns and replacing the storage buffer with 0.1% NaCl. The protein concentration was determined based on the spectrophotometric absorbance value at a wavelength of 280 nm, utilizing the Beer-Lambert Law equation [11]:

$$A = \epsilon cl.$$

Where:

A: the absorbance value measured at a wavelength of 280 nm.

ϵ : the molar absorptivity (or extinction coefficient) of the protein, typically expressed in units of [mg/mL]⁻¹ cm⁻¹ or [M]⁻¹cm⁻¹.

c: the protein concentration, expressed in mg/mL or M (Molar).

l: the path length of the cuvette (cm).

2.5 Antibacterial assay

The antimicrobial efficacy of recombinant Colicin N was evaluated against a panel of *Vibrio* species, including *V. parahaemolyticus*, *V. alginolyticus*, *V. cholerae*, *V. vulnificus*, *V. harveyi*, *V. owensii*, and *V. mimicus*. All strains were isolated from the pond water of whiteleg shrimp (*Litopenaeus vannamei*) aquaculture facilities. Minimum inhibitory concentration (MIC) was determined using the resazurin microtiter assay to provide a quantifiable and metabolic endpoint. The assay was performed in plates with 96 wells using Mueller Hinton Broth (MHB) adjusted with cations supplemented with 1.5% NaCl to support the optimal growth of *Vibrio*. Bacterial inocula were standardized to approximately 10^6 CFU/mL. A serial dilution of Colicin N was prepared in the culture medium, resulting in ten final concentrations: (25, 12.5, 6.25, 3.125, 1.563, 0.781, 0.39, 0.195, 0.098, and 0.049) μ g/mL. The plates were incubated at 35 °C for 24 hours. Following incubation, 0.02% (w/v) resazurin solution was added to each well and incubated for an additional 2 hours. The MIC was defined as the lowest concentration of Colicin N that prevented the color transition from blue to pink, indicating a complete

inhibition of bacterial metabolic activity. To ensure reproducibility and robustness, all experiments were conducted in three independent biological replicates, with each sample tested in triplicate.

3 Results and Discussion

3.1 Cloning, expression, and purification of Colicin N

The PCR product of the Colicin gene, using NdeI-colicin and XhoI-colicin primers, was double-digested with the restriction enzymes NdeI and XhoI. The pET-28b(+) plasmid was processed similarly. Both the digested gene fragments and the vectors were then subjected to agarose gel electrophoresis for size verification and purification prior to the ligation step (Figure 1A). After correct sequence verification, the Colicin gene was cloned into the pET-28b(+) plasmid. The Colony PCR method was used to verify the presence of the recombinant plasmid pET-28b(+)-colicin (Figure 1B). The observed size of the Colicin fragment on the electrophoresis gel was approximately 1.2 kb. The recombinant plasmid, following purification, was transformed into *E. coli* BL21 (DE3) for Colicin protein expression and purification.

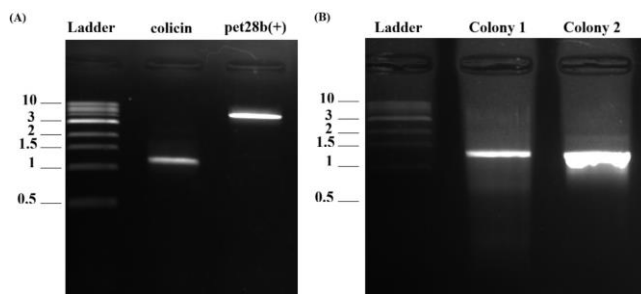


Figure 1 Agarose gel electrophoresis of Colicin fragment, recombinant plasmid pET-28b(+), and colony PCR products. (A) The colicin fragment and pET-28b(+) were double-digested with NdeI and XhoI. (B) PCR amplification of the colicin gene from two *E. coli* BL21 (DE3) colonies.

Recombinant Colicin was overexpressed using (0.1, 0.2, 0.4, 0.8, and 1) mM of IPTG for 18 h at 20 °C (Figure 2A). The results of induction across various concentrations of IPTG showed no significant differences, indicating that Colicin N can be expressed efficiently at a concentration as low as 0.1 mM. Therefore, all subsequent experiments to produce

recombinant Colicin N utilized a concentration of 0.1 mM. The wash fraction using the NPI-10 buffer did not contain Colicin (Figure 2B, lane 7), indicating that the protein was bound to the Ni-NTA column. Then, Colicin was eluted with NPI-60, NPI-90, NPI-120, NPI-200, NPI-300, and NPI-500 buffer as a single band. Purified recombinant Colicin was quantified using the Beer-Lambert Law equation. The purified Colicin was concentrated using a Vivaspin 20 ml concentrator (50K MWCO) and subsequently diluted to specific concentrations for the MIC test to evaluate its antimicrobial activity.

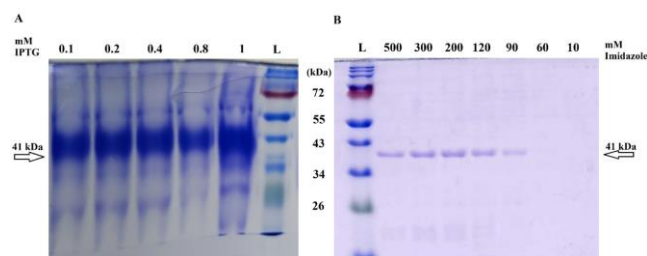


Figure 2 Colicin was analyzed by SDS-PAGE electrophoresis. (A) Colicin overexpression. (B) Colicin purification using Ni-NTA agarose.

3.2 Assessment of Colicin's inhibitory activity against *Vibrio* strains

The susceptibility of the seven isolates of *Vibrio* to Colicin N is presented in Figure 3. The assay used serial dilutions with a ratio of one to two, starting from a maximum concentration of 25 µg/mL down to 0.05 µg/mL. Among the tested isolates, Colicin exhibited inhibitory activity against four species: *V. parahaemolyticus* and *V. owensii* showed a MIC of 25 µg/mL, while *V. vulnificus* and *V. harveyi* demonstrated higher sensitivity with a MIC of 12.5 µg/mL. In contrast, no inhibitory effect was observed for *V. alginolyticus*, *V. cholerae*, and *V. mimicus* even at the highest concentration tested (Table 1). While the purification process included a secondary desalting step using PD-10 columns to eliminate residual imidazole and low molecular weight components, the absence of an empty vector protein control is a recognized limitation of this study. However, it is noteworthy that *E. coli* BL21 (DE3) is not known to endogenously produce bacteriocins or peptides with

specific inhibitory activity against *Vibrio* species under the tested conditions [12-13]. Therefore, the observed antimicrobial effects are highly likely attributable to the purified recombinant Colicin N. Nonetheless, future studies should incorporate an empty vector control to provide a more definitive validation of protein specificity.

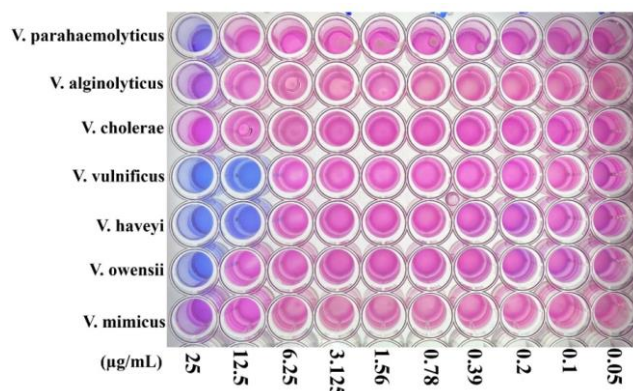


Figure 3 MIC for Colicin N on seven strains of *Vibrio*

This variation in susceptibility suggests that the antibacterial action of Colicin N is dependent on specific biological interactions with each *Vibrio* species. The resistance observed in *V. alginolyticus*, *V. cholerae*, and *V. mimicus* base on compatible receptors on the outer membrane, or a low efficiency in the translocation machinery required for the protein to reach its target [10]. Conversely, the higher sensitivity of *V. vulnificus* and *V. harveyi* might reflect a more effective binding affinity to their surface proteins or a more rapid penetration through the periplasm. These differences highlight the importance of the molecular structure of the cell envelope in determining the range of activity for Colicin N across the *Vibrio* genus.

Acknowledgement

This research is funded by Nguyen Tat Thanh University, Ho Chi Minh City, Viet Nam, under grant number 2025.01.214.

Table 1 Determination of the MIC by Resazurin aided microdilution method of Colicin N against seven *Vibrio* strains

Bacteria	MIC reported in this study (µg/mL)
<i>V. parahaemolyticus</i>	25
<i>V. alginolyticus</i>	-
<i>V. cholerae</i>	-
<i>V. vulnificus</i>	12.5
<i>V. harveyi</i>	12.5
<i>V. owensii</i>	25
<i>V. mimicus</i>	-

4 Conclusion

In conclusion, recombinant Colicin N was successfully expressed in *E. coli* BL21 (DE3) and purified using Ni-NTA affinity chromatography. The purified Colicin N showed specific *in vitro* inhibitory activity against four different *Vibrio* spp.. The MIC values of 12.5 µg/mL for *V. vulnificus* and *V. harveyi*, 25 µg/mL for *V. parahaemolyticus* and *V. owensii* were recorded. However, no inhibitory activity was observed against *V. alginolyticus*, *V. cholerae*, and *V. mimicus* at the maximum concentration (25 µg/mL).

This study also has certain limitations, particularly the absence of a negative control in the MIC assays. Further research is needed in considering its potency in combating Colicin N *in vivo* experiments, also checking its safety concerning microbiota in the host. These investigations are necessary to determine its potential applicability in shrimp farming.

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Tạo dòng và biểu hiện Colicin N có hoạt tính kháng một số loài *Vibrio* gây bệnh phân lập từ môi trường nuôi thủy sản

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Tóm tắt Kháng kháng sinh ở vi khuẩn gây bệnh trong ngành thủy sản là một mối đe dọa phổ biến, phát sinh từ việc lạm dụng kháng sinh ngày càng thiếu kiểm soát. Bacteriocin hiện đang được quan tâm nghiên cứu như một giải pháp thay thế khả thi cho kháng sinh trong ngành thủy sản nhờ vào tính an toàn cao. Trong nghiên cứu này, Colicin N nguồn gốc từ vi khuẩn *Escherichia coli* đã được nhân dòng, biểu hiện và tinh sạch. Trong số bảy chủng *Vibrio* nguồn gốc từ ao nuôi tôm được thử nghiệm, Colicin N tái tổ hợp ở nồng độ 25 µg/mL cho thấy hoạt tính ức chế đối với *Vibrio parahaemolyticus* và *Vibrio owensii*. Đối với hai chủng *Vibrio vulnificus* và *Vibrio harveyi* thì Colicin N thể hiện hoạt tính ức chế ở nồng độ 12,5 µg/mL. Kết quả của nghiên cứu này chứng minh hoạt tính ức chế *in vitro* của Colicin N đối với *Vibrio* gây bệnh, gợi mở tiềm năng ứng dụng trong việc chống lại bệnh do *Vibrio* trong nuôi tôm.

Từ khóa AMR; Colicin N; *Vibrio*; AHPND; *Escherichia coli*.

Experimental Evaluation and Modeling of a Regenerative Braking System for an Electric Motorcycle

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Abstract

This research presents an experimental evaluation, modeling and simulation of a regenerative braking system for an electric motorcycle based on the VinFast Klara S platform equipped with a 1.2 kW brushless direct current motor. An experimental test bench was established to examine the effects of battery state of charge and braking power on energy recovery efficiency. The results indicate that the maximum recovery efficiency reached 22.5% at a braking power of 200 W, within an optimal state of charge range of (50-60)%. A Matlab/Simulink model was developed and validated using experimental data to assess the impact of the RBS on the motorcycle's driving range. Under the proposed urban driving cycle, the driving range per full charge increased from 63.64 km to 67.06 km, corresponding to an improvement of 5.4%. The study also clarified the influence of the rectifier-boost converter efficiency, battery internal resistance and proposed directions to enhance the performance of regenerative braking systems for small electric motorcycles.

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Keywords

RBS system; EM;
energy recovery;
simulation; efficiency.

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1 Introduction

In recent years, electric vehicles (EVs) have been considered as one of the efficient means to decrease greenhouse gas emissions and to improve energy efficiency of the transportation sector. Electric powertrain efficiency and environmental effects have been studied in numerous studies in the context of on road urban mobility [1]. One of the major constraints of EVs is the restricted driving range which depends heavily on the battery capacity and the energy management strategy. Regenerative braking system (RBS) have been widely used in electric and hybrid electric vehicles to utilize electric energy efficiently.

RBS enables the recovery of a portion of the vehicle's kinetic energy during braking and converts it into electrical energy stored in the battery of the vehicle [2]. Therefore, RBS have been reported to help increasing the vehicle's energy efficiency and extending the driving range which is highly beneficial for urban driving conditions as they involve large amounts of braking and acceleration [3].

At present, most studies on RBS are concentrated on EVs and hybrid electric vehicles (HEVs), where sophisticated control algorithms and power electronics are designed to enhance the conversion of braking energy via the traction motor [4]. Intelligent braking control and motor control strategies for enhancing the

braking torque and stability of the entire vehicle during RBS have also been studied [5]. In recent years, RBS technology has been increasingly focused on application in light-weight vehicles such as electric motorcycles and electric bikes. A few modeling and simulation techniques have been proposed to study the implications of RBS on vehicle performance and battery state of charge (SOC) [6, 7]. These results suggest that the energy efficiency of such vehicles can be improved via regen braking. However, the realistic efficiency of braking energy recovery is generally limited by losses in power electronics processing and limitations related to batteries performance. There are few studies on RBSs specifically designed for electric motorcycles (EMs), and most of the research is concentrated on simulation or conceptual design without experimental validation [8, 9]. Hence, integrated studies combining experimental testing and simulation modeling for EMs are still necessary. This study was initiated to address the lack of RBS in the literature for Vietnamese EVs. We introduce an analytical RBS for the VinFast Klara S electric motorcycle (EM) powered by a brushless direct current (BLDC) hub motor. This study investigated the performance of the RBS for the motorcycle test with the identified parameters implemented in Matlab/Simulink.

2 Theoretical Background

2.1 Operating principle of the RBS

EMs have a drivetrain system that converts the electrical energy from the battery to initiate movement, basically into a mechanical force that turns the wheels of the motorcycle. Among various types of motors applied, BLDC motor is the most common type of motor used in EMs due to several advantages, including light-weight design and high energy efficiency. However, the BLDC motor still has several limitations that may affect its suitability for use in EM operation [10].

In traction mode, a permanent magnet BLDC motor acts as the prime mover. The throttle signals are received by the controller, and it controls the battery current flow using a DC-AC inverter and feeds the

three-phase currents to the stator windings. These currents produce a rotating magnetic field, which interacts with the rotor magnets to produce torque; and back electromotive force (EMF) is used to detect rotor position and perform correct commutation.

In decelerating or braking operation (regenerative mode), the BLDC motor acts as a generator. As a result of inertia, the rotor is turned due to the weight of the motorcycle and induce an EMF in the stator windings, which creates a regenerative current. This current is fed back into the battery converting some of the motorcycle's kinetic energy into electrical energy. The electromagnetic torque also acts to slow down the motorcycle, assisting with the deceleration process.

2.2 Selection of base motorcycle

Several major manufacturers are currently active in the Vietnamese EM market, including VinFast, Pega, and Yadea. Among them, VinFast leads the market with the largest share in recent years, as reported by the International Council on Clean Transportation (ICCT) [11]. In this study, VinFast Klara S is used as a representative platform to develop and evaluate a RBS. Its technical specifications are mainly employed for model calibration and driving cycle development, ensuring realistic operating conditions.

Table 1 The VinFast Klara S EM's technical parameters [13]

Motor Type	12 inch BLDC
Rated Power/Peak power (w)	1,200 / 1,700
Battery Type	Lithium 22 Ah 60 V
Full Charging Time (h)	3 h (90% SOC) or 4,8 h (100% SOC) per battery
Max Speed (km/h)	48
Load Capacity (kg)	150
Motorcycle and Battery Mass (kg)	108

The objective is not to replicate the original motorcycle configuration, but to assess the effectiveness of RBS in terms of energy recovery and driving range improvement. Therefore, while a 1.2 kW BLDC motor is retained, a 60 V/20 Ah lead-acid battery was adopted. This choice reflected the widespread use of lead-acid

batteries in low-cost electric motorcycles, which typically lack RBS. Thus, the study focuses on evaluating the practical benefits of RBS for this vehicle category. The differences between lithium-ion and lead-acid batteries, particularly in terms of internal resistance and charge-discharge behavior, are acknowledged as a limitation and will be addressed in future work.

2.3 Development of driving cycle

Furthermore, the motorcycle speed can be different by changing the velocity input signal. This input was random and depended entirely on the rider's will, according to the driving cycle. From this, the total resistive force at a given speed defined by the cycle can be obtained ($F_{Load}(V_{demand}(t))$). A PI control algorithm was then applied to regulate the actual motor speed ($V_{act}(t)$). Consequently, the simulation yielded the final outcomes, namely the demanded power (P_{k_demand}) and the demanded traction force (F_{k_demand}) as functions of time-varying motorcycle speed.

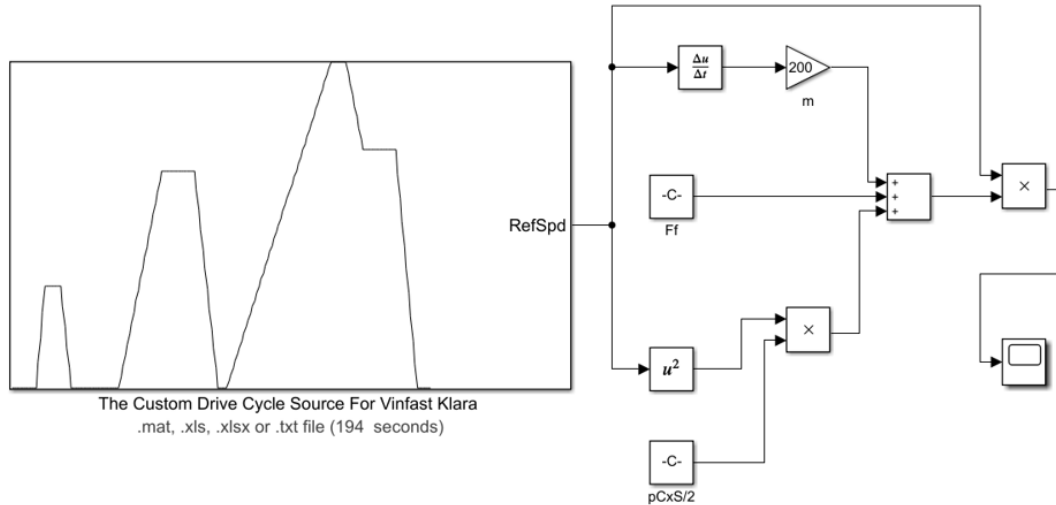


Figure 1 Dynamic model of EM movement

In this research, based on the technical parameters shown in Table 1, a motorcycle dynamics model was built by using Matlab/Simulink. A corresponding test cycle that suits the operating ability of the motorcycle was selected while keeping the traction requirement below the peak power of the motor. The simulation analyses the performance of the motorcycle in terms of traction force, which is determined by the balance of

$$F_{demand}(t) = F_c(t) + F_{PI}(t) \quad (1)$$

$$F_c(t) = F_f(t) + F_w(t) + F_j(t) \quad (2)$$

$$F_{PI}(t) = K_p e(t) + K_i \int_0^t e(\tau) \quad (3)$$

$$e(t) = V_{demand}(t) - V_{act}(t) \quad (4)$$

$$P_{demand}(t) = F_{demand}(t) \times V_{demand}(t) \quad (5)$$

A dedicated test cycle must be developed specifically for EMs in Viet Nam to objectively assess the effectiveness of RBS during simulation. Standard test cycles such as ECE-R40 and Japan 10-Mode are commonly used for urban performance simulations of passenger motorcycles. However, these cycles are not suitable for the Klara S model, which has a rated power of only 1,200 W (1,700 W peak) and falls under the EM category according to regulation for motorcycles with motor power below 4 kW [12]. In addition, the maximum speed of the VinFast Klara is limited to 48 km/h (see Table 1)

traction and resistance forces through motorcycle dynamics equations.

$$F_f + F_w \pm F_j \pm F_i \leq F_k \leq F_{k_{max}} \quad (6)$$

The quantities F_f , F_w , F_j , and F_i represent the rolling resistance, aerodynamic drag, inertial resistance, and gradient resistance, respectively. In this study, it was assumed that the motorcycle operated on flat roads within urban areas; thus, the gradient resistance was

neglected, and only rolling resistance, aerodynamic drag, and inertial resistance were considered.

$$F_f = G \times f = m \times g \times f \quad (7)$$

$$F_W = \frac{1}{2} \times C_d \times A \times \rho \times V^2 \quad (8)$$

$$F_{max} = m \times j_{max} \quad (9)$$

Once the total resistive forces were determined, they were combined with motorcycle velocity to obtain the required traction power according to the following equation:

$$P_k \geq P_c = F_c \times V \quad (10)$$

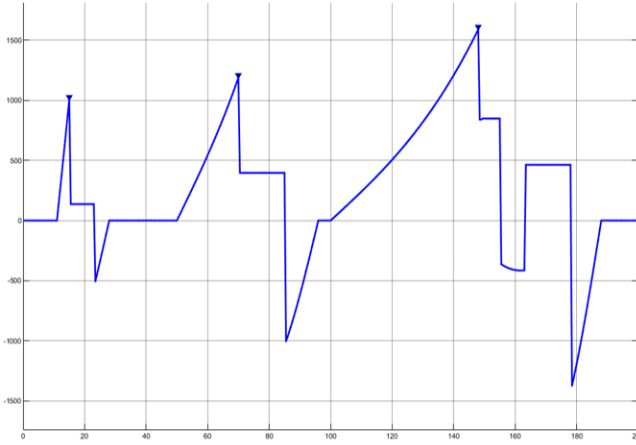


Figure 2 Motorcycle power demand during the adjusted driving cycle

A motorcycle dynamics simulation model was developed to examine the suitability of the proposed test cycle. In this model, the resistive force components were represented as separate functional blocks, thereby enabling an accurate assessment of the cycle's required power and verifying its feasibility according to equation. Simulation results in Figure 2 showed that the maximum traction power was about 1,589 W, below the motor peak power of 1,700 W. Therefore, the test cycle is appropriate.

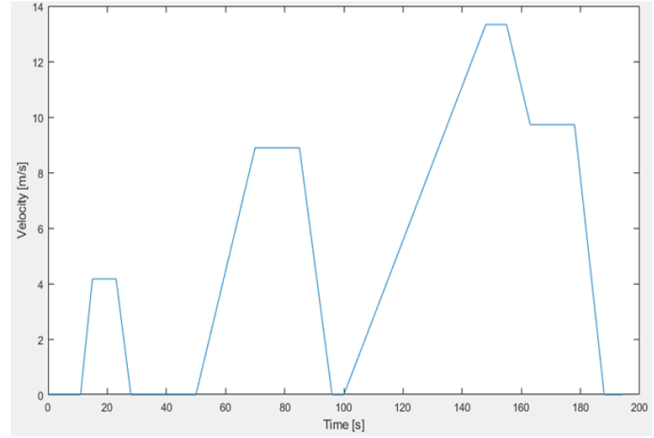


Figure 3 Adjusted driving cycle based on the ECE-R40 standard

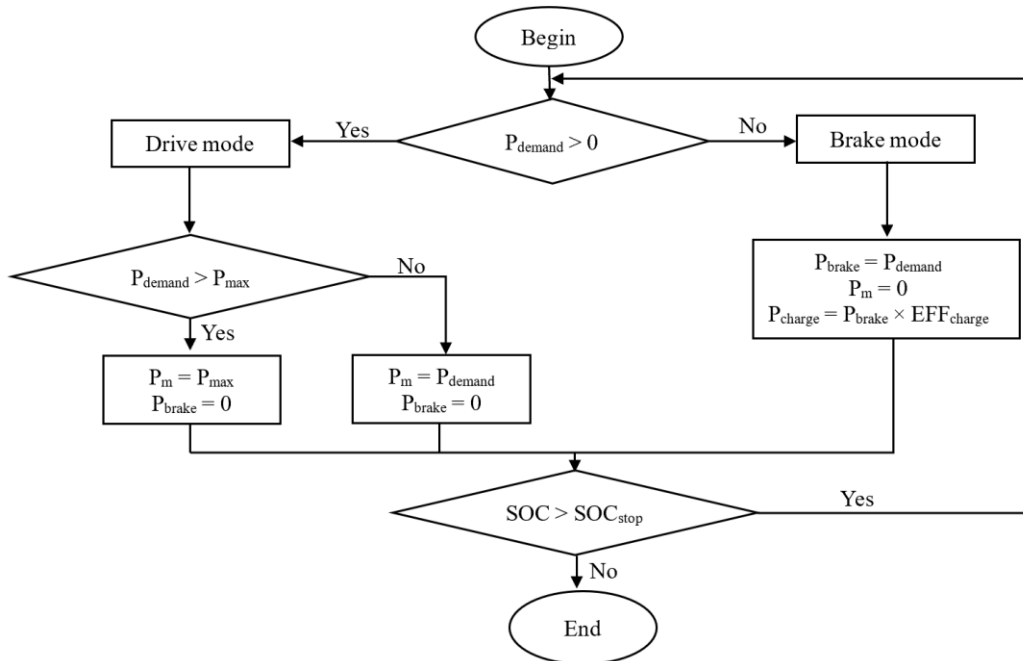


Figure 4 Control algorithm flowchart for traction and RBS modes

2.4 Modeling of power source

In an EM powertrain, the battery is the main energy source. Its output voltage varies with operating conditions, particularly charge/discharge current and temperature, which directly affects its ability to supply voltage and current.

$$SOC(t) = SOC_0 - \frac{1}{3600} \int_{t_0}^t \frac{I(\tau) \times \eta_{batt}}{Q_i} d\tau \quad (11)$$

$$SOC(t) = SOC_0 \quad (12)$$

The simulation is in line with the control algorithm flowchart as depicted in Figure 4. The demanded power from the driving cycle (P_{k_demand}) was used to decide whether the motorcycle ran in drive mode or brake mode. The controller power managed the power distribution to the hub motor and constrained the traction power to the permissible amount. In regeneration, charging power to battery was distributed and corresponding electric power (P_{elec}) was calculated during RBS.

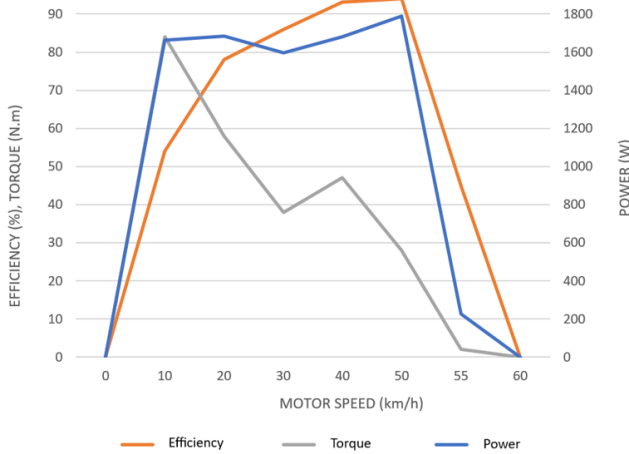


Figure 5 Hub motor 12-inch 1200W characteristic curve versus speed [14]

In the traction mode ($P_{k_demand} > 0$), the braking power (P_{brake}) was set to zero, and the BLDC motor power demand (P_{m_demand}) equaled P_{k_demand} , but it was always constrained not to exceed the maximum motor power (P_{m_max}). In each operating mode, the motor power was associated with corresponding efficiency (η_m). Based

on the Hub motor characteristic curve shown in Figure 5, the required input power for the Hub motor was determined.

$$P_{elec} = \frac{P_m}{\eta_m} \quad (13)$$

In the braking mode ($P_{k_demand} < 0$), corresponding to the deceleration phases of the cycle, the BLDC motor operated as a generator and initiated the regenerative current recovery process. In this case, $P_{m_demand} = 0$ and $P_{brake} = P_{k_demand}$. The charging power (P_{charge}) was then supplied back to the battery pack. In practice, RBS in EMs did not function entirely independently but was typically combined with a mechanical braking system. However, in this study, the model was simplified to consider only the RBS in order to evaluate the system's effectiveness. By conducting experimental measurements on the RBS model, the charging recovery efficiency (EFF_Charge) was determined at SOC of 50%, 70%, and 90% for a 60 V/20 Ah lead-acid battery pack. Combined with the P_{brake} , the recovered charging power was calculated according to the following equation:

$$P_{elec} = P_{charge} = P_{brake} \times EFF_{Charge} \quad (14)$$

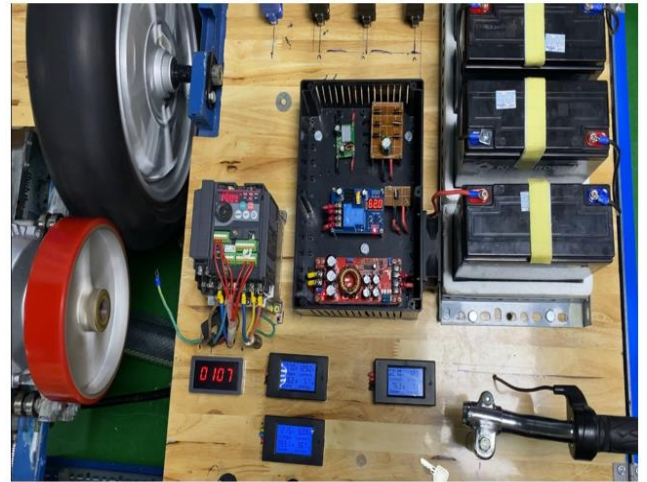


Figure 6 Operation of the RBS test bench for experimental measurement of regenerative current recovery efficiency

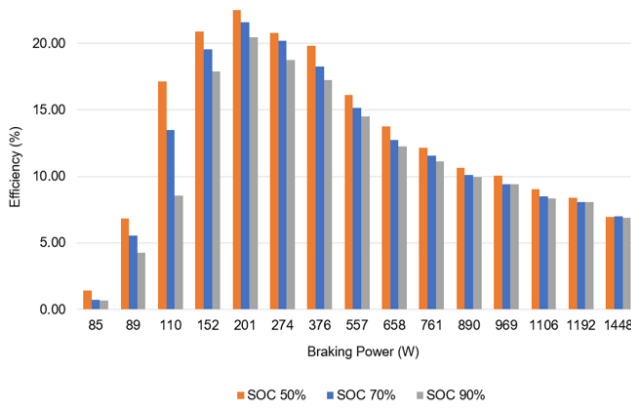


Figure 7 RBS recovery efficiency at different SOC levels after the boost circuit

To model the voltage characteristics of the battery pack, a simplified equivalent circuit model was commonly employed. In this model, the battery was represented as an ideal voltage source with an open-circuit voltage (U_{OC}), connected in series with an internal resistance that accounted for inherent losses. When a charging or discharging current flows through the battery, this internal resistance caused a voltage drop (U_r). Accordingly, the actual high-voltage output of the battery pack (U_{HV}) was determined by the following equation:

$$U_{HV} = U_{OC} - U_r \quad (15)$$

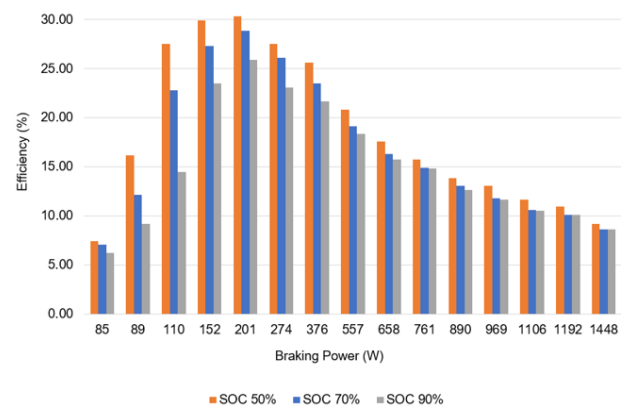


Figure 8 RBS recovery efficiency at different SOC levels before the boost circuit

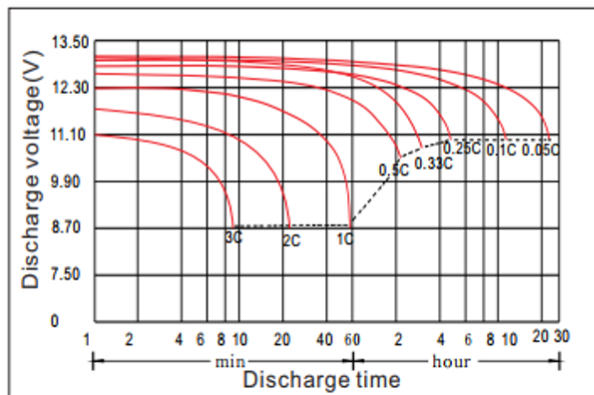
The value of U_{OC} was modeled as a nonlinear function of SOC, which is typically determined through experimental measurements or approximation methods.

$$U_{OC} = f(SOC) \quad (16)$$

The internal resistance of the battery varied depending on the operating temperature, SOC, and whether the battery was in charging or discharging mode. The voltage drops caused by internal resistance were determined according to Ohm's law:

$$U_r = i \times r \quad (17)$$

Discharge characteristic



Charging characteristic

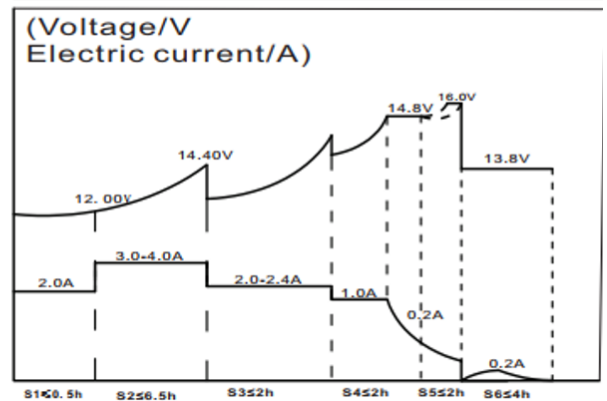


Figure 9 Characteristic curve of 6-DZF-20L lead-acid battery [15]

This model provided a simple yet effective approach to simulate the behavior of the battery pack under real operating conditions such as acceleration, deceleration, RBS or operation in varying temperature

environments. Its application also supported the design process and the evaluation of energy efficiency, as well as the prediction of the battery pack's durability throughout its lifecycle.

3 Results and Discussion

The complete model was constructed using functional blocks of the EM in Matlab/Simulink. The simulation was carried out with the input velocity obtained from the driving cycle shown in Figure 3 as a function of

time t . Battery SOC should remain above 50% to prevent damage and prolong its lifespan. Therefore, in the simulation, $SOC_{stop} = 0.5$ was used as the stop condition; the driving cycle terminated when SOC reached 0.5.

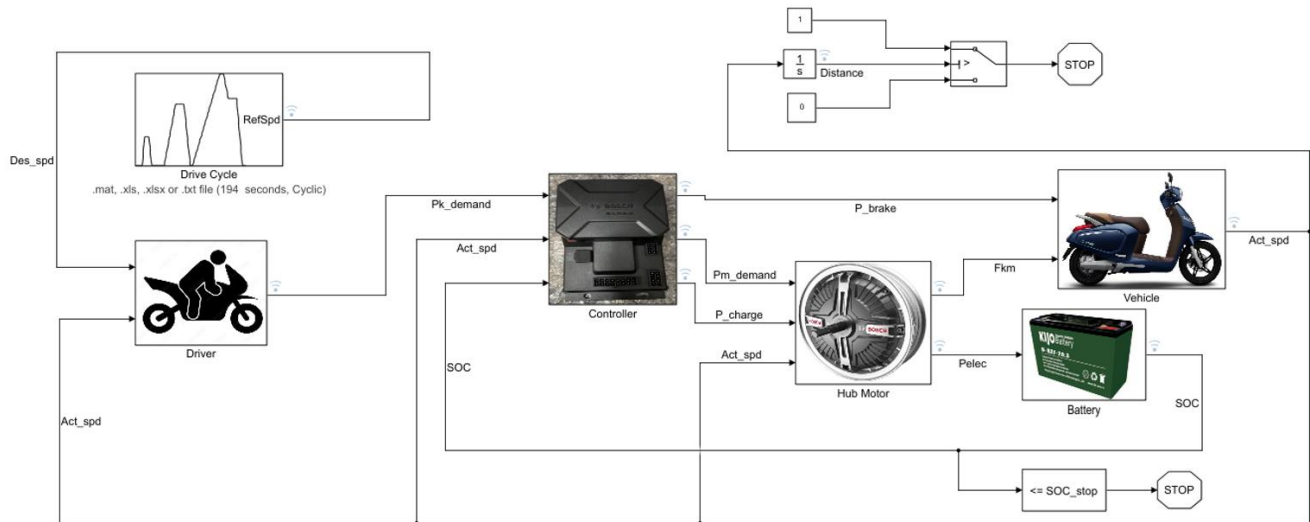


Figure 10 The model of the EM was simulated in Matlab/Simulink

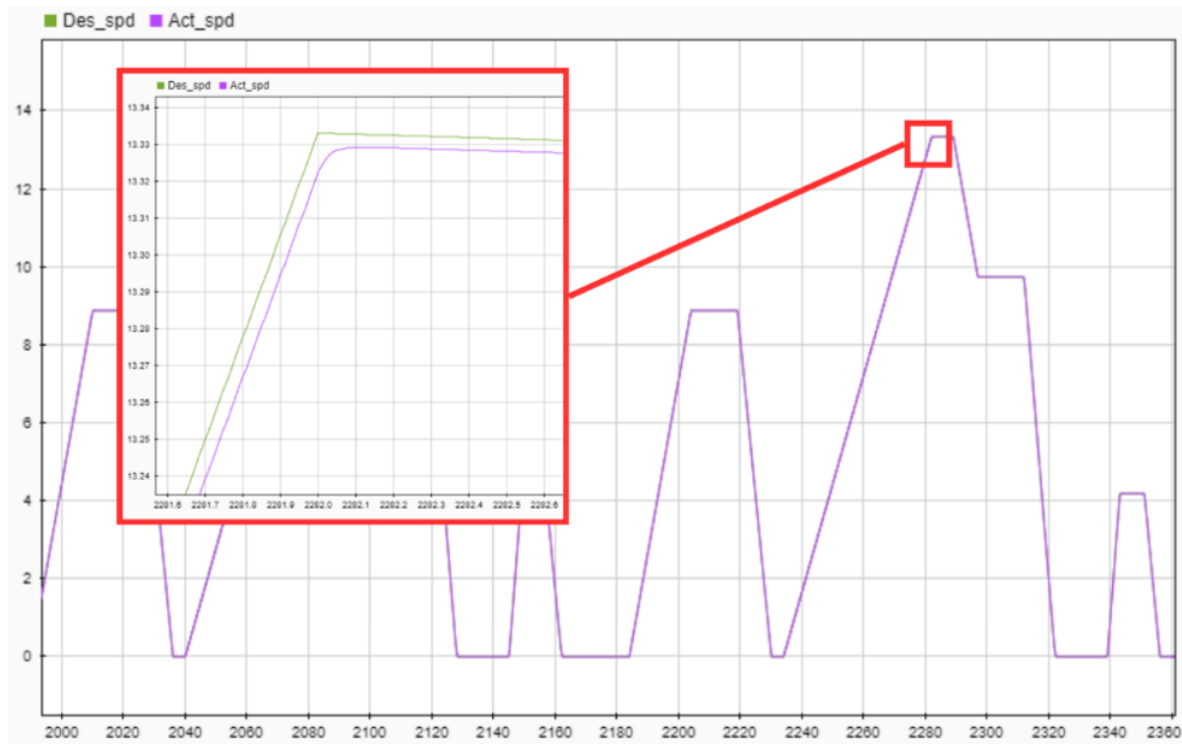


Figure 11 Result of simulation for response of PI algorithm at EM speed

Figure 11 showed the comparison of actual speed with the command speed in simulation. It can be seen that the difference was very small, indicating that the PI

controller can stably and effectively control the motor speed. Yet, a lagged system response was observed, in particular at high vehicular speed (48 km/h). The error

is largely due to the application of a plain linear PI control without any special consideration on fast transient response. But the real speed gradually converged and held at the reference after the transient,

which means that the system can keep good control accuracy under steady state, and the errors were too small to be taken into consideration.

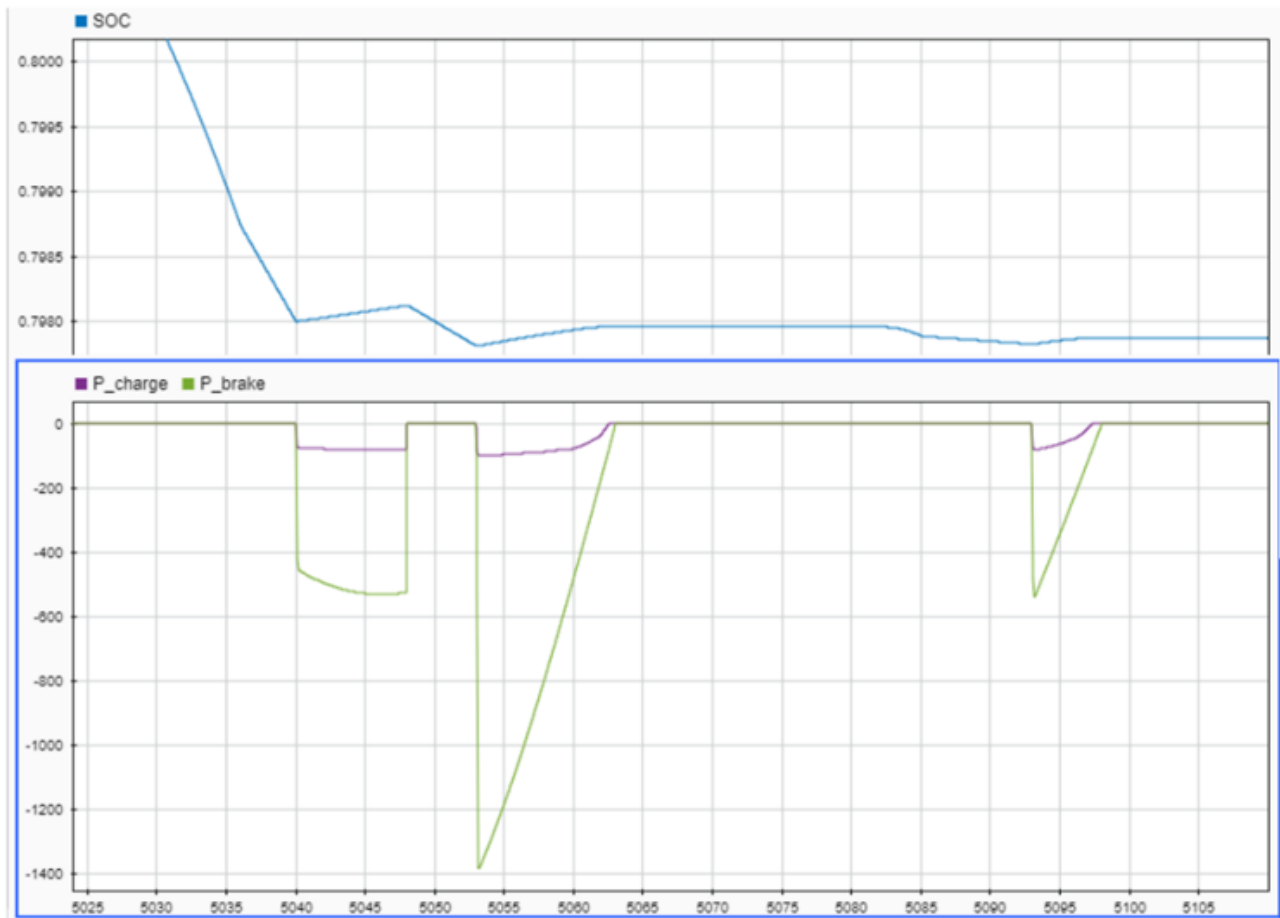


Figure 12 RBS energy recovery by decelerating with 60 V/20 Ah lead-acid battery

Figure 12 illustrates the change in SOC during brake. The SOC increased when P_{charge} was recovered via RBS by returning energy to the battery, and this is how the system regained braking energy.

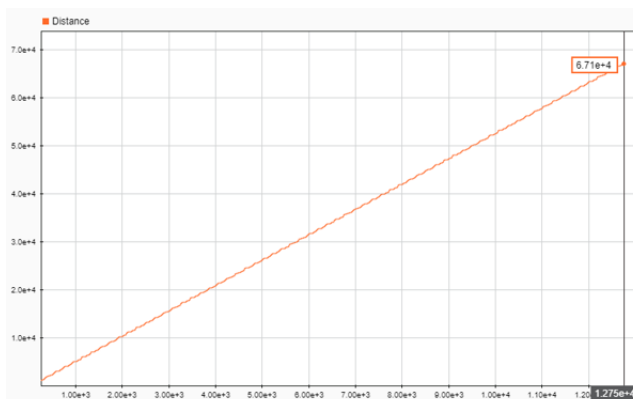


Figure 13 Driving range of motorcycles with RBS at SOC_{stop} limit 50%

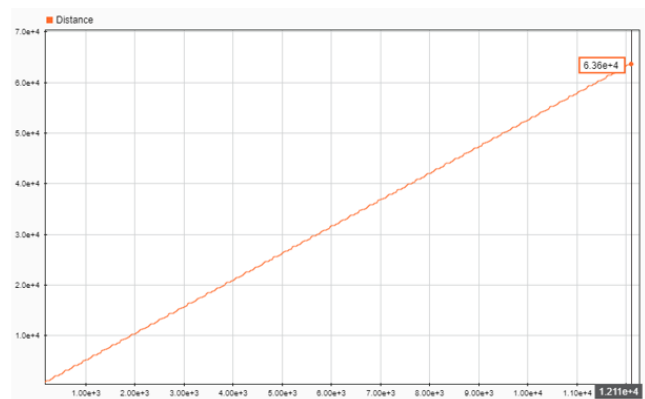


Figure 14 Driving range of the motorcycle without RBS at SOC_{stop} limit = 50%

Figure 13 and Figure 14 demonstrated the vehicle distance travelled in the simulation cycle with and without RBS, respectively. The ranges of motorcycles with RBS were 67.06 km and 63.64 km respectively, corresponding to an improvement of 5.4% in driving range. The extra simulations were run for different battery discharge limits, with $SOC_{stop} = 50\%$, 60% , and 70% . Ideal and practical energy recovery efficiencies (Figure 7 and Figure 8) have been considered in the analysis.

Table 2 Driving range simulation at various operating conditions of VinFast Klara S powers by 60 V/20 Ah lead-acid battery

Driving Range (Km)	SOC_{stop} 70%	SOC_{stop} 60%	SOC_{stop} 50%
Ideal Recovery Efficiency	41.01	54.69	68.09
With RBS	40.44	53.81	67.06
Without RBS	38.47	51.1	63.64

As shown in Table 2, the travel distance achieved under the test cycle varies according to the battery discharge limit and the efficiency of the energy recovery system. At $SOC_{stop} = 60\%$, the distance of motorcycle travel with regenerative braking was 54.69 km, while without RBS was 51.1 km, which led to an

improvement of 7.02%. Nonetheless, because of the efficiency of the regenerative energy recovery device, which converted the generated AC during braking into DC and feeds the battery, the total benefit was still rather small.

4 Conclusion

This research presents an experimental investigation combined with simulation of an RBS for an electric motorcycle based on the VinFast Klara S platform using Matlab/Simulink. The results demonstrate that the RBS can recover a portion of the motorcycle's kinetic energy during braking. The recovery efficiency was influenced by the battery SOC and braking power, reaching a maximum value of approximately 22.5%. The integration of the RBS led to an increase in the driving range of about 5.4%, indicating its potential for application in low-power electric motorcycles. The findings also indicated that the energy recovery performance is constrained by the efficiency of the rectifier-boost converter and the relatively high internal resistance of the lead-acid battery. Future work will focus on the implementation of high-efficiency power electronic converters and lithium-ion batteries, as well as real-world testing to further improve energy recovery capability and overall system reliability.

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Đánh giá thực nghiệm và mô hình hóa hệ thống phanh tái sinh cho xe máy điện

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Tóm tắt Nghiên cứu này trình bày việc đánh giá thực nghiệm, mô hình hóa và mô phỏng hệ thống phanh tái sinh cho xe máy điện dựa trên nền tảng VinFast Klara S sử dụng động cơ điện một chiều không chổi than 1,2 kW. Một mô hình thử nghiệm được xây dựng để đánh giá ảnh hưởng của trạng thái sạc pin và công suất phanh đến hiệu suất thu hồi năng lượng. Kết quả cho thấy hiệu suất thu hồi cực đại đạt 22,5 % tại công suất phanh 200 W, trong vùng SOC tối ưu (50-60)%. Mô hình Matlab/Simulink được phát triển và kiểm chứng bằng dữ liệu thực nghiệm nhằm đánh giá ảnh hưởng của hệ thống phanh tái sinh đến quãng đường di chuyển. Trong chu trình lái đô thị được xây dựng, quãng đường di chuyển trong một lần sạc đầy tăng từ 63,64 km lên 67,06 km, tương ứng cải thiện 5,4 %. Nghiên cứu làm rõ ảnh hưởng của hiệu suất bộ chỉnh lưu-khuếch đại điện áp và điện trở trong của pin, đồng thời đề xuất hướng nâng cao hiệu quả hệ thống phanh tái sinh cho xe máy điện cỡ nhỏ.

Từ khóa Phanh tái sinh; xe máy điện; thu hồi năng lượng; mô phỏng; hiệu suất.

Impact of Artificial Intelligence on Education, Employment, and Global Social Issues

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Abstract

The rapid advancement of artificial intelligence changes the paradigm shift across the global economy and society. This study analyzes, syndicates findings from experts in AI, educators, and policy makers to comprehensive assess the impacts of AI on the education, employment and important social issues. Qualitative thematic synthesis and data extraction were applied to compare and establish a consolidates view of consensus or contradictions. The findings indicate that application of AI has outpaced previous technological revolutions, expect over 80% of companies to use AI or develop related tools in 2027; AI threatens codified knowledge jobs but increase jobs requiring creativity and strategic thinking; accelerating enable AI in learning while protecting academic integrity; job displacement, data security and ethical issues are important problems in the AI era. Research recommends providing mandatory, interdisciplinary AI knowledge, re-designing curriculum toward foundational and human-centered skills, and establishing robust governance for responsible AI use.

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Keywords

Artificial intelligence; AI era; educating orientation in the AI era; social changes in the AI era; AI ethics.

1 Introduction

The rapid advancement of Artificial Intelligence (AI) is no longer a marginal concept but an industrial revolution driving change across industries, higher education and social behaviors. The AI tools such as Large Language Models (LLMs) integrated into operations every day and the core questions have changed from “Will AI replace us?” to “How will people use AI redefine talent?” [1]. The adopting AI has moved faster than the early of internet and personal computer age, more achieving and more pervasive usage in study, research, and administration. Evidence has suggested that over 80% of Fortune 500 companies

have applied AI, although only 1% feel they have reached maturity in its implementation [2]. Furthermore, the global growth of AI is projected to reach \$4,8 trillion by 2033 – increase 25-fold in only one decade [3].

The global context suggests that AI adoption has outpaced previous technological shifts. The forecasting that over 80% companies will either use Generative AI (GAI) or develop related tools by 2027 [4]. The AI acceleration creates a twin imperative for higher education and policy makers have to leverage AI for enhanced productivity while mitigating significant risks related to academic integrity, job displacement, and social inequality [5].



This study provides a consolidated analysis of the multifaceted impacts of AI on three domains, including employment, education, and the global social issues. This synthesis aims to inform policy makers, educators, and researchers on necessary strategic adjustments in the AI era.

2 Methodology

Researching methodology of this paper applied qualitative thematic synthesis and comparative data extraction approach.

2.1 Data extraction

The data in this study are referring employment statistics, AI performance metrics and social impacts were extracted directly from the original reports without external modification. A systematic extraction process was performed:

Thematic coding: information was categorized into core themes: global AI application, Job Risk and New Roles, Essential Skills, Higher Education Curriculum Strategy, Social and Ethics Concerns (including social impacts).

Quantitative data extraction: all numerical facts, percentages, metrics (such as decline rates, important skills, percentages, figures, ...) were extracted and recorded with their specific source citation (Author, Report, page reference). Job market data collected from World Economic Forum (WEF), and Stanford University, and Forbes 2025 reports cited within [6-8]. Model and method reproduction: details on AI models, evaluation pipelines, or empirical study descriptions were extracted verbatim or concisely summarized, citing the source precisely.

Table 1 GAI score summary [9]

Model	Knowledge (%)	Reading & writing (%)	Math (%)	Reasoning (%)	Working memory (%)	Total GAI score (%)
GPT-4 (2023)	8	6	4	0	2	27
GPT-5 (2024)	9	10	10	7	5	58

Additionally, the reporting data indicates that 40% of interactions of user to tools like Microsoft Copilot involve the AI playing role as a coach or an advisor than executing tasks, including a shift toward augmentation [10].

Comparative data generation: where multiple reports provided the relation or overlapping metrics (such as job risk categories, skill work requirements), a synthesis table was constructed for comparison, and each cell's data traceable to its original report.

2.2 Synthesis method

This synthesis identifies three primary pillars of study: the transformation of labor-market (cover Displacement vs Augmentation), current technical and ethical limitations of AI, and strategic responses for higher education.

3 Results

The results of the synthesis, the quantitative and qualitative analysis from reports reveals significant of AI's impacts on three domains

3.1 The Capabilities and Limitations of AI

Despite of the widespread adoption of AI, significant technical limitations remain. The reports highlight that LLMs (such as ChatGPT) often operate on approximate retrieval rather than true reasoning. The paper "GSM-Symbolic" [1] is cited as noting that LLMs struggle with basic mathematical reasoning when irrelevant information is introduced. While proficient in knowledge-intensive domains, the current AI systems have serious deficits in the foundational cognitive machinery, particularly in long term memory storage. Table 1 explores that the result of GAI scores (GPT-4 at 27%, GPT-5 at 58%) specifically quantify both the rapid process and the significant gap remaining before GAI [9].

3.2 Impact on Employment and Transformation of the Labor-Market

The threat to the workforce posed by AI is significant. The integration of AI is fundamentally redefining job requirements, particularly for white-collar workers (such as staffs, officers) and new graduates entering

administrative and analytical fields. A Stanford University Report indicates that AI is reshaping the labor market. This substitution risk is especially acute for young workers who rely more heavily on systematized knowledge. Employment among people aged 22-25 in AI exposed occupation has fallen by 13% relative to less exposed jobs between late 2022 and mid-2025. In contrast, older workers who possess tacit

expertise and contextual understanding remain relatively protected [8]. This phenomenon suggests a Death Valley for career starts, where young workforce struggle to gain the experience required to become the experts who survive automation by the AI era. Table 2 show the job roles identified across the reports as being high risk of substitution versus those expected to survive or emerge to AI augmentation.

Table 2 Job roles at risk and emerging/surviving roles in the AI era

Category	Roles identified as at high risk	Roles identified as surviving or emerging
Middle management	Routine decision-makers, supervisors in logistics/retail	Strategic leaders, change management experts, cross-disciplinary innovators
Legal & finance	Paralegals, legal researchers, accountants (compliance focus), tax preparers, financial advisors (low-tier), credit analysts, auditors	Strategic financial planners, risk managers, complex deal making, legal technologist, ethics consultant, legal AI checker
Healthcare/ad min	Medical coders, insurance claim processors, Appointment schedulers, billing staff	Not reported in the provided documents
Education/ research	Standardized test graders, basic tutors for factual subjects, administrative academic staff, consultants performing data/market research	Not reported in the provided documents
Technology	The best programmers are not those who write code the fastest; AI is disrupting its own creators.	Expertise in AI orchestration (connecting AI models, tools, data, and services)

Data from the World Economic Forum (WEF) Future of Jobs Report 2025 [6] indicated a net positive but structurally altered job market.. The routine administrative roles face the highest decline, while technology-centric and human-service professions grow. This aligns with the concept that AI automates codified tasks (e.g. customer service) while

augmenting complex tasks. It explores that relative to office or systematized jobs are being replaced quickly, while creative or critical thinking jobs are experiencing the highest growth. Table 3 project net growth and decline for few occupations over the next five years (see more in [6]).

Table 3 Top fastest-growing and fastest-declining jobs, 2025-2030 [6]

Rank	Occupation	Net growth (%)	Occupation	Net decline (%)
1	Big Data Specialists	113	Postal Service Clerks	-34
2	FinTech Engineers	94	Bank Tellers and Related Clerks	-31
3	AI and Machine Learning Specialists	82	Data Entry Clerks	-26
4	Software and Applications Developers	57	Cashiers and Ticket Clerks	-20
5	Security Management Specialists	55	Administrative Assistants and Executive Secretaries	-20
6	Data Warehousing Specialists	49	Printing and Related Trades Workers	-19
7	Autonomous and Electric Vehicle Specialists	48	Accounting, Bookkeeping and Payroll Clerks	-18
8	UI and UX Designers	47	Material-Recording and Stock-Keeping Clerks	-16

9	Light Truck or Delivery Services Drivers	46	Transportation Attendants and Conductors	-15
10	Internet of Things Specialists	42	Door-To-Door Sales Workers, News and Street Vendors, and Related Workers	-14
11	Data Analysts and Scientists	41	Graphic Designers	-12
12	Environmental Engineers	40	Claims Adjusters, Examiners, and Investigators	-11

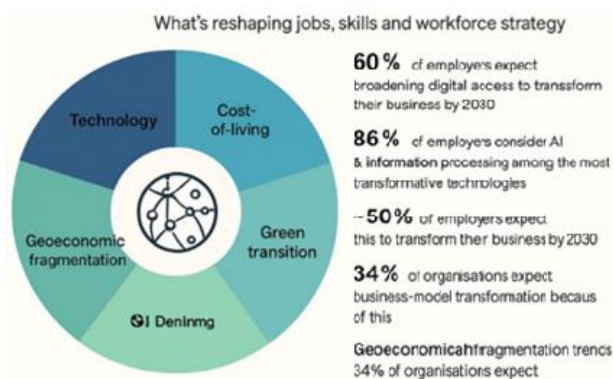


Figure 1 Major drivers of labor-market change (2025-2030) [3]

Technology trends partly contribute to the growth of security-related roles such as Security Management Specialists. Green and energy transition roles, including Autonomous and Electric Vehicle Specialists, Environmental Engineers and Renewable Energy Engineers appeared within the top 15 fastest growing roles. By contrast, the fastest declining roles such as Cashiers and Ticket Clerks, Secretaries, Printing Workers, Accountants and Auditors, etc (Figure 1). Technological skills are projected to grow in importance more rapidly than any other skills in the next five years. AI and big data are at the top of the list,

followed by networks and cybersecurity and technological literacy. Figure 1 show prediction about leading the fastest growing and fastest declining jobs from 2025-2030.

Contrary to the fear of total replacement, the consensus across is that “AI will not replace human, but human using AI will shape the future” [5]. This human – AI collaboration is evidenced by a finding of 25% higher productivity [11]. Consequently, the focus shifts to careers where AI acts or influence as an assistant, such as data science, business analysis, user interface/user experience design, and healthcare technology. The adaptation requires a shift in core workforce competencies. An analysis of employer expectations for 2030 skill indicates essential top 5 skills of worker for difference in AI era, including real-world experience, global mindset, interaction skills, critical thinking, creativity, AI literacy, ethics, and communication [7].

3.3 Impact on education and university’s responses

The speed of application of AI tools has outpaced the early personal computer and internet era. The AI is now widely used in learning-teaching activities, research, and administration. The results of the Digital Education Council’s 2024 survey are presented in Table 4 [12].

Table 4 Impact to students of AI in Teaching and Learning [12]

Top interested contents of student	Percent of student (%)
Already use AI in their studies	86
Use AI at least on a weekly basis	54
Student perception of university’s AI integration: AI integration does not fully meet their expectations	80
Expect their universities to increase the use of AI in teaching and learning	59
Agree that universities should provide training for faculty on the effective use of AI tools	73
Agree that universities should provide training for students on the effective use of AI tools	72

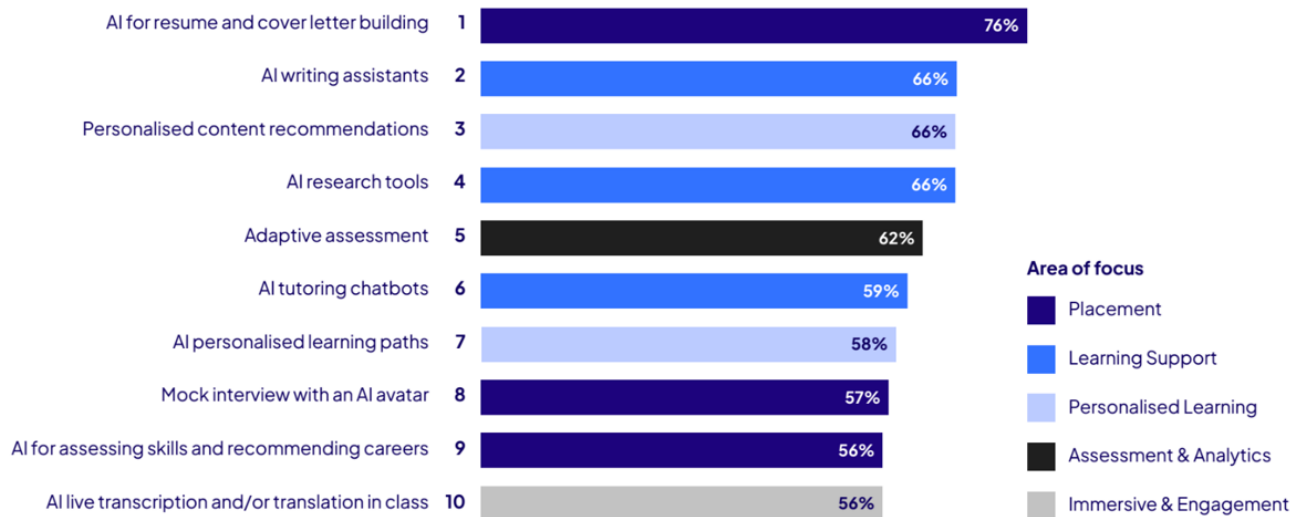
Agree that universities should offer more courses on AI literacy	72
Expect and percept of involvement in AI decision making	71
Feel that there is still room for improvement in their university AI guidelines	93

Currently, the higher education faces two major demands: integrating AI into educating activities, and addressing the needs of the labor-market in line with the AI era. Many assisted AI tools, such as adaptive learning system, intelligent tutoring system, can provide a flexible, personalized learning environment,

while also support faculty automated grading and provide insights into learning data [5]. However, it also presents significant integrity challenges. Figure 2 presents the top 10 cases of AI tools use in university that are most student-driven.

Top 10 AI use cases in higher education most positively perceived by students

Question: To what extent do you agree the following AI use case is useful/ will improve your learning journey/outcome



Note: rankings are based on unrounded percentages

Source: Digital Education Council Global AI Student Survey, 2024

Figure 2 Top 10 cases of AI use in higher education most positively perceived students [14]

The required educational strategy involves two key pillars: teaching AI and teaching with AI. All disciplines should integrate core AI thinking and teaching students how to use, build, and criticize AI systems. Integrating AI in education brings many perspectives: provides foundational AI literacy for all students and to domain specific depth; curriculum oriented on foundational skills (e.g. learn to learn, Algebra, Mathematics, and Physics) [13]; string partnerships with socio-economic sectors; faculty continuous development.

However, the integrating AI requires to adapting curriculum and programs [10]:

- **Mandatory AI Literacy:** foundational AI literacy is required for all students to prevent inequity, alongside domain specific depth.
- **Core Modules:** new core modules should include prompt engineering, data fluency, AI ethics/safety, and legal/regulatory basics.
- **Assessment Redesign:** must shift to authentic tasks, process evidence, oral defenses, and tool transparency to maintain integrity.

For instance, universities in Poland are responding with new programs (e.g. Computer Science AI at Wroclaw University of Science and Technology, AI and Data Analysis at AGH University) and introducing interdisciplinary programs, such as the Cognitive Science

and AI which integrates psychology, linguistics, and neuroscience [14].

Beyond academics, AI transforming university administration through applications such as student retention/success prediction, enrolment forecasting, chatbots, and academic integrity monitoring.

3.4 Impact on social issues and ethics

The reports indicate that three major societal concerns related to AI adoption: job loss and replacement; privacy and data security; ethical issues and bias.

Generative AI is capable of creating new content but carries risks, including hallucination and bias. Research examining the mathematical reasoning limits of LLMs (e.g. GMS-Symbolic) suggests that LLMs can be easily distracted by irrelevant information, raising questions about trust and reliability [1]. Human review and testing are thus crucial for AI – generated artifacts, such as code.

However, the developing of AI creates significant physical footprints. The energy consumption and carbon footprint of AI are a global social concern. Statistics from 2018 to 2025 showed that the training of 369 GAI models is estimated to have consumed (24.97-41.104) Terawatt-hours of electric, generated (10.67-18.61) million tons of carbon emissions. It is estimated that by 2035, global CO₂ emissions attributable to GAI could reach up to 246 million tons annually [15].

4 Discussion

The synthesis of these reports that AI is a “paradigm shift” rather than a temporary trend. The consolidated results underscore a consensus that AI represents a fundamental redefinition of the human role in the workforce, thus necessitating a rapid and profound educational overhaul.

The reports strongly align on the direction of impact. It focuses on the immediate threat to young, white – collar workers in codified, routine roles (e.g. low – tier finance, administration) and emphasize the shift from replacement to augmentation and subsequent productivity gains (25% higher with human – AI collaboration). This shared observation leads to a unified call for prioritizing non-automatable, human-

centric skills, creative thinking, critical thinking, ethics, and emotional intelligence [14]. The reports also focused heavily on the institution’s response, advocating a holistic “import AI into curriculum” approach rather than simply integrating AI only to computer science programs.

The crucial gap identified in the reports is a detailed framework (or regulation) for equity and accessibility of AI-related education, beyond the general call for leveraging AI’s knowledge to prevent inequity. The high environmental cost of GAI training is a significant social issue not elaborated upon in the context of university strategy development or curriculum transformation by the other reports, representing a thematic gap.

Implication for Education: The curriculum must mandate foundational AI literacy for all students, not only in STEM majors. Teaching has to focus on how to use, build, and critique AI systems. Priority should be given to teaching “tacit knowledge”, strategic thinking, creative skills, and interpersonal communication skills that AI cannot easily replicate. Educational institutions must adopt the transparent assessment tools to address integrity challenges in AI-driven learning.

Implication for Employment Transformation and Policy Makers: policies must contrast the stagnation of entry level job-market growth. Strategies should focus on lifelong learning and re-skilling for workers at risk middle management and administrative roles towards augmentation more focused and career-oriented. Establish governance frameworks that ensure transparency and human oversight. Address the “Responsible AI Gap” by mandating rigorous checks on AI systems for bias and safety.

Implication for Social Issues: focus on applying AI to solve specific problems of various fields (e.g., agriculture, healthcare) rather than relying on current purely theoretical models. The top risks in the AI era, such as career transformation, privacy, and ethics, are interconnected. Addressing AI ethics and bias issues require the greater transparency and ongoing audited mechanisms. The significant quantifiable environmental impacts of GAI demand a responsible

policy of sustainable energy development and consumption for developing AI technology. Currently, this topic has remained under research.

This study is limited to the reliance solely on provided reports, expert's opinions, empirical data from specific sources, cited specific figures, and university's references (including Australia, Poland, The United States of America, France, and Viet Nam). It may not fully cover all the impacts of AI on the global issues.

5 Conclusion and Recommendation

5.1 Conclusions

The AI has a profound impact on all aspects of society. It has not only transformed the landscapes of education and employment but also opened an era of human-AI collaboration and the imperative adaptation continuously. The AI is defined by risk of replacing routine tasks, systematized jobs and leading to employment growth at the entry-level for young workers. The future careers of workers require AI to join as an augments and an agent, and without a substitution.

In the context of a rapidly expanding global digital economy, Viet Nam faces a significant opportunity for a strategic breakthrough. Therefore, cultivating AI literacy must be prioritized equally essential as digital competency. With projections indicating that over 80% of enterprises will integrate GAI by 2027, it is imperative for Vietnamese enterprises to comprehensive digital and technological re-skilling strategy, specifically targeting knowledge-intensive sectors such as IT, business administration, and services.

Higher education must integrate AI literacy and ethics into all curricular. Re-designing the assessing method towards practicality and transparency by using educational tools is also crucial. Critical concerns such as employment transformation, data security, AI ethics, and environmental costs, are the massive quantitative factors related to GAI training.

As a multi-disciplinary and highly oriented application institution, Nguyen Tat Thanh University (NTTU) should implement a "Smart University" model that is not only teaches AI but also integrates into institutional

operations such as predicting student dropout rates, personalizing learning pathways, and optimizing academic management. Lessons from the successful frameworks of Macquarie University and Bordeaux University have shown that establishing a robust industry partnership network is the key of ensuring NTTU's graduated students are not left behind in an AI-driven labor market.

5.2 Recommendations

For policy makers/ government regulatory agencies: It is imperative to establish comprehensive legal and regulatory frameworks that controls the ethical and socially responsible deployment of AI. AI should be integrated into the national training curriculum to synchronize whole the educational system. Digital technology and AI competency frameworks must be supplemented to reflect the evolving requirements for workforce across all professional sectors of society.

For institutions/ administrators: It is imperative to reconfigure the strategic roadmap for AI' development and establish policy framework that ensure responsible AI's usage and academic integrity within institution. Strategic investments should be focused to technological infrastructure, data, and interdisciplinary AI researches that prioritize adaptive learning platforms and AI-driven student support systems to elevate educational quality. The institutions should integrate AI across the curriculum from foundational literacy to advance into all training industries. Universities must strengthen collaborations with industries, job-markets, and real projects in order to prepare students for augmenting focused roles.

For faculty/lectures: educators must recalibrate pedagogical and assessment methods to adapt using of AI tools by students. Emphasis should be applied on the authentic assessment methodologies such as viva voce, oral defenses, process-based evaluation, while mandating student transparency regarding AI utilization. Lecturers should utilize AI tools for lecture content and provide guidance for students these technologies to broaden their knowledge and enhance critical learning.

For students: Beyond mastering AI tools and developing AI critical literacy, students must cultivate creative thinking, critical reasoning and lifelong learning abilities. These human-centric core skills are essential for safeguarding them against obsolescence in increasingly automated and AI-driven labor market.

5.3 Limitations and Future Researches

Despite efforts to cover pivotal aspects, certain limitations remains in terms of technological currency due to unprecedented pace of AI evolution and current scope of available data within specific statistical

domains surveyed at the time of this study. Future researches should focus on a detailed reproducible evaluating pipeline or dataset from a large-scale empirical study, including (1) depth comparative studies on university-level AI curriculum reform, using metrics such as graduated employment rates and student's AI proficiency across disciplines and (2) developing and evaluating reproducible assessment models to evaluate effectiveness of learning activities and learning processes of student in the GAI era.

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Tác động của trí tuệ nhân tạo đối với giáo dục, việc làm và các vấn đề xã hội toàn cầu

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Tóm tắt Sự phát triển nhanh chóng của trí tuệ nhân tạo (AI) làm thay đổi sự thay đổi mô hình trong nền kinh tế và xã hội toàn cầu. Nghiên cứu này phân tích, tổng hợp các phát hiện của các chuyên gia về AI, các nhà giáo dục và các nhà hoạch định chính sách để đánh giá toàn diện tác động của AI đối với giáo dục, việc làm và các vấn đề xã hội quan trọng. Phương pháp nghiên cứu áp dụng tổng hợp chuyên đề định tính và trích xuất dữ liệu định lượng để so sánh và thiết lập quan điểm thống nhất về sự đồng thuận hoặc mâu thuẫn. Kết quả chỉ ra rằng ứng dụng AI đã vượt xa các cuộc cách mạng công nghệ trước đây, dự kiến hơn 80 % công ty sẽ sử dụng AI hoặc phát triển các công cụ liên quan vào năm 2027; AI đe dọa các công việc tri thức được hệ thống hóa nhưng làm tăng công việc đòi hỏi sự sáng tạo và tư duy chiến lược; tăng tốc cho phép AI trong học tập đồng thời bảo vệ tính liên chính trong học tập; mất việc làm, bảo mật dữ liệu và các vấn đề đạo đức là những vấn đề quan trọng trong kỷ nguyên AI. Nghiên cứu khuyến nghị về kiến thức AI bắt buộc, liên ngành, thiết kế lại chương trình giảng dạy theo hướng các kỹ năng cơ bản và lấy con người làm trung tâm, đồng thời thiết lập quản trị mạnh mẽ để sử dụng AI có trách nhiệm.

Từ khóa Trí tuệ nhân tạo; kỷ nguyên AI; định hướng giáo dục trong thời đại AI; thay đổi xã hội trong kỷ nguyên AI; đạo đức AI.

Using Digital Learning Platforms To Enhance Medical Students' Motivation in EMP Courses

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Abstract

This study aimed to examine the impact of digital learning platforms on medical students' motivation in their English for Medical Purposes courses, using Self-Determination Theory as the theoretical framework. The research employed an explanatory sequential mixed-methods design with three research tools, including a structured questionnaire, classroom observations of three classes and two focus group interviews with 12 volunteering. Results have shown that digital learning platforms significantly boosted the students' motivation in learning medical English. The present study also reported how digital learning platforms were utilized in the English for Medical Purposes classes, the interaction patterns of the students when they learned with the tools and psychological factors that motivated the students in the learning. These findings highlighted the need for incorporating digital learning platforms into the English for Medical Purposes curriculum to enhance medical students' motivation in learning medical English and increase the learning outcomes as well.

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Digital learning platforms; motivation; Self-Determination Theory; English for Medical Purposes.

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1 Introduction

As a subset of English for Specific Purposes (ESP), English for Medical Purposes (EMP) plays a crucial role in contemporary medical education. Medical students are required to achieve sufficient medical English proficiency as English is the primary language in scientific publications, international conferences and worldwide healthcare communication [1].

However, students still encounter substantial difficulties in learning medical English. The main challenge emerges from Greek and Latin roots that create complex medical vocabulary [2]. This poses another significant difficulty when most students

experience language anxiety and low confidence in learning the specialized language [3]. Moreover, pedagogical limitations and inadequate learning resources also present obstacles to successful learning of the specialized language among students [3]. Besides these challenges, Vietnamese medical students, in particular, also perceive that their low level of general English competency causes considerable difficulty in comprehending medical texts written in English [4].

Research have shown that motivation serves as the main factor which determines how well students learn languages [5]. Students who lack motivation show lower levels of active participation and regular practice

which leads to their failure in retaining vocabulary for long. For this reason, there is a practical need to develop effective teaching methods that will increase students' learning outcomes while maintaining their interest in learning medical English.

At Nguyen Tat Thanh University, the Medical English department has been introduced since 2021, aiming to equip medical students with sufficient knowledge of medical English to meet the required standards of English level mandated by the university. The university has consistently encouraged both teachers and students to use digital learning platforms including Learning Management Systems (LMS), online interactive tools, multimedia resources and mobile applications for their teaching and learning activities in order to achieve desired learning outcomes and boost greater engagement among students.

2 Literature Review

2.1 Digital learning platforms

Digital learning platforms are defined as systems that integrate technology with the Internet to deliver educational content and create interactive learning spaces outside the classrooms [6]. On the other hand, digital platforms are also defined as software tools to manage educational content and track students' progress in a blended classroom environment [7]. Furthermore, digital platforms utilize digital technologies that enable communication among students and teachers as well as assist teachers in evaluating students' progress through online assessments and quizzes [8]. The combined definitions

show that digital learning platforms function as interactive educational systems which allow students to participate in assessment activities while interacting with teachers and peers in both online and blended learning environments.

Digital learning platforms have various types which provide support for different instructional purposes. Adaptive learning platforms represent a major category which uses algorithms together with learning analytics to create personalized instruction for students based on their performance and progress. The systems can adjust the content difficulty and provide specific feedback while predicting learning results [8]. In other words, adaptive learning platforms can create personalized learning pathways which enable students to progress based on their competencies, meeting the needs of diverse learners. Moreover, digital platforms which combine assessment features with gamification elements also gain increasing popularity. The tools provide online quizzes and classroom response systems together with game elements which include points and badges and leaderboards to support formative assessment and students' engagement [8].

2.2 Underlying theories of motivation in language learning

The Self-Determination Theory (SDT) developed by Deci and Ryan provides a complete motivation framework that differentiates between intrinsic motivation and extrinsic motivation. The cognitive performance and engagement patterns of students are clearly shown through the core components of Self-Determination Theory [9].

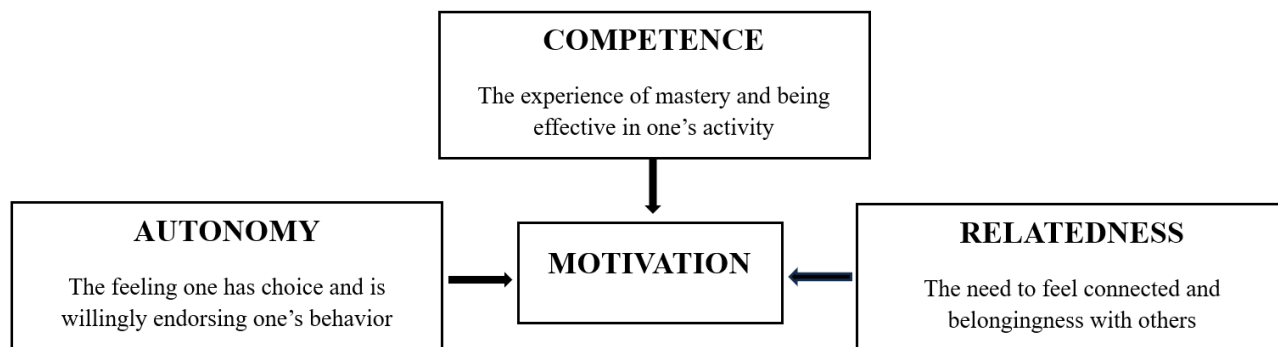


Figure 1 Self-Determination Theory framework [9].

Accordingly, SDT consists of three fundamental dimensions, including autonomy, competence, and relatedness. From the theory, people experience autonomy when they can actively take control over their own learning and have the freedom to choose their actions. In terms of competence, learners demonstrate their ability to perform academic tasks successfully. Meanwhile, relatedness enables them to build bonds with other people during the implementation of activities [9]. It has been found that when these requirements are met, students develop self-directed learning and remain motivated in their studies [9].

In the digital era, digital learning platforms offer multiple features which have proven to students develop the three fundamental aspects of SDT [10]. Digital learning platforms provide opportunities for students to develop autonomy through their self-paced modules and flexible learning paths and optional tasks so that students can decide on their learning preferences. The systems also use immediate feedback and adaptive quizzes and progress-tracking dashboards to help learners build their competencies through performance assessment which they need to maintain their motivation to learn. Online learning spaces enable students to develop social connections through interactive activities which include discussion forums and peer collaboration practices.

2.3 Previous studies

Earlier studies demonstrated that digital learning platforms boost students' motivation because they create interactive learning environments which allow students to learn independently. It was found that digital educational platforms and applications used in higher education helped students develop learning motivation and academic knowledge when they enabled students to participate actively and manage their own learning process [11]. Another research study showed that interactive online course technologies help students learn better because they deliver instant assessments and offer multimedia content and opportunities for group work which keep students interested in learning [12]. In the field of ESP, using digital technology tools creates positive effects on

students' learning attitudes and their motivation to learn. According to them, digital technology implementation in ESP classrooms created better learning outcomes through interactive educational content which improved students' access to learning resources [13]. The systematic review also revealed that mobile learning platforms for English for Specific Purposes academic programs helped students learn better through their ability to study language in different real-life situations which results in higher motivation levels [14]. Strikingly, medical students experienced motivational links to their perceived professional relevance because technology-based learning environments provide authentic materials which prepare students for their future medical careers [15]. The research shows that digital learning platforms increase students' motivation through two main factors, which include technological interactivity and the creation of learning experiences that match students' educational goals and expectations.

In the Vietnamese context, a study conducted at a private university discovered that simulation-based digital learning enhanced medical students' satisfaction and confidence in healthcare education [16]. In addition, the students at Da Nang University of Medical Technology and Pharmacy, who participated in the survey of 631 students, found that online learning platforms provided two main benefits which included time savings and convenience, leading to their increased motivation. Nevertheless, the students admitted facing three main problems which included unstable internet access, their limited digital abilities and their tendency to get distracted while using the internet [17]. Moreover, a study conducted at School of Medicine - Vietnam National University Ho Chi Minh City revealed that three factors which included perceived usefulness, convenience and accessibility of the platforms determined students' satisfaction and motivation in learning [18].

From these previous studies, it is clear that existing research on digital learning platforms for medical education does not address essential gaps. First, most studies focus on medical education in general rather

than English for Medical Purposes (EMP). Second, existing studies often emphasize general perceptions of e-learning without clearly identifying specific instructional methods or pedagogical practices, which creates challenges for assessing how these platforms boost students' motivation. Moreover, most studies use self-reported data from surveys and interviews for their findings with a lack of classroom observations as a tool to get a general picture of students' actual learning behavior and interaction in real settings. The current study, therefore, was conducted to fill in these gaps.

3 Methodology

3.1 Research site and sampling

The research took place at Nguyen Tat Thanh University, Ho Chi Minh City, Viet Nam.

Regarding sampling, the required sample size has to be calculated using a formula ($n = N / (1 + Ne^2)$) based on the total population and a specific error tolerance level [19]. As there were around 600 students in the entire cohort of second-year students in the department, a sample size of about 240 students at a 5% margin of error is required. In the current study, there were 272 students taking part in the survey and this number surpassed the estimated minimum sample size. Therefore, the quantitative results from this sample size are considered reliable.

Among 272 participants, there were 117 males and 155 counterparts aged between 19 years old to 21 years old, all of whom had finished their General English courses in the previous semesters and just started their first specialized course of English Medical Terminology.

3.2 Research design and instruments

The study employed an explanatory sequential mixed-methods design with the collection of both quantitative and qualitative data. An explanatory sequential design is defined as an approach with two phases: quantitative data is collected first before qualitative data is gathered to give more detailed explanations of the quantitative findings [20]. This process aims to provide statistical evidence of an effect or a phenomenon, which is followed by the analysis of qualitative data to further interpret the findings [20]. This approach is suitable for

the current study as it aligns with the aims of the study. The initial quantitative phase would help the researcher to quantitatively assess the extent to which digital learning platforms affect students' motivation. Then qualitative data is collected to clarify the students' experiences and perceptions towards these tools, which is used to elaborate their effects on the students' motivation in EMP courses.

There were three different research tools used to gather data for the study. The first instrument was a structured Likert-scale questionnaire aiming to preliminarily collect quantitative data about the students' perceptions of the use of digital platforms and the effects of these tools on their motivation and engagement. The questionnaire consisted of two parts. The first part gathered the participants' background information including class, age and duration of English learning. The second part contained 12 question items, which evaluated the students' motivation based on three aspects of the Self-Determination Theory: autonomy, competence and relatedness. The questions were translated into Vietnamese so as to avoid students' misunderstanding and delivered online via Google form to the whole cohort's Zalo group after the course ended, which allowed them to reflect on their learning experience.

Table 1 Cronbach's alpha Reliability for STD subscale

No.	STD subscale	Cronbach's alpha	Interpretation
1	Autonomy	0.77	Acceptable reliability
2	Competence	0.89	Good reliability
3	Relatedness	0.81	Good reliability

Cronbach's alpha (α) values between 0.7 and 0.8 meet acceptance standards [21]. The table shows that all three subscales which measure autonomy, competence and relatedness present internal consistency results which range from 0.77 to 0.89, meaning that the questionnaire was reliable to be used in the study.

Next, three EMP classes were directly observed to investigate how digital learning platforms were used and how the students interacted with these platforms during their lessons. Then the researchers conducted

two focus group interviews with 12 volunteer students to collect detailed qualitative data about their experiences and challenges with the digital learning platforms to complement the data collected from the questionnaire.

For data analysis, quantitative data collected from the questionnaire was analyzed using SPSS version 22 to examine descriptive statistics including frequencies, means and standard deviations to assess students' motivation in relation to the platforms. Then the researchers conducted an analysis of classroom observations by examining detailed field notes which revealed how digital learning platforms were utilized in classroom activities and how students participated. Next, focus group interview responses, on the other hand, were transcribed and conducted based on themes to identify the students' experiences and perceptions regarding the use of the digital learning tools in their learning process.

The qualitative findings helped the researcher to better understand the students' reasons, justifications and experiences that led to survey results, which allows for the combination of both types of data for complete comprehension of the real practice of using digital platforms in EMP classrooms and their effects on the students' motivation in learning medical English.

The researcher decided to employ only descriptive analysis because the study is merely exploratory and does not involve formal hypothesis testing nor draw conclusions about a broader population[22]. Instead, the researcher used descriptive statistics including means, max, min and standard deviations to summarize and present a clear description of the students' responses.

4 Findings

4.1 The use of digital learning platforms in EMP classes

Six classroom observations were conducted in three Medical English classes which the three teacher participants were in charge of. The findings revealed numerous digital learning platforms were widely used throughout the pre-class, in-class period and the post-class stages.

4.1.1 Pre-class activities

During the course, the three teachers employed multiple digital platforms to communicate with their students through Zalo, Gmail, the institutional Learning Management System and Google Classroom. The platforms were chosen for different functions but shared a common purpose. In all three classes, the teachers uploaded the lecture slides and medical terminology lists as well as reading materials to the institutional LMS system one or two days before the class meetings. The students were required to preview the vocabulary and relevant materials for the upcoming lessons before classes. At this stage, there were no pre-class assignments or activities.

4.1.2 In-class activities

The teachers used digital platforms during their teaching to create lessons that actively engaged the students. First, the teachers used PowerPoint slides as their main teaching resource to demonstrate medical concepts together with key medical terms. Then interactive platforms like Kahoot! and Quizizz, Blooket, Gimki, Padlet, etc. Were employed by the teachers to create active learning environments through vocabulary comprehension check and review activities and games. These activities and games came in a variety of forms, either requiring the students to work on their own or compete with others. This created an interactive learning environment so that the students had the opportunities to consolidate their knowledge and have fun at the same time.

4.1.3 Post-class activities

Digital platforms were also used in the post-class stage to enable the students to continue their learning through assigned tasks and review activities as homework. The teachers used the LMS and Google Classroom to share supplementary materials and create assignments which the students were required to do through online quizzes from Quizizz and Blooket. The quizzes assessed the students' knowledge of medical terms and essential learning concepts which they had learned in the previous lessons. The students could receive instant feedback and scores after they finished the quizzes, which helped them to reinforce their understanding of the knowledge through

the errors and wrong answers. In some cases, the students were also required to record role-play conversations of doctor-patient dialogues or oral presentations and submit their work through the digital platforms. The activities provided the students with opportunities to develop their

communication skills and apply the medical vocabulary in practical situations.

4.2 Students' perceptions towards the use of digital platforms in their EMP classes

4.2.1 Students' autonomy development

Table 2 Perceived impact of digital learning platforms on Autonomy

Items	N	Min	Max	Mean	S.D
<i>Self-Determination – Autonomy</i>					
A1. The digital learning platforms align well with my choices and interests.	272	3	5	4.21	0.781
A2. The digital learning platforms fit perfectly the way I prefer to study medical English.	272	3	5	3.97	0.684
A3. I feel that my use of digital learning platforms is an expression of my own learning choices.	272	2	5	3.64	0.857
A4. I feel that I have sufficient freedom to choose how and when to use digital learning platforms for my studies.	272	2	5	3.72	0.901

The Table 2 shows that the students held positive perceptions about the digital platforms used in their courses. The scores range from 2 to 5, with a mean between 3.64 and 4.21 on a five-point Likert scale. The standard deviation values were generally low, ranging from 0.781 to 0.901. This shows that the students' responses were consistent with minimal disparity among the items.

Complementing the findings, the data from the focus groups also reveal that the students developed greater independence in their learning through the use of digital platforms. The students reported that platforms such as Google Classroom and the LMS allowed them to manage their own learning pace and prioritize content based on their individual needs. One participant stated that *"On Google Classroom, I can review the vocabulary before class and focus more on the words I find difficult"* (FG1-S2). Another

participant further stated: *"If I miss something in class, I can check the slides again on the LMS instead of asking the teacher immediately"* (FG1-S4). Interactive gaming platforms like Blooket and Kahoot! and Quizizz were also perceived to enable the students to learn independently through an instant feedback system which allowed the students to evaluate their own progress. One student commented: *"When we play Blooket, I realize which medical terms I still don't remember, so I review them again later because the link for the game review is available on Google classroom"* (FG2-S4). Another student mentioned, *"Quizizz shows my mistakes right away, so I know what I need to improve without waiting for the teacher"* (FG2-S3). This means the students demonstrated their autonomy in their learning through their responses.

4.2.2 Students' competence development

Table 3 Perceived impact of digital learning platforms on Competence

Items	N	Min	Max	Mean	S.D
<i>Self-Determination – Competence</i>					
C1. I feel that I am making significant progress toward my learning goals through digital learning platforms.	272	3	5	4.07	0.885
C2. I feel that I use digital learning platforms effectively to achieve my academic goals.	272	2	5	3.88	0.974

C3. I feel confident in my ability to use digital learning platforms successfully.	272	2	5	3.96	0.883
C4. I feel capable of managing digital learning platforms to enhance my learning outcomes.	272	3	5	4.22	0.614

The students showed positive views towards the use of digital learning platforms, with the mean score of all items reaching above 3.0 and a minimum score of 2. Strikingly, most of them agreed that they could confidently navigate the platforms to achieve their desired learning outcomes. The standard deviation values showed a range from 0.614 to 0.974 which indicated that the students answered with both low levels of variability in their responses.

The responses from the focus groups supported these findings. The students reported that they found Google Classroom and the institutional LMS platform simple to use and convenient (FG1-S1, FG1-S2, FG1-S4, FG2-S1, FG2-S3, FG2-S5). The students also agreed that they achieved their learning objectives through the digital platforms which helped them learn medical terminology successfully. As they shared, “When the

teacher uploads vocabulary lists and quizzes online, I can review many times, so I feel my medical English improves gradually” (FG2-S3, FG2-S4). Some other participants noted, “*Doing quizzes on Quizizz or Blooket helps me realize how much I understand. I keep working until I remember all the terms in the lesson*” (FG1-S1, FG1-S2). Interestingly, some students appreciated the use of digital platforms in enhancing their confidence in learning medical English (FG1-S2, FG1-S3, FG1-S5, FG2-S1, FG2-S4, FG2-S5). From their sharing, the digital tools helped them reduce their embarrassment in learning because no one would notice when they made mistakes. This helped to improve their confidence by providing them with the opportunities to learn at their own pace.

4.2.3 Students’ relatedness development

Table 4 Perceived impact of digital learning platforms on Relatedness

Items	N	Min	Max	Mean	S.D
<i>Self-Determination – Relatedness</i>					
R1. I feel comfortable interacting with others through digital learning platforms.	272	3	5	4.12	0.659
R2. I feel connected to my classmates when using digital learning platforms.	272	1	5	3.67	1.126
R3. I feel that digital learning platforms provide open and supportive communication channels with my peers.	272	2	5	3.83	0.971
R4. I feel at ease collaborating with others through digital learning platforms.	272	3	5	4.03	0.744

It is shown that the students generally agreed that the digital learning platforms had a positive impact on their relatedness with others. This category received scores varying between 1 and 5, and mean scores ranging from 3.67 to 4.12. The standard deviation values appeared consistent throughout most items which demonstrated that the participants shared similar opinions. However, the item R2 showed a higher standard deviation (SD = 1.126) compared to the other items. The higher dispersion demonstrates that the students responded with greater

variation because they viewed digital platforms as providing different levels of peer connection.

According to the data from the focus groups, some participants highlighted that the interactive tools such as Padlet created meaningful opportunities for peer interaction. One student demonstrated this through his sharing: “On Padlet, I can post my ideas and read my classmates’ answers. I also comment on their posts” (FG2-S5). Another participant further explained: “*When someone writes a good answer on Padlet or*

Mentimeter, I can hit the heart icon, so it feels more interactive and encouraging” (FG2-S3). Moreover, some others admitted the increased interactivity through the game-based activities on platforms like Bloocket, Quizlet or Kahoot!. In their opinion, there were times when the teachers allowed them to play the games in groups, either to support or compete with one another. This, in turn, created opportunities for them to interact with others while enjoying the fun together.

However, some participants reported limited experiences with peer connection on certain platforms. One participant noted, *“On the LMS, I mostly download materials. The platform lacks any real social interaction”* (FG1-S4). Another participant stated: *“We rarely communicate with others when doing the activities assigned on the platforms. It is because the teachers mostly let us play individual games or do self-review activities”* (FG2-S4).

5 Discussion

The research findings show both conformity with existing literature and provide new information about the way digital tools operate in medical English instruction.

On the one hand, the findings demonstrate that digital learning platforms increase students' motivation by enabling them to learn at their own pace while delivering personalized learning support based on their individual needs [8, 16-18]. The students reported that platforms such as LMS systems, Quizizz, and Padlet allowed them to review materials in advance, revisit vocabulary lists, and complete quizzes multiple times. This insight aligns with the claim about the accessibility of learning resources offered by digital learning platforms, resulting in students' achievement of better learning outcomes [13]. The study also supports previous findings which demonstrate that instant feedback improves students' level of engagement. The automated quizzes together with online exercises delivered immediate feedback to students which helped them discover their errors and develop their skills without needing to wait for teacher's support [8, 12].

However, there are several new findings revealed, contributing to the existing literature. First, with the implementation of classroom observations, the study shows structured integration of digital platforms into the EMP courses throughout pre-class, in-class and post-class activities with specific tasks and assignments for students in each stage of the class. Moreover, the findings of the study demonstrate that the digital platforms used in the courses significantly reshape the practice of social interaction among the students instead of increasing it. In other words, these platforms are perceived to be more suitable for self-paced learning and review while there are other forms of interaction, such as comment and “liking” features offered by Padlet in which the students reported feeling encouraged when their classmates reacted positively to their work. Another interesting finding from the study is the psychological safety resulting from the use of digital platforms in doing quizzes to review medical terms. This, according to the students, helps them to avoid embarrassment and fear of negative evaluations when they make mistakes. Finally, the study brings a fresh perspective through its examination of medical English which represents a specialized field that demands more advanced cognitive abilities than General English. This next context promisingly paves way for future research on innovative approaches to enhance learners' outcomes in the ESP field.

6 Conclusion

This research investigated the impact of digital learning platforms on medical students' motivation to study EMP courses. The research used three methods for gathering data which included classroom observations, questionnaires and focus group interviews, revealing that digital platforms enabled the students to develop three important aspects of SDT which, in turn, enhanced their motivation in learning medical English. The research demonstrates that self-paced learning, along with immediate feedback and accessible study materials help to enhance students' ability to learn independently while developing their competence and maintaining their interest in the learning process. The

research also introduces several novel findings which further contributes to the existing literature on the field. Further studies could be conducted to examine whether the motivational effects of digital learning platforms are sustained throughout extended periods. Moreover, the relationship between motivation and actual learning outcomes, such as improvements in medical vocabulary, pronunciation, or professional

communication skills can also be worth a thorough investigation. In addition, future research should expand the research scope by including different ESP disciplines or institutions to enhance generalizability. The study may compare medical English with other ESP fields to determine whether decreased embarrassment and anxiety have special importance in medical environments.

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Sử dụng nền tảng học tập kỹ thuật số để nâng cao động lực học tập của sinh viên y khoa trong các học phần tiếng Anh chuyên ngành y

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Tóm tắt Nghiên cứu này được thực hiện nhằm mục đích điều tra ảnh hưởng của các nền tảng học tập kỹ thuật số đến động lực học tập của sinh viên y khoa trong các khóa học tiếng Anh chuyên ngành y khoa, sử dụng Lý thuyết Tự quyết làm khung lý thuyết. Nghiên cứu sử dụng thiết kế phương pháp hỗn hợp tuần tự giải thích với ba công cụ nghiên cứu, bao gồm bảng câu hỏi có cấu trúc, quan sát lớp học của ba lớp và hai cuộc phỏng vấn nhóm tập trung với 12 sinh viên tình nguyện tham gia. Nghiên cứu cho thấy các nền tảng học tập kỹ thuật số đã thúc đẩy đáng kể động lực học tập tiếng Anh chuyên ngành y khoa của sinh viên. Nghiên cứu cũng đưa ra những phát hiện mới về cách thức sử dụng các nền tảng học tập kỹ thuật số trong các lớp học tiếng Anh chuyên ngành y, các mô hình tương tác của sinh viên khi học tập với các công cụ này và các yếu tố tâm lý thúc đẩy sinh viên trong học tập. Những phát hiện này nhấn mạnh sự cần thiết phải tích hợp các nền tảng học tập kỹ thuật số vào chương trình giảng dạy tiếng Anh chuyên ngành y để nâng cao động lực học tập tiếng Anh chuyên ngành y của sinh viên y khoa và đồng thời cải thiện kết quả học tập.

Từ khóa Nền tảng học tập kỹ thuật số; động lực; thuyết tự quyết; tiếng Anh chuyên ngành Y.



Correlation Between Inhibition and Speaking Performance Among Vietnamese EFL Teenage Learners

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Abstract

Inhibition, defined by anxiety or fear of unfavorable judgement, is a outstanding psychological obstacle to enhancing speaking skills. This quantitative research, together with multiple regression analysis, aimed to find out predictors of inhibition as well as explored the interrelation between inhibition and speaking performance among Vietnamese teenage learners of EFL (English as a Foreign Language). The participants were 190 Vietnamese EFL teenagers at the age of 13 to 18. Inhibition was examined via a 20-item Likert-scale questionnaire survey, while speaking ability was judged based on IELTS Academic Speaking Test scores. The findings indicated a notable negative correlation between inhibition and speaking ability, where increased inhibition correlates with decreased IELTS speaking scores. Multiple regression analysis revealed that cultural factors and peer pressure were prominent predictors of inhibition. After all, the outcomes proposed some helpful solutions for EFL teenagers to overcome these psychological barriers and to perform speaking better.

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EFL teenagers;
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1 Introduction

Speaking skills play a key role in language acquisition, particularly in teaching and learning EFL (English as a Foreign Language) context. Regarding psychological factors, inhibition, widely known as anxiety, self-awareness, and fear of negative judgment, can dramatically prevent a learner from getting involved in speaking activities [1]. In Viet Nam's culture, the significance of preserving face and avoiding making mistakes in the crowd could restrict one's confidence to speak out naturally [2]. Although proficiency in English language is getting more and more pivotal for academic and career growth in Viet Nam, the majority

of EFL learners find the oral session of high-stakes tests challenging due to objective and subjective reasons.

Inhibition is widely recognized as a significant psychological barrier in language learning, especially in speaking. Anxiety and self-esteem influence a lot on how well students can learn a new language [3]. When being too shy, learners feel more anxious and fail to communicate well, which hinders their speaking skills [4]. Teenagers are especially sensitive to what others think of them and get demotivated about being judged. In Viet Nam, these feelings are made worse by the pressure to fit in and not stand out from the group. As

a result, Vietnamese EFL teenagers may feel self-conscious, which can be reflected through the way they perform in class discussions and other speaking activities [5]. By understanding these emotional factors, teachers can find ways to help their students feel more comfortable and confident to speak better.

The main purpose for conducting this study is to examine the interconnection between psychological inhibition and the speaking performance of Vietnamese teenagers as well as to find out possible predictors of inhibition using multiple regression analysis. By means of the IELTS Academic Speaking Test scores, speaking performance is carefully evaluated to provide insights into how psychological barriers affect language acquisition. Particularly, regarding psychological aspect, the research aims to find out the exact predictors of inhibition as well as to link psychological theory with pedagogical practice, recommending a variety of potential solutions to overcome such barriers and improve EFL teaching in Viet Nam.

2 Inhibition in Language Learning

2.1 Inhibition, language anxiety and communication apprehension

Inhibition, in the context of second language acquisition, is defined as the "psychological defenses or mental walls" that are set up to protect one's ego or self-identity [3]. Inhibition is different from language anxiety in the sense that, unlike language anxiety, which is a specific situational feeling of tension and apprehension related to a second language context, inhibition is a more general psychological mechanism that is characterized by "being self-conscious or fearful of negative evaluation." [3, 4]. In addition, communication apprehension is a general trait that is related to an individual's level of fear or anxiety about real or anticipated interaction with others [4], but in this study, inhibition is specifically defined in terms of the affective factors that prevent teenage learners from being willing to take risks in their oral production, particularly in a collectivist culture such as Viet Nam, in which the "face-saving" concept is related to the internal motivation to avoid mistakes in front of others, more so than the motivation to practice the language [5].

2.2 Predictors of inhibition

When it comes to learning a new language, there are numerous unavoidable factors that can challenge people. Elements like the norms of their culture, the influence of those around them, and how they feel about themselves can all play a major role [5]. In Viet Nam, where the group is more important than the individual, people may feel even more pressure to fit in and avoid standing out. This can make it harder for learners to speak up and get engaged in class. Understanding what is behind this hesitation is a must for applying effective strategies to help learners feel more comfortable and confident [2]. By recognizing these different factors, teachers can have some interventions that address the specific needs of their students and help them overcome their inhibitions.

2.3 IELTS academic speaking test as a measure of performance

The IELTS Academic Speaking Test is a standardized test assessing how well the test taker can speak in English, used widely in schools and at work. The international-valued test evaluates its applicants' speaking skills regarding fluency, coherence, lexical resource, grammatical range and precision, and pronunciation, with scores spanning from 1 to 9. Research indicates that the IELTS Speaking Test is a trustworthy and valid assessment of speaking skills [10].

2.4 Previous studies

The impact of inhibition on speaking performance has been a concern of many scholars and researchers. Students who get extremely nervous and want everything to be perfect often feel bad when they are forced to speak and that makes them do worse. For instance, research shows that those who have much anxiety and perfectionism have a tendency to feel negative when speaking, which affects their performance [6]. Similarly, Chinese learners of English who are very shy fail to be willing to communicate much. That leads them not to be very good at speaking [7]. Creating a friendly and fear-free speaking atmosphere where students feel confident is extremely essential to helping them feel comfortable to speak. One study found that when feeling at ease and

supported, learners are more likely to gain confidence and speak up better [8]. In Viet Nam, this is especially paramount since students might be afraid to speak if they think they will make mistakes in front of their friends [9].

2.5 Research gap

A number of previous studies have proved a significantly inverse relationship between psychological obstacles and language production. Particularly, those who always get nervous and want to do everything best often feel bad about themselves and that makes it hard for them to speak in front of others [6]. This is also true for EFL students in general and Vietnamese students in particular since when they are shy, they do not want to talk to others [7]. In addition, Viet Nam researchers have found that students get much more anxious when they think their friends are making a judgment on them and they also worry about what their culture thinks is right, leading them to be quite nervous in class. Consequently, learners fail to do well on speaking tests because of these psychological obstacles and language production issues [5, 9].

There are a variety of studies on language anxiety around the globe but not many focus on how cultural expectations affect speaking performance among Vietnamese EFL teenagers. Most research has been done on adults or in a particular school, which may not apply to Viet Nam's unique culture in which people value "face" and have a tendency to get along with the group. Few studies have used standardized tests like the IELTS Speaking Test to see how inhibition affects speaking skills in this age group. This study helps bridge that gap by looking at how cultural factors affect EFL teenagers' speaking test scores. The IELTS test is selected as a tool because it is widely recognizedly standardized and used. To be specific, the IELTS Academic Speaking Test is utilized to get a measure of the learners' speaking skills. By doing so, how much cultural norms and inhibition affect their speaking ability is revealed. Specifically, their test scores are compared with to the extent how inhibited they feel when speaking, to see how much psychological factors affect their speaking skills.

3 Research Method

This study was designed as quantitative research with the aim of figuring out the interconnection between inhibition and speaking performance among Vietnamese EFL teenagers. A 20-item Likert-scale questionnaire survey to measure inhibition as well as the IELTS Academic Speaking Test to assess speaking performance served as research instruments to collect data. A multiple regression model was also used to measure the impact of three main independent variables (cultural expectations, peer pressure and self-esteem) on the dependent variable (inhibition scores) based on the three hypotheses as follows:

- (H1) Cultural expectations, including face-saving, have a significant and positive effect on inhibition;
- (H2) Peer pressure from the classroom environment positively influences levels of inhibition;
- (H3) Self-esteem is a significant predictor of the learner's mental state.

The study employed a purposive sampling design to sample 190 Vietnamese learners from a range of secondary schools and language centers in Ho Chi Minh City. Purposive sampling was chosen as the most appropriate design because it enabled the researcher to select a group of people who were "information-rich" [11] in the sense that the participants were teenagers between the ages of 13 and 18, and were either in the process of preparing to sit the IELTS exam or had done so in the recent past. The learners in this particular research were described as EFL learners. This is a unique categorization of the learners that underscores the gravity of their learning of the language in the context of the Vietnamese socio-economic environment. Although the learners in this research were from a traditionally EFL environment, they were classified as EFL learners because they were often in intensive learning programs where the medium of instruction in a number of subjects was English or where the learning of the language was a requirement in their academic or migration pursuits. These EFL learners were teenagers ranging from the age of 13 to 18 and were in a learning environment where the learning of the language was a critical requirement in

their future professional or academic success and was thus a psychological pressure on their speaking performance as gauged by standard measures such as the IELTS exam.

This is a non-probability sampling technique, and it is important because it ensures the participants have certain characteristics in common, such as a cultural background and academic aspirations. These participants were diverse in the range of English proficiency but homogeneous in the demographic category because the participants were adolescents, a category that is psychologically more susceptible to peer group pressure and self-consciousness.

The primary means of data collection was a Likert scale questionnaire of 20 items, designed to measure the extent of inhibition felt by the participants. This questionnaire was adapted from Dörnyei's theory of questionnaire design in second language research to guarantee it effectively addressed the anxiety, self-consciousness, and fear of negative evaluation experienced by the participants [12]. This questionnaire is also based on Krashen's Affective Filter Hypothesis, which supposes that emotional factors such as high anxiety and low self-confidence, serve as obstacles to the language acquisition [3]. Through assessing the level of psychological obstacles to language learning on a scale of 1 to 5, ranging from strongly disagree to strongly agree, this survey offered a means for investigating the correlation between mental states and language proficiency, as evidenced by a high Cronbach's alpha of 0.85.

The data collection lasted for three months. The questionnaire was given offline in every EFL classroom, whereas the IELTS Academic Speaking Test was performed individually by authorized examiners. The speaking performance was obtained from the IELTS Academic Speaking Test, a globally recognized standardized examination. Participants were requested to provide their scores from their most recent official test sessions in the IELTS exam, which had been organized by British council or IDP

Education. This procedure was strictly employed to guarantee that the performance measure was both objective and reliable, regarding the four key areas of evaluation: fluency and coherence, lexical resource, grammatical range and accuracy, and pronunciation. Utilizing the official test scores rather than the results from a class test allowed the researcher to have a strong level of validity and reliability in evaluating the learners' true speaking abilities.

Data analysis was done by means of SPSS software. Cronbach's Alpha was computed to guarantee the reliability of the questionnaire. Descriptive statistics outlined the inhibition levels and IELTS speaking scores, while Pearson's correlation coefficient was used to examine the relationship between inhibition and speaking performance. Moreover, multiple regression analysis revealed key predictors of inhibition [13].

4 Findings and Discussion

In regard to the participation of 190 samples, the quantitative analysis was based on the responses from a 20-item Likert-scale questionnaire with Cronbach's alpha of 0.85, indicating high internal reliability. The results showed that the participants had a notable level of fear of speaking with an average score of 3.7, and their average IELTS English speaking score of 5.6. The figures showed that there was a connection between fear of speaking and speaking performance. Particularly, those who were more afraid of speaking had lower scores. This connection was indicated by Pearson's correlation analysis with $r = -0.52$ and $p < 0.01$. The study also found that expectations and pressure from friends were two major elements predicting fear of speaking. With a high score of 0.42 and 0.35 respectively, both cultural expectations and peer pressure had a remarkable impact on speaking performance. The study's model showed that 58% of the participants had a variety of inhibition levels, as shown by the equation: $R^2 = 0.58$. In other words, cultural and social factors are the prominent reasons for fear of speaking.

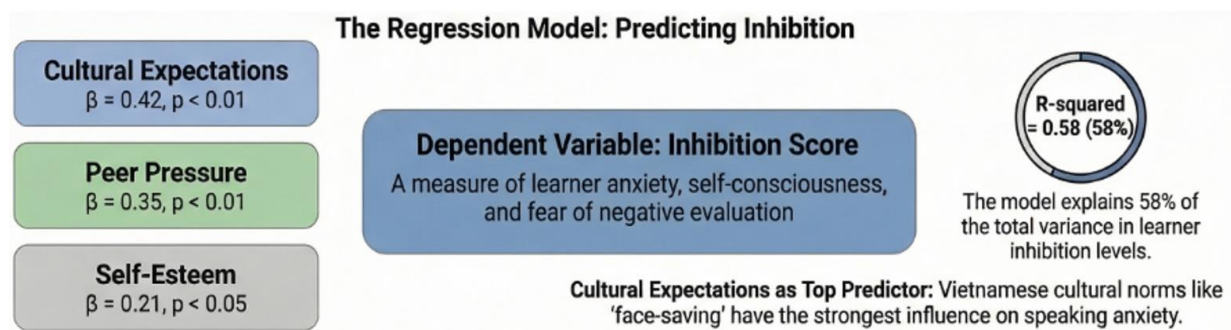


Figure 1 The Regression model

4.1 Reliability analysis

As an instrument for examining the participants' inhibition, the questionnaire was quite reliable, considering Cronbach's alpha value of 0.85, demonstrating strong internal consistency and reliability [14].

Table 1 Cronbach's alpha for inhibition questionnaire

Scale	Number of Items	Cronbach's alpha
Inhibition questionnaire	20	0.85

4.2 Descriptive statistics

Table 2 presented the descriptive statistics for inhibition and IELTS speaking scores. With the mean inhibition score of 3.7 ($M = 3.7$), the levels of inhibition among participants were moderate. Meanwhile, speaking proficiency was also at the moderate level with the mean IELTS speaking score of 5.6 ($M = 5.6$).

Table 2 Descriptive statistics for inhibition and IELTS speaking scores

Variable	Mean	Standard deviation	Minimum	Maximum
Inhibition Score	3.7	0.68	2.0	5.0
IELTS Speaking Score	5.6	0.95	4.0	7.5

4.3 Correlation analysis

Pearson's correlation analysis indicated a noticeable negative correlation between inhibition and IELTS speaking scores ($r = -0.52, p < 0.01$). In other words, higher levels of inhibition were associated with lower speaking performance.

Table 3 Correlation between inhibition and IELTS speaking scores

Variable	Inhibition Score	IELTS Speaking Score
Inhibition Score	1	-0.52**
IELTS Speaking Score	-0.52**	1

Note: Correlation is significant at the 0.01 level (2-tailed).

4.4 Multiple regression analysis

To identify prominent predictors of inhibition, a multiple regression analysis was taken advantage. Three variables including cultural expectations, peer pressure, and self-esteem were considered in the model. The results showed that cultural expectations ($\beta = 0.42, p < 0.01$) and peer pressure ($\beta = 0.35, p < 0.01$) were top predictors of inhibition, accounting for 58% of the variance in inhibition levels.

Table 4 Significant predictors of inhibition

Predictor	B	T	p-value
Cultural Expectations	0.42	5.68	< 0.01
Peer Pressure	0.35	4.89	< 0.01
Self-Esteem	0.21	2.85	< 0.05

4.5 Discussion and limitations

The findings spotted out a consequential negative correlation between inhibition and speaking performance among Vietnamese EFL teenagers. Specifically, cultural expectations and peer pressure were identified as striking predictors of inhibition due to the multiple regression analysis. This finding corresponds with previous research reporting how destructive psychological barriers are to language

learning, particularly in speaking skills [6]. Vietnamese teenagers may face increased shyness owing to cultural expectations ranging from the attention to group harmony to fear of losing face in front of peers. Such cultural norms may prevent learners from speaking activities involvement, resulting in ineffective performance in speaking [2, 5]. These results align with those of previous researchers [6], proposing some helpful implications for EFL educators. Firstly, cultural expectations and peer pressure must be taken into account to help EFL learners overcome inhibition and improve their speaking skills. To do so, English teachers are supposed to build a friendly classroom atmosphere, encourage constructive peer interactions, and giving constructive feedback [15].

However, the research is limited due to the fact that it solely relies on the self-reported IELTS results from the participants, leading to the remarkable impact on the results' overall accuracy. As a matter of fact, future studies should ask for more authentic official score reports directly from test centers to achieve stronger data validity. Furthermore, longitudinal study should be applied to examine long-term effectiveness of

targeted interventions in reducing psychological inhibition among learners.

5 Conclusion

The findings of this study support the hypothesis that inhibition such as the fear of embarrassment and under high-pressure learning environments prevents Vietnamese EFL teenagers from good speaking performance. In other words, the more learners feel anxious, the less they speak well. Meanwhile, cultural and social factors have a paramount contribution to inhibition due to the analysis of multiple regression. The research proposes proposed some practical solutions for EFL teachers to promote a fear-free and supportive teaching and learning environment where learners feel free to speak out and receive constructive feedback from their peers and teachers.

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Mối tương quan giữa sự tự ti và khả năng nói của thanh thiếu niên Việt Nam học tiếng Anh như ngoại ngữ

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Tóm tắt Sự rụt rè, được định nghĩa là sự lo lắng hoặc sợ hãi bị đánh giá tiêu cực, là một rào cản tâm lý đáng kể đối với việc phát triển kỹ năng nói. Nghiên cứu định lượng này, cùng với phân tích hồi quy đa biến, nhằm tìm ra các yếu tố dự báo sự rụt rè cũng như tìm hiểu mối tương quan giữa sự tự ti và khả năng nói của thanh thiếu niên Việt Nam học tiếng Anh như ngoại ngữ (EFL). Đối tượng nghiên cứu là 190 thanh thiếu niên người Việt Nam ở độ tuổi từ 13 tuổi đến 18 tuổi. Bảng hỏi 20 câu, thang đo Likert 5 cấp độ đo sự tự ti, và điểm số bài kiểm tra nói IELTS học thuật đo khả năng nói có vai trò là công cụ nghiên cứu. Kết quả cho thấy mối tương quan nghịch đáng kể giữa sự tự ti và khả năng nói, với mức độ tự ti cao hơn liên quan đến điểm nói IELTS thấp hơn. Phân tích hồi quy đa biến cho thấy các yếu tố văn hóa và áp lực từ bạn bè là yếu tố dự báo nổi bật của sự tự ti. Cuối cùng, kết quả đề xuất một vài giải pháp hữu ích cho thanh thiếu niên học tiếng Anh như ngoại ngữ vượt qua rào cản tâm lý và nói tốt hơn.

Từ khóa Người học tiếng Anh như ngoại ngữ ở độ tuổi thiếu niên; sự tự ti; phân tích hồi quy đa biến; khả năng nói.

A Personalized Method for Computing Course and Program Learning Outcome Attainment in Higher Education

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Abstract

This study proposed a personalized outcome evaluation method for Outcome Based Education that traced assessment evidence from course components to Course Learning Outcomes and aggregated them into Program Learning Outcomes to support accreditation oriented quality assurance such as ABET and AUN-QA. A binary linkage matrix identified assessment components contributing evidence to each Course Learning Outcome, ensuring alignment with constructive alignment principles and Bloom's taxonomy. Course Learning Outcome scores were calculated on a normalized scale from zero to one hundred using course grading weights and outcome specific evidence weights, and attainment was determined using a pre-defined threshold. Program Learning Outcome scores were obtained through weighted aggregation of related Course Learning Outcome scores, and results were reported using the Percentage of Program Learning Outcome Completion indicator for clear interpretation. A pilot implementation in one master's program using data from five courses demonstrated high outcome attainment and identified weaker areas, thus providing auditable evidence and practical support for continuous program improvement.

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Course Learning Outcomes (CLO); Program Learning Outcomes (PLO); Personalized Assessment; Quality Assurance.

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1 Introduction

Outcome-Based Education (OBE) emphasizes demonstrating students' attainment of intended learning outcomes through explicit alignment between outcomes, teaching activities, and assessment evidence. In this paradigm, courses define measurable Course Learning Outcomes (CLOs) while programs define broader Program Learning Outcomes (PLOs) quality assurance and continuous improvement require that a traceable link from assessment results to outcome attainment must be established [1, 2].

Quality Assurance (QA) frameworks oriented towards international accreditation such as ABET [3] and AUN-QA [4] require programs to define outcomes, and provide systematic auditable evidence of attainment at course and program levels, which feeds back to curriculum improvement cycles.

However, in many implementations, assessment results are aggregated at class or course-grade level [5], and normalization practices can obscure individual students' strengths and weaknesses with respect to specific outcomes. Therefore, instructors could not have any



actionable diagnostics for targeted support and program managers could lack consistent evidence for evaluating how CLO evidence accumulates into PLO attainment. This refers to a methodological gap: a transparent and reproducible computation which maps heterogeneous assessment components to CLO attainment and then aggregates CLO evidence across courses into PLO attainment at the level of an individual.

To fill the gap in the existing system, the present study proposed a student-level achievement calculation approach which (i) assigned a linkage between the components of student assessment and the CLOs using a pre-defined linkage matrix, (ii) calculated scores for each student using weighted values, and (iii) combined student performance of CLOs to determine scores in PLOs using specific rules. This approach adopted Bloom's Taxonomy to satisfy the constraint of the designed level of cognition of CLOs, enhancing the levels of validity of the approach [6]. This approach aimed to be executed using data provided in the learning management system, providing a mean of reportability in relation to teaching support and quality assurance [2, 7].

The contributions of this paper are:

- (1) a unitary computation process to transform the scores of the assessment components into the achievement of the CLOs by the students through the use of linkage matrices and normed weights;
- (2) an aggregation rule for determining PLO scores and levels of achievement using weighted average or threshold/proportion methods based on CLO evidence across courses;
- (3) outcome-level indicators for the purpose of transparent and continuous quality improvement procedures.

The rest of this document gives the related work and the methodology followed in the project along with the pilot results and implications.

2 Literature Review

OBE focuses upon constructive alignment and involves assessment evidence traceable to learning outcomes rather than the course grades alone [1]. Under the focus of accreditation-based quality assurance, there is also

the need to specify learning outcomes through systematic assessment procedures to encourage a correspondence between the course learning outcomes (CLOs) and the learning outcomes in the programs (PLOs) [3, 4].

To ensure the results are valid, it is necessary to design the learning outcomes with cognitive expectations. The revised Bloom taxonomy is a common framework used to align the learning activities to the intended cognitive levels of the learning outcome to avoid misrepresentation of the learning level in the assessment results [6]. In the process of implementing the Outcome-Based Education model, it is essential to ensure that the achievement of CLOs includes valid evidence rather than just the completion results in the test format.

On a practical level, achievement of the outcome can be calculated either by threshold rules (pass/fail) or by a continuous process, which provides a deeper diagnosis on the outcome level. Because assessors are often combining a range of different elements such as examinations, assignments, or projects, there must be a clear process of weighing and normalizing in order to be able to arrive at a pass/fail measure of each component, a necessary first step toward determining achievement on a CLO. Therefore, it follows on a program level, PLO achievement can be aggregated across a range of courses, either by a process of weighted averages or a formulaic measure, which seems to support a process for continuous improvement toward a quality process required for and anticipated by accreditation. Nevertheless, a lack of clear specification in the computation process may make it difficult to achieve a consistent measure.

Some universities in Viet Nam have also put in practice the tools using the OBE framework for assessments and reports of CLOs. For example, the information given in the CLO framework is offered at a class or test group level in the case of Ton Duc Thang University [8]. The indicators are given in a graphical format but can be aggregated in a manner that affects the ability to make student- or CLO-specific interpretations in the case of Lac Hong University [9].

Indeed, there emerged a methodological gap in current practices. There remained a need for a transparent and reproducible calculation in which divergent scores for the assessment components are converted into CLO achievement scores for individuals before summarizing CLO achievement scores in courses to PLO achievement scores based on weighted schemes. This problem is addressed in this study by proposing a method.

3 Materials and Methods

3.1 Notation and data model

$$i \in 1, \dots, N, c \in 1, \dots, C, j \in 1, \dots, J_c, k \in 1, \dots, K, a \in 1, \dots, A_c. \quad (1)$$

For each student i , course c , and assessment component a , the observed component score is denoted by:

$$s_{i,c,a} \in [0,100]. \quad (2)$$

To explicitly represent which assessment components provide evidence for each CLO, we introduce a binary linkage variable (CLO – component linkage matrix):

$$\gamma_{c,j,a} = \begin{cases} 1, & \text{if component } a \text{ provides evidence for CLO } j \text{ in course } c, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

The linkage $\gamma_{c,j,a}$ is derived from the course assessment design (Table 1) and reflects constructive alignment, i.e., only assessment components that are intended to measure a given CLO are allowed to contribute evidence to that

This study models outcome attainment at the individual student level by tracing assessment evidence from course assessment components to CLOs, and subsequently aggregating CLO evidence across courses to PLOs. Let i denote a student, c a course, j a CLO in course c , k a PLO, and a an assessment component in course c (e.g., midterm, final, assignment, project). All assessment scores are normalized to a common scale of 0-100 to ensure comparability across components and courses.

We define the index sets as follows:

CLO. In addition, $\gamma_{c,j,a}$ is constructed to respect the intended cognitive level of each CLO based on Bloom's taxonomy, thereby supporting the validity of interpreting component scores as outcome evidence.

Table 1 The matrix between CLO and assessment groups.

CLO	Midterm (30%)	Final (40%)	In-class (20%)	Presentation (10%)
CLO1	X	X	X	-
CLO2	X	X	-	X
CLO3	-	X	X	X

In summary, the minimal data required to apply the proposed method consist of: (i) normalized component scores $s_{i,c,a}$, (ii) the CLO – component linkage matrix $\gamma_{c,j,a}$, and (iii) the course and program outcome definitions (CLO and PLO sets). The subsequent sections build upon this data model to compute CLO scores and attainment at the student level, and to aggregate CLO evidence into PLO attainment in a transparent and reproducible manner.

3.2 Computing CLO score for each student

In order for an attainment of a CLO, on an individual student basis, to be calculated, a structured consolidation of evidence taken from various

assessment components must occur. In light of various assessment components having different contributions towards a course grade and towards different CLOs, there will now be two levels of weighting involved: course grading weights, and evidence weights for individual CLOs.

First, for each course c , a set of course grading weights is defined for its assessment components:

$$\alpha_{c,a} \geq 0, \sum_{a=1}^{A_c} \alpha_{c,a} = 1, \quad (4)$$

where $\alpha_{c,a}$ represents the proportion of the total course grade allocated to assessment component a , as

specified in the course syllabus (e.g., midterm 30%, final 40%, assignments 20%, project 10%).

Second, for each CLO j in course c , CLO-specific evidence weights are defined over the assessment components that provide evidence for that CLO:

$$\beta_{c,j,a} \geq 0, \sum_{a: \gamma_{c,j,a}=1} \beta_{c,j,a} = 1, \quad (5)$$

where $\beta_{c,j,a}$ reflects the relative importance of assessment component a in measuring CLO j . These weights allow instructors to distinguish components that provide stronger or weaker evidence for a given CLO, even when multiple components are linked to the same outcome.

Given the normalized component score $s_{i,c,a}$ and the CLO – component linkage $\gamma_{c,j,a}$ defined in Section 3.1, the CLO score of student i for CLO j in course c is computed as:

$$\text{CLO}_{i,c,j} = \frac{\sum_{a=1}^{A_c} (\alpha_{c,a} \beta_{c,j,a} \gamma_{c,j,a}) s_{i,c,a}}{\sum_{a=1}^{A_c} \alpha_{c,a} \beta_{c,j,a} \gamma_{c,j,a}}. \quad (6)$$

In this formulation, $\alpha_{c,a}$ captures the contribution of each assessment component to the overall course evaluation, $\beta_{c,j,a}$ captures the strength of the component as evidence for CLO j , and $\gamma_{c,j,a}$ ensures that only assessment components explicitly aligned with CLO j contribute to its score. The normalization in the denominator guarantees that $\text{CLO}_{i,c,j}$ remains on the same 0-100 scale regardless of the number of contributing components.

This weighted and normalized computation provides a transparent mechanism for translating heterogeneous assessment scores into CLO-level attainment that is both interpretable and auditable. In contexts where simplicity is preferred, either the course grading weights $\alpha_{c,a}$ or the CLO-specific weights $\beta_{c,j,a}$ may be omitted; however, the combined formulation is retained in this study to preserve consistency with course grading policies while allowing fine-grained control over CLO evidence allocation.

3.3 CLO attainment decision (threshold rule)

After finding the CLO score for the student as discussed in Section 3.2, the ensuing step would be to

decide whether the student has satisfied the related CLO. In outcome-oriented evaluation, this can be done by comparing the evaluated CLO score with the learning outcome threshold.

For each course c and CLO j , an attainment threshold is defined as:

$$\tau_{c,j}^{\text{CLO}} \in [0,100], \quad (7)$$

where $\tau_{c,j}^{\text{CLO}}$ represents the minimum score required for a student to be considered as having achieved CLO j . The threshold value is typically determined by the program or faculty in accordance with institutional quality assurance policies and the expected difficulty level of the course. In practice, common threshold values include 50/100 or 60/100.

Based on this threshold, a binary CLO attainment indicator is defined for each student i :

$$I_{i,c,j}^{\text{CLO}} = \mathbb{1}(\text{CLO}_{i,c,j} \geq \tau_{c,j}^{\text{CLO}}), \quad (8)$$

where $\mathbb{1}(\cdot)$ is the indicator function, returning 1 if the condition is satisfied and 0 otherwise. Thus, $I_{i,c,j}^{\text{CLO}} = 1$ indicates that student i has achieved CLO j in course c , while $I_{i,c,j}^{\text{CLO}} = 0$ indicates non-achievement.

As a generally accepted practice for quality assurance, the threshold-level decision rule leads to a clear interpretable standard for the attainment of the CLOs, which facilitates consistent reporting and aggregation of the CLO evidence at the program level. Moreover, the binary output facilitates the easy calculation of the proportions for the attainment of the outcomes.

3.4 Aggregating CLO evidence into PLO score

PLOs, as stated, encapsulate higher-order skills, which are usually the result of aggregation of various learning outcomes from numerous courses. Therefore, the achievement of the PLO has to be inferred based on the aggregated performance of the various relevant CLOs. To explicitly model the relationship between CLOs and PLOs, a CLO-PLO linkage variable is defined as:

$$m_{c,j,k} \in \{0,1\}, \quad (9)$$

where $m_{c,j,k} = 1$ indicates that CLO j of course c contributes evidence toward PLO k , and $m_{c,j,k} = 0$ otherwise. This linkage is derived from the curriculum mapping process and reflects the intended alignment between course-level outcomes and program-level

outcomes, as typically documented in accreditation materials.

Since not all CLOs contribute equally to a given PLO, a non-negative contribution weight is assigned to each CLO-PLO linkage:

$$w_{c,j,k} \geq 0, \quad (10)$$

where $w_{c,j,k}$ represents the relative importance of CLO j in course c as evidence for PLO k . These weights are

$$PLO_{i,k} = \frac{\sum_{c=1}^C \sum_{j=1}^{J_c} w_{c,j,k} m_{c,j,k} CLO_{i,c,j}}{\sum_{c=1}^C \sum_{j=1}^{J_c} w_{c,j,k} m_{c,j,k}}. \quad (11)$$

In fact, a more correct formulation for defining CLO includes only those CLOs that can be k add to the score, while the normalization part of the denominator ensures that the results remain comparable across students as well as programs irrespective of the number of contributing CLOs. This approach enables a fine-grained analysis of the results based on the performance of the students at the Program level by having the attainment of the PLOs as well as the CLOs on the same scale with respect to scores.

3.5 PLO attainment decision

After computing the continuous PLO score $PLO_{i,k}$ in Section 3.4, the next step is to determine whether student i has achieved PLO k . In practice, programs may adopt different attainment rules depending on the strictness required by the curriculum and quality assurance policies. This study supports two commonly used decision rules: a score-threshold rule and a proportion-based rule.

3.5.1 Score-threshold rule

Under the score-threshold rule, each PLO k is assigned an attainment threshold:

defined by the program council or quality assurance (QA) team to ensure consistency with program objectives and accreditation requirements [3, 4].

Given the CLO scores computed in Section 3.2, the PLO score of student i for PLO k is calculated as a weighted average of the contributing CLO scores:

$$\tau_k^{PLO} \in [0,100]. \quad (12)$$

A student is considered to have achieved PLO k if the computed PLO score meets or exceeds this threshold:

$$I_{i,k}^{PLO} = \mathbb{1}(PLO_{i,k} \geq \tau_k^{PLO}), \quad (13)$$

where $\mathbb{1}(\cdot)$ is the indicator function that returns 1 if the condition is satisfied and 0 otherwise. This rule provides a direct and interpretable attainment decision while retaining $PLO_{i,k}$ as a continuous metric for benchmarking and longitudinal analysis.

3.5.2 Proportion-based rule (CLO achievement proportion)

In some programs, a PLO may be considered achieved if a student attains a sufficient proportion of the CLOs linked to that PLO, acknowledging that certain CLOs may play a supplementary role. Let $\rho_k \in (0,1]$ denote the minimum required proportion of achieved CLOs for PLO k . Using the CLO attainment indicator $I_{i,c,j}^{CLO}$ defined in Section 3.3, the proportion-based attainment decision is defined as:

$$I_{i,k}^{PLO} = \mathbb{1}\left(\frac{\sum_{c=1}^C \sum_{j=1}^{J_c} m_{c,j,k} I_{i,c,j}^{CLO}}{\sum_{c=1}^C \sum_{j=1}^{J_c} m_{c,j,k}} \geq \rho_k\right). \quad (14)$$

This rule is particularly useful when a program aims to emphasize breadth of competency attainment across multiple CLOs, rather than relying solely on the weighted average PLO score. In practice, typical values for ρ_k include 0.6 or 0.7, depending on program expectations.

3.5.3 Practical usage and consistency

The score-threshold rule (Eqs. (12)-(13)) supports continuous reporting and comparison across cohorts, while the proportion-based rule (Eq. (14)) supports flexible attainment decisions when PLO evidence consists of multiple CLOs with varying instructional roles. Both rules are compatible with accreditation-oriented documentation practices, provided that thresholds τ_k^{PLO} and proportions ρ_k are defined consistently and approved through the program's QA governance process [3, 4].

3.6 PLO attainment decision

Although immediate scores of CLO and PLO assessments result in precise quantitative data of result achievement, there would also need to be additional summary metrics in order for transparent reporting and interpretation of outcome data at both individual and programmatic assessments. In this particular research study, there would also need to be a "Percentage of PLO Completion" metric.

For each student i and PLO k , PPC is defined as the percentage of CLOs linked to PLO k that have been achieved by the student. Using the CLO – PLO linkage $m_{c,j,k}$ (Section 3.4) and the CLO attainment indicator $I_{i,c,j}^{\text{CLO}}$ (Section 3.3), PPC is computed as:

$$\text{PPC}_{i,k} = 100 \times \frac{\sum_{c,j} m_{c,j,k} I_{i,c,j}^{\text{CLO}}}{\sum_{c,j} m_{c,j,k}}. \quad (19)$$

By construction, $\text{PPC}_{i,k} \in [0,100]$, where higher values indicate a greater proportion of required CLO evidence satisfied for PLO k . Unlike weighted PLO scores, PPC provides an intuitive measure of outcome coverage, making it suitable for visual reporting and diagnostic analysis.

To provide an overall view of outcome completion at the individual level, an aggregate PPC across all PLOs is defined as:

$$\text{PPC}_i = \frac{1}{K} \sum_{k=1}^K \text{PPC}_{i,k}. \quad (20)$$

This program-level indicator provides an overall assessment of progress towards program outcomes for a student, and it will be used in results for comparing patterns of attainment.

Transparency in reporting at the level of outcomes based on PPC indicators enables feedback for students and teachers as well as helping in acquiring continuous quality improvements during the accreditation process.

4 Results

4.1 Pilot setting and report completeness

The proposed outcome computation was piloted in one master's program using assessment data from five courses delivered to a single class. The cohort size for the exported report is 26 students (26 records). Among these, 6 students have missing PPC entries ("–") because they transferred to another program.

The program specifies 10 PLOs (PLO1–PLO10), each decomposed into indicators. For each student i and outcome k , PLO completion is summarized using the Percentage of PLO Completion $\text{PPC}_{i,k}$ and the overall completion index PPC_i defined in Eqs. (19)-(20). This enables results to be directly traceable to the CLO attainment evidence and CLO-PLO mapping described in Section 3.

4.2 Student-level PLO completion profiles (PPC)

For the 20 students with complete PPC values, the overall completion index PPC_i exhibits high attainment with notable dispersion:

- Mean $\bar{\text{PPC}} = 86.0$
- Median = 100.0
- Minimum = 0.0, Maximum = 100.0

A majority of students show full outcome evidence coverage: 11/20 students (55%) achieved $\text{PPC}_i = 100$. Meanwhile, a small subset presents substantial gaps: 2/20 students (10%) have $\text{PPC}_i < 50$, and one additional student is at approximately $\text{PPC}_i \approx 70.2$. These cases demonstrate that the PPC-based reporting can identify students who require targeted support at the outcome level, without conflating missing data from transferred students with underachievement.

At the PLO level, cohort mean PPC values remain above 80 across all outcomes, but differences are observed:

- Highest mean PPC: PLO5 (90.0); PLO1 and PLO3 (both 89.7)
- Lowest mean PPC: PLO6 (82.1) and PLO7 (82.1)

This indicates that evidences linked to PLO6 and PLO7 are the least complete on average in the pilot class, suggesting these outcomes as priority targets for diagnostic review.

4.3 Outcome-level diagnosis for program improvement
Because $PPC_{i,k}$ is computed from CLO attainment indicators $I_{i,c,j}^{CLO}$ and the CLO-PLO linkage $m_{c,j,k}$ (Eq. (19)), low PPC values can be traced back to specific CLO evidences and, ultimately, to assessment components through $\gamma_{c,j,a}$ and the CLO score computation (Eq. (6)). In practical QA workflow, the program council can start from the weakest PLOs (here, PLO6 and PLO7 by mean completion), then identify which linked CLOs contribute most to incompleteness, and review whether the issue is due to assessment design, insufficient evidence coverage in the five pilot courses, or threshold settings.

At the student level, PPC profiles support targeted advising by identifying which outcomes remain incomplete. At the cohort level, PPC summaries provide an interpretable measure of outcome evidence coverage that can be tracked across future intakes, supporting continuous improvement and accreditation-aligned reporting.

5 Conclusion

This paper presented a transparent, auditable method to compute learning outcome attainment at the individual student level by tracing assessment evidence from course assessment components to Course Learning Outcomes (CLOs), and aggregating CLO evidence across courses to Program Learning Outcomes (PLOs). The method was grounded in Outcome-Based Education and constructive alignment, where only assessment components explicitly linked to a CLO contribute to its computed score through the linkage matrix $\gamma_{c,j,a}$ and a two-layer weighting scheme $\alpha_{c,a}$ and $\beta_{c,j,a}$. CLO attainment was then determined using a threshold rule, and PLO scores were

derived as weighted aggregations of CLO evidence using the curriculum mapping $m_{c,j,k}$ and contribution weights $w_{c,j,k}$. To support interpretable reporting, the paper further defined outcome-level indicators such as $PPC_{i,k}$, enabling direct linkage between reported completion patterns and underlying CLO attainment evidence.

Pilot testing was done in a masters level course by using data from five courses and an exported file of 26 student records. The PPC outcome data was incomplete in six student records because the students transferred to another class, and the other records included complete PPC data at the cohort level. The pilot test suggests that it is possible and feasible to generate outcome data at the student level and also identify those outcomes that have relatively low rates of completion. There are two implications of the proposed approach for practice. Firstly, the approach offers an auditable and reusable computation process, which is supported by documentation principles formalized for accreditation purposes, including proof of achievement of outcomes and progress towards continuous improvement. Secondly, the approach offers outcome-level diagnostics at the individual student and cohort level which will allow the teacher and the respective program councils to trace back the identified gaps to the respective CLOs and components.

However, there are also certain limitations that need to be taken into consideration. First, only one program is targeted by this pilot, as well as only certain numbers of courses, due to restrictions imposed by the lack of data available regarding students who transfer into the institution. Currently, only the quantitative component scores and achievement levels are emphasized by this implementation, even where certain outcomes are better assessed by more qualitative methods, such as rubrics, portfolios, or observation-based assessment. Future development will target full implementation across multiple cohorts, programs, as well as involve more qualitative measures regarding data governance rules, such as privacy as well as LMS tools.

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Phương pháp cá nhân hóa trong tính toán mức độ đạt được chuẩn đầu ra môn học và chuẩn đầu ra chương trình trong giáo dục đại học

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Tóm tắt Nghiên cứu này đề xuất phương pháp đánh giá chuẩn đầu ra theo hướng cá nhân hóa trong Giáo dục dựa trên chuẩn đầu ra (OBE), truy vết dữ liệu từ các thành phần đánh giá của học phần đến chuẩn đầu ra học phần (CLO), sau đó tổng hợp CLO qua nhiều học phần để suy ra chuẩn đầu ra chương trình (PLO), phục vụ đảm bảo chất lượng và kiểm định (ABET, AUN-QA). Ma trận liên kết nhị phân xác định thành phần nào thực sự cung cấp bằng chứng cho từng CLO, phù hợp nguyên tắc liên kết dạy học, đánh giá và mức nhận thức theo Bloom. Điểm CLO được tính trên thang 0-100 theo công thức chuẩn hóa, kết hợp trọng số theo đề cương và trọng số bằng chứng cho CLO, rồi áp dụng ngưỡng để quyết định đạt/không đạt. Điểm PLO được tổng hợp có trọng số từ các CLO liên quan; báo cáo dùng chỉ số PPC để trực quan hóa mức hoàn thành. Thử nghiệm trên một chương trình thạc sĩ với dữ liệu 5 học phần (26 bản ghi) cho thấy mức hoàn thành nhìn chung cao và giúp nhận diện các PLO yếu. Phương pháp tạo bằng chứng minh bạch, hỗ trợ chẩn đoán cá nhân và cung cấp đầu vào cho cải tiến liên tục.

Từ khóa Chuẩn đầu ra học phần (CLOs); chuẩn đầu ra chương trình (PLOs); đánh giá cá nhân hóa, đảm bảo chất lượng.



Impact of Grit and Self-Control on the Academic Success of Pharmacy Students

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Abstract

This study examines Grit and Self-control and identifies predictors of pharmacy students' GPA. A cross-sectional study of 417 students (October 2021) utilized the Grit-S scale (assessing Consistency and Perseverance) and the Self-control scale. Data analysis was performed using R language. Students reported moderate levels of Perseverance, Consistency, and Self-control (mean: 3.00-3.40, score range: 1.00-5.00). A Bayesian Modeling Averaging-optimized regression analysis ($R^2 = 0.6161$) confirmed all three traits ($p < 0.001$), and religious affiliation ($p = 0.003$) as positive predictors of GPA, contributing 40.10%, 37.51%, 20.50%, and 1.89%, respectively, of the R^2 . As the first empirical study in Viet Nam validating the strong predictive capacity of these traits ($R^2 > 61\%$), this work establishes a critical baseline by characterizing student cohorts' GPA assessed in 2021. These data suggested that students with higher levels of grit could achieve higher GPAs. Therefore, school administrators should implement strategies to improve students' grit for academic success.

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1 Introduction

Education is critical in developing qualified human capital, accelerating economic development and solving social challenges. Students are also expected to devote a significant amount of time to education and graduate with high academic standing. Grade point average (GPA) is the most widely used indicator of academic success, accounting for 54.8% of the total studies analyzed in a comprehensive analytic review [1]. Grit is defined as a person's ability to maintain passion and perseverance for long-term goals in the face of setbacks and obstacles [2, 3]. Self-control is defined as the ability to deliberately adjust or repress internal

reactions, as well as the ability to interrupt and refrain from engaging in unwanted behavioral tendencies [4]. Unlike cognitive intelligence (IQ), which is relatively fixed, these non-cognitive traits are malleable and open to pedagogical interventions, making them critical targets for educational research [2, 5]. In physical therapist students, grit was reported to play a crucial role as an independent predictor of academic success and clinical performance [6]. Additionally, a positive association has been found between grit and higher levels of achievement among nursing students [7]. A study of 1 377 Vietnamese medical and health students at Medicine and Pharmacy Faculty, Hue University found that self-control was the most critical component



in students' ready network structures for online learning [8]. However, there is a paucity of research examining these traits within the specific context of pharmacy education. Unlike general undergraduate programs, the pharmacy curriculum imposes a unique “dual demand”: the need for short-term regulation (Self-control) to manage high-volume memorization and long-term stamina (Grit) to navigate rigorous clinical training. Therefore, disentangling the specific influence of these two constructs on academic performance is essential.

This study aimed to (1) examine the level of grit and self-control of pharmacy students and (2) evaluate the impacts of these two intrinsic factors on the academic success represented by the grade point average of Vietnamese pharmacy students. Findings from this research would provide insights into the role of grit and self-control in shaping academic outcomes and offer recommendations for educational institutions to enhance the learning experiences and performance of pharmacy students.

2 Research methods

2.1 Study design and participants

We conducted an online cross-sectional study at Nguyen Tat Thanh University, Ho Chi Minh City, Viet Nam from October 11th to October 20th, 2021.

Inclusion criteria: Pharmacy students engage in the survey voluntarily and anonymously.

Exclusion criteria: incomplete responses and repetitive answers, and zigzagging response patterns to the entirety of the questionnaire.

The sample size for this cross-sectional study was determined using Cochran's formula [9] for an infinite population. Based on a 95% confidence level ($Z = 1.96$) and a 5% margin of error ($e = 0.05$), a minimum of 385 participants was required. A total of 454 students from second to forth year participated; after excluding outliers, the final sample consisted of 417 students.

2.2 Research instruments

The survey questionnaire had 30 items, which comprised three parts:

Socio-demographic characteristics: (nine questions) to gather data of gender, year of study, GPA, religion, financial pressure, rigorous curriculum, enjoyment of studying pharmacy, intention to pursue another degree or a postgraduate degree, and adequate sleep.

The Short Grit Scale (Grit-S): The scale was selected due to its established validity and suitability compared to other scales [3]. The Grit-S scale was divided into two four-item subscales: Perseverance of Effort and Consistency of Interest. The mean score was calculated by dividing the total score by 8. This average score lied between 1 point (low grit) and 5 points (high grit). Higher mean scores indicate higher degrees of grit.

The Brief Self-Control Scale (BSCS) (13-items) was designed to evaluate dispositional self-regulatory behaviors utilizing a 5-point Likert scale that ranged from 1 (not like me at all) to 5 (very much like me). The scale has demonstrated comparable efficacy to full-length version [4]. This mean score can vary from 1 point (weak self-control) to a maximum of 5 points (strong self-control). The total score was divided by 13 to derive the mean score. Higher mean scores on this scale imply greater self-control.

2.3 Validity and data collection procedures

The questionnaire was prepared in Vietnamese and distributed to students via convenience sampling through their Google Meet classrooms.

Pilot phase: the purpose of the pilot study was to determine the viability and applicability of the Vietnamese research tools and procedures as well as the participants' level of comprehension. The responses acquired from pilot study participants were omitted from the final data analysis (official phase). The eight-item Grit Scale was analyzed through factor analysis, confirming two factors: “Perseverance of Effort” and “Consistency of Interest”, which aligned with the previous research [3]. The scale showed high reliability for the total scale and subscales (all Cronbach's alphas > 0.8). Similarly, the 13-item Brief Self-Control Scale was examined through factor analysis, revealing a single component as found in previous research [4]. The scale exhibited high reliability (Cronbach's alpha > 0.8).

Official phase: the survey was validated and distributed to participants using an online platform due to the COVID-19 pandemic in Ho Chi Minh City, Viet Nam, at the time of the research, which required remote data collection methods for safety reasons. An online version of the questionnaire was developed using Google Forms

for data collection. The survey was distributed through online classes where participants received a hyperlink to access the questionnaire. The estimated completion time was 10 min to 15 min. The reliability of the scales and subscales was acceptable, with all Cronbach's alpha coefficients exceeding 0.7 (Table 1).

Table 1 Factor analysis, reliability statistics of the instruments

Scale/Subscales	Pilot test (in-person) (n = 60)		Official test (online survey) (n = 417)	
	Number of items	Cronbach's alpha	Number of items	Cronbach's alpha
Total Grit-S	8	0.822	8	0.787
Perseverance of Effort	4	0.806	4	0.825
Consistency of Interest	4	0.854	4	0.816
Total Brief Self-Control	13	0.934	13	0.852

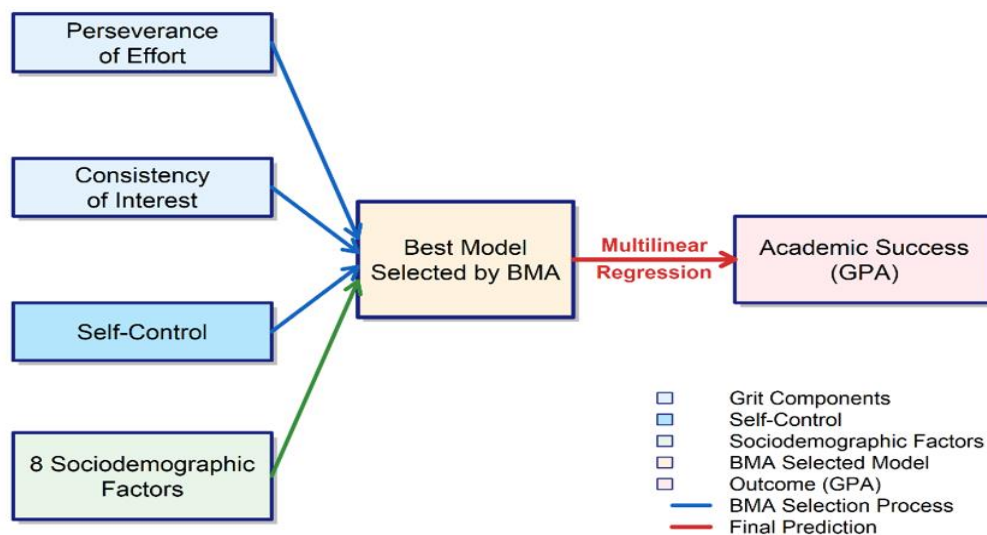
2.4 Data analysis

Data analysis and visualization were performed using the R statistical environment (version 4.3.1). Prior to analysis, outliers were checked. The difference was considered significant at a p -value ≤ 0.05 .

The Kruskal-Wallis test was used to examine differences in the scores of students' grit or self-control scores across different GPA levels.

Multiple linear regression was used to identify the predictors of the students' GPA. The dependent

variable was GPA. The independent variables included eight socio-demographic indicators, two Grit subscale scores, and a Self-control scale score. The regression model was constructed based on the theoretical structure illustrated in the Conceptual Framework (see Figure 1). The highest posterior probability model proposed by Bayesian Model Averaging (BMA) was used in a multivariable linear analysis. According to the Lindeman, Merenda, and Gold definition, the relative importance of each GPA predictor was established.



Note: 8 Sociodemographic Factors include: Gender, Academic year, Religion, Financial pressure, Rigorous curriculum, Enjoy studying pharmacy, Intention to pursue further education, and Adequate sleep

Figure 1 The conceptual framework of the study illustrating the hypothesized relationships between Grit (Perseverance of Effort and Consistency of Interest), Self-control, Sociodemographic factors, and Academic Success (GPA).

2.5 Ethical consideration

Students' involvement in the study was entirely voluntary and anonymous. We further emphasized that students were also free to withdraw from this study at any time. Prior to beginning data collection, the study

questionnaire was approved by Nguyen Tat Thanh University's Pharmacy Faculty scientific Committee.

3 Results

3.1 Socio-demographics of participants

Table 2 Characteristics of participants

Characteristics	Classification	Number of students (n = 417)	Rate (%)
Gender	Female	334	80.1
	Male	83	19.9
Year of study	Year 2	136	32.6
	Year 3	149	35.7
	Year 4	132	31.7
Grade point average (GPA)	Mean (SD)	2.68	(0.40)
	Fail (Under 2.00)	15	3.6
	Average (2.00-2.49)	118	28.3
	Fairly good (2.50-3.19)	238	57.1
	Good (3.20-3.59)	39	9.4
	Excellent (3.60-4.00)	7	1.7
Religion	(Yes)	121	29.0
Financial pressure	(Yes)	213	51.1
Rigorous curriculum	(Yes)	337	80.8
Enjoy studying pharmacy	(Yes)	370	88.7
Intend to pursue another degree/ a postgraduate degree	(Yes)	248	59.5
Get enough sleep	(Yes)	281	67.4

The validated response rate was 91.85% (417/454). Of the 417 participants, 334 were female (80.1%). There were 32.6% second-year students, 35.7% third-year students, and 31.7% fourth-year students. The grade point average on a 4.0 scale (GPA) was 2.68 (0.40), with a fairly good 57.1%, an average 28.3%, a good 9.4%, and an excellent 1.7%, whereas 3.6% of students had a failing GPA. Approximately 29.0% of participants had a religious affiliation. Financial

pressure was reported by 51.1% of participants. A significant proportion considered the curriculum rigorous (80.8%), and most students enjoyed studying pharmacy (88.7%). The intention to pursue another degree or postgraduate education was expressed by 59.5% of participants. In terms of sleep, most participants reported sufficient sleep (67.4%).

3.2 Pharmacy students' grit and self-control level

Table 3 Students' grit and self-control scores

Scale/Subscales	Possible range	Observed range	Mean (SD)
Total Grit-S	1-5	1.25-4.88	3.07 (0.73)
Perseverance of Effort	1-5	1.00-5.00	3.14 (0.91)
Consistency of Interest	1-5	1.00-5.00	3.00 (0.94)
Total Brief Self-Control	1-5	1.54-4.77	3.40 (0.76)

Participants showed moderate levels of grit and self-control, with mean scores of 3.07 (0.73) (score range of 1 to 5) and 3.40 (0.76), respectively. Regarding the Grit scale, students had higher score on Perseverance of effort 3.14 (0.91) than Consistency of Interest 3.00 (0.94).

3.3 Grit and self-control of pharmacy students by GPA level

Figure 2A illustrates a clear increasing trend in Grit levels as students' GPA increases. Specifically, Grit scores were significantly higher than the overall mean (3.07) in the Fairly good ($p < 0.05$), Good and Excellent groups ($p < 0.001$). Conversely, Grit scores in the Failing and Average groups were significantly lower than the overall mean ($p < 0.001$).

Figure 2B indicates a different pattern for Self-control. Scores were significantly higher than the overall mean (3.4) in the Fairly Good and Good (GPA) groups ($p < 0.001$). However, unlike Grit, the Excellent group did not show a significantly higher score. Meanwhile, Self-control scores in the Failing and Average groups remained significantly lower than the mean ($p < 0.001$).

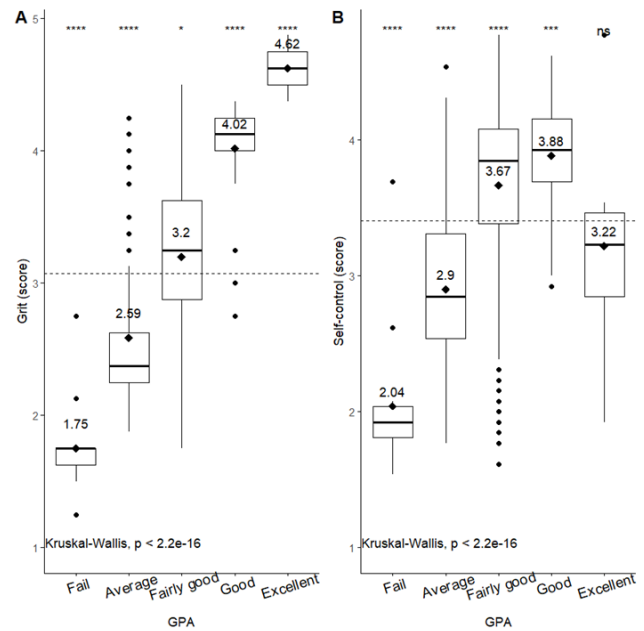


Figure 2 Pharmacy Students' Grit and Self-Control scores by GPA level (Notes: compare each of the five groups against "all", ---- base mean score, ns: non-significance)

3.4 Predictors of Students' GPA

Table 4 Predictors of Vietnamese pharmacy students' academic performance (GPA)

Predictors	Regression coefficient (SE)	p-value	Relative important metrics of R square (%)
(Constant)	1.155 (0.064)	< 0.001	
Religion (Yes)	0.082 (0.027)	0.003	1.89
Self-Control	0.105 (0.018)	< 0.001	20.50
Grit-Perseverance of Effort	0.190 (0.015)	< 0.001	40.10
Grit-Consistency of Interest	0.181 (0.014)	< 0.001	37.51

$R^2 = 0.6161$, Residual standard error = 0.2517, posterior probability = 0.488

Equation: $GPA = 1.155 + 0.082 \times \text{Religion (Yes)} + 0.105 \times \text{Self-Control} + 0.190 \times \text{Grit-Perseverance of Effort} + 0.181 \times \text{Grit-Consistency of Interest}$

According to the BMA prediction model with the highest posterior probability (48.80%), the multiple linear regression identified Consistency ($p < 0.001$), Perseverance ($p < 0.001$), Self-control ($p < 0.001$), and Religion ($p = 0.003$) as significant positive predictors of GPA. These factors explained 61.61% of the variance in students' GPA. The remaining 38.39% is likely attributable to other variables not observed in this study. Perseverance was the most significant predictor,

contributing 40.1% of the R-square. Consistency, Self-control, and Religion made up (37.5, 20.5, and 1.89)%, respectively, of the R-square. Notably, the Grit subscales combined accounted for 77.61% of R-square. These data suggest that students with higher levels of grit and self-control, especially grit, could achieve higher GPA.

4 Discussion

This study investigated the features of grit and self-control in pharmacy students and how these factors could help predict a student's GPA. In this study, we have shown that (1) pharmacy students exhibited a moderate level of grit, with both subscales, Perseverance of Effort and Consistency of Interests, also at a moderate degree. In terms of self-control, the participants also demonstrated moderate levels. There were significant differences in grit and self-control among different students' GPA levels (both p -values < 0.001). Indeed, students with GPA ranging from fairly good to higher scales had higher scores than the grit mean score and the opposite of the remaining GPA levels. In the case of self-control, students with fairly good and good GPA scales had higher self-control scores than the mean score, whereas students with excellent GPA had no significant differences. However, it should be noted that the sample in this study included only seven excellent students, which limited the statistical power. A critical theoretical distinction explains why both constructs were included in the model despite their correlation. While Grit and Self-control are related, they operate on different temporal scales. Self-control, as defined by [4], refers to the capacity to regulate momentary impulses and resist immediate temptations (e.g., ignoring distractions to study for an evening). In contrast, Grit entails the enduring stamina to maintain interest and effort toward a superordinate goal over years, despite failure or adversity [3]. For pharmacy students, Self-control is required for daily study discipline, while Grit is essential for navigating the long-term rigor of the multi-year clinical curriculum. (2) After adjusting for socio-demographic factors in the linear regression equation, grit and self-control (in scores) were the main positive predictors of students' GPA.

The current study's findings on pharmacy students' grit score (3.07 (0.73)) were consistent with prior research on four-year degree Malaysian pharmacy students, where the grit mean score was (3.1 (0.4)) [10], and undergraduate pharmacy students in 14 Asian and Middle Eastern nations had a grit mean score of (3.15 (0.54)) [11]. Our

results on the relationship between grit level and the GPA scale also share similarities with previous studies' findings. Specifically, while our study showed that grit level had a relationship at all GPA scales, previous studies reported that students with self-reported GPA ≥ 3.50 had a higher mean grit (3.3 (0.4)) and (3.7 (0.5)), respectively, than any others with lower GPA [10, 12].

In terms of Self-control, pharmacy students' level of self-control (3.4 (0.76)) was aligned with a previous study on university students (3.16 (0.6)) [5], indicating that students both had moderate levels of self-control. In addition, self-regulation of effort was positively and statistically significantly associated with students' GPA at a moderate level [13].

According to the linear regression model, students' grit was a positive predictor of the GPA, and a higher grit level increased the students' GPA. This finding is aligned with previous findings, which provided evidence that heightened levels of grit were associated with enhanced academic performance among pharmacy students [11]. The greater the grit and self-efficacy, the greater the academic success [14]. Besides, students' self-regulation of effort strongly predicted students' GPA [13]. In terms of religious affiliation, the current study is in accordance with a study conducted in Pakistan, which also reported a significant correlation between religion and academic performance among undergraduate students [15].

From an educational perspective, these findings have practical implications. Instead of viewing Grit and Self-control as fixed innate traits, recent research suggests that they are malleable qualities open to pedagogical interventions [2]. Therefore, pharmacy schools should move beyond traditional lecturing to implement strategies that foster these attributes. For instance, incorporating complex problem-based learning and rigorous clinical simulations can require students to practice perseverance. Furthermore, mentorship programs can help students reframe academic setbacks as learning opportunities rather than failures, thereby explicitly cultivating their psychological endurance alongside professional knowledge.

Finally, regarding the data collection timeline (October 2021), we acknowledge the temporal gap. However, the positive correlation between these non-cognitive traits and academic performance is a strong positive relationship, which is consistent with previous studies [11, 14]. This implies that the basic association between grit, self-control, and GPA holds true for current pharmacy cohorts, regardless of the specific academic year.

5 Conclusion

This study provides empirical evidence that grit and self-control are positively associated with the academic achievement of pharmacy students. Specifically, our results showed that the students possessed moderate levels of these traits. The multivariable linear regression model successfully explained 61.61% of the variance in their GPAs. Among the key predictors, perseverance of effort had the strongest impact by accounting for 40.1% of the R^2 followed by consistency of interest (37.5%) and self-control (20.5%). Higher

scores on measures of perseverance, consistency of interests, and self-regulation were linked to superior cumulative GPAs. Consequently, fostering these non-cognitive skills should be considered a valuable component of pharmacy training. To operate this, educators could design curricula that feature complex, semester-long projects to develop sustained effort. Training modules focused on practical time-management and specific learning techniques could enhance students' self-control. Furthermore, establishing a dedicated faculty mentorship system would allow for personalized guidance, helping students cultivate a growth mindset by learning from setbacks. Communicating the value of effort-based achievement and providing avenues for skill development are crucial for improving both academic results and long-term professional resilience.

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Tác động của sự kiên trì và sự tự chủ đến thành tích học tập của sinh viên dược Việt Nam

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Tóm tắt Nghiên cứu này khảo sát Sự bền bỉ (Grit) và Khả năng tự chủ (Self-control) và xác định các yếu tố dự báo điểm trung bình tích lũy (GPA) của sinh viên ngành Dược. Nghiên cứu cắt ngang (tháng 10 năm 2021) với sự tham gia của 417 sinh viên, sử dụng thang đo Grit-S (đánh giá Sự nhất quán và Sự kiên trì) và thang đo Tự chủ rút gọn (Brief Self-Control scale). Dữ liệu được phân tích bằng ngôn ngữ R. Kết quả cho thấy sinh viên Dược có mức độ Kiên trì, Nhất quán và Sự tự chủ ở mức trung bình (điểm số trung bình từ 3,00 đến 3,40, khoảng điểm: 1,00 đến 5,00). Phân tích hồi quy tuyến tính đa biến sử dụng mô hình dự báo tốt nhất theo phương pháp Bayesian Modeling Averaging ($R^2 = 0,6161$) xác định cả ba đặc điểm ($p < 0,001$) và có theo tôn giáo ($p = 0,003$) là những yếu tố dự báo tích cực của GPA, và lần lượt đóng góp 40,10 %, 37,51 %, 20,50 % và 1,89 % vào giá trị R^2 . Đây là nghiên cứu thực nghiệm đầu tiên tại Việt Nam kiểm chứng năng lực dự báo mạnh mẽ của các đặc điểm này ($R^2 > 61$ %). Công trình thiết lập một cơ sở nền tảng quan trọng cho đặc điểm kết quả học tập (GPA) của các nhóm sinh viên được đánh giá trong năm 2021. Kết quả nghiên cứu cho thấy sinh viên có mức độ bền bỉ và tự chủ cao hơn có khả năng đạt được thành tích học tập tốt hơn. Do đó, nhà trường cần xem xét triển khai các chiến lược nhằm bồi dưỡng những phẩm chất này cho sinh viên để góp phần nâng cao kết quả học tập.

Từ khóa Sự kiên trì, sự nhất quán, khả năng tự chủ, GPA, đào tạo dược

Solutions to Promote Innovative Work Behavior: case of Ho Chi Minh City's Software Sector

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Abstract

Amid rapid growth and rising competition in Ho Chi Minh City's software sector, innovative work behavior is essential for sustaining competitive advantage. This study investigates its key antecedents and examines the mediating role of functional flexibility, grounded in social exchange theory and the dynamic theory of organizational knowledge creation. Using a quantitative design, data were collected through non-probability sampling, combining on-site convenience surveys in selected software firms with online snowball sampling among software engineers, producing 380 valid responses. SmartPLS3 was used to assess structural relationships. Results showed that transformational leadership and knowledge sharing positively influenced innovative work behavior, while functional flexibility mediated these relationships and converted leadership and knowledge resources into innovative action.

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Keywords

Transformational leadership;
knowledge sharing;
functional flexibility;
infrastructure capability;
innovative work behavior.

1 Introduction

In the rapidly expanding and highly competitive software sector in Ho Chi Minh City (HCMC), employees' innovative work behavior (IWB) has become a critical determinant of organizational adaptability and long-term performance. However, despite increasingly strong technological and organizational infrastructures, innovative outcomes at the employee level remain uneven. A key paradox in HCMC's software sector is that available knowledge resources and managerial support do not automatically translate into IWB when employees lack the functional flexibility (FF) needed to move across tasks, absorb new knowledge, and respond to rapidly changing project demands. In this sense, FF can be viewed as a crucial mechanism that helps convert organizational resources alone into IWB in practice. As technological

complexity accelerates, numerous firms increasingly depend on knowledge-based resources and effective leadership mechanisms to stimulate employees' capacity to generate, promote, and implement novel ideas. Transformational leadership (TL) plays a pivotal role in shaping innovative outcomes by cultivating psychological safety, intrinsic motivation, and proactive engagement among employees [1]. Simultaneously, knowledge-intensive environments require a robust knowledge management infrastructure capability (KMIC), which includes organizational culture, structure, and information systems that support knowledge creation, exchange, and utilization [2]. Complementing this infrastructure, knowledge sharing (KS) enhances employees' cognitive resources, facilitates collaboration, and serves as a key antecedent of innovation.

Recent studies have also highlighted FF as a critical adaptive capability that enables employees to apply diverse skill sets, responds effectively to dynamic task demands, and translates accumulated knowledge into innovative behaviors. Despite its importance, empirical studies investigating FF as a mediating mechanism linking leadership and knowledge-related factors to IWB remains limited, particularly within Viet Nam's software sector. Therefore, the present study examined the effects of TL, KS, and KMIC on IWB and the mediating role of FF using data from 380 software engineers working in major software firms in HCMC. By integrating TL, KMIC, KS, and FF into a unified analytical framework, the study advanced theoretical understanding and provided practical insights for fostering innovation in knowledge-intensive organizations.

2 Literature Review and Hypotheses Development

2.1 Transformational leadership (TL)

TL is a leadership style that motivates employees to move beyond personal interests toward collective goals through four key components: *idealized influence*, *inspirational motivation*, *intellectual stimulation*, and *individualized consideration* [3]. Through these components, TL fosters positive changes in followers' values, beliefs, and emotions

2.2 Knowledge management infrastructure capability (KMIC)

The KMIC reflects an organization's competence in establishing a foundational framework comprising a supportive culture, a flexible structure, and effective information systems. These components enable efficient creation, sharing, and utilization of knowledge, thereby fostering organizational learning, stimulating innovation, and strengthening core competencies.

Specifically, a flexible structure promotes interaction and empowerment. A supportive culture nurtures collaboration, learning, and KS [3], while information technology systems provide the basis for rapid knowledge exchange and connectivity, reinforcing core capabilities [5].

2.3 Knowledge sharing (KS)

KS is defined as the process of exchanging skills, experiences, and information among individuals within

an organization, encompassing both explicit and tacit knowledge. This process occurs through communicative interactions where individuals acquire and actively disseminate knowledge. Consequently, recipients are able to apply the acquired knowledge effectively in their work, thereby fostering innovation at both individual and organizational levels .

2.4 Functional flexibility (FF)

According to [4], FF denotes employees' capacity to leverage their knowledge, skills, and expertise to undertake a range of tasks beyond their routine duties. This capability enables them to adapt swiftly, participate in varied activities, and respond effectively to shifting demands in dynamic work environments. Such flexibility is underpinned by a holistic set of competencies, encompassing skills, attitudes, and behaviors.

2.5 Innovative work behavior (IWB)

IWB is regarded as a purposeful sequence of individual actions aimed at generating, promoting, and realizing new ideas. It encompasses the exploration, creation, dissemination, and implementation of creative ideas in the workplace, thereby enhancing individual, team, and organizational effectiveness. This behavior reflects proactivity and the capacity to solve problems through novel approaches [5].

2.6 Hypotheses development

2.6.1 TL on IWB

According to SET [6], individuals are more likely to engage in constructive and proactive behaviors when their contributions are acknowledged and appreciated by the organization. TL, which includes intellectual stimulation, individualized consideration, and inspirational motivation, fosters employee trust and enhances organizational commitment. This leadership approach contributes to the development of a psychologically safe environment that encourages innovation by reducing fear of failure and promoting open communication. When employees feel supported and recognized, they are more inclined to demonstrate creative behaviors and pursue continuous improvement [1]. In the context of the software sector, such recognition and encouragement from leaders increase employees' willingness to experiment, generate novel ideas, and embrace the risks associated with innovation.

H1: Transformational leadership positively influences IWB in the software sector in HCMC.

2.6.2 KS and IWB

SET posits that individuals seek to maintain a balance of benefits within relationships. When benefits are received, a norm of reciprocity tends to emerge prompting individuals to engage in positive reciprocal behaviors. In organizational settings, knowledge contributors often anticipate future rewards, while recipients are inclined to reciprocate, fostering mutual support that promotes innovation [7]. KS enhances employees' depth of understanding, facilitates the application of insights, and stimulates creativity, it also initiates novel ideas and increases motivation for innovation through peer learning. Empirical research across manufacturing, automotive, hospitality, and education sectors has consistently demonstrated a positive relationship between KS and innovative behavior. In the software sector in HCMC, where the environment is both dynamic and innovation-intensive, KS facilitates effective team collaboration, problem-solving, and development of novel solutions. Therefore, the following hypothesis is proposed:

H3: Knowledge sharing positively influences IWB in the software sector in HCMC.

2.6.3 FF and IWB

According to the theory of organizational knowledge creation, organizations generate new knowledge through the interaction between tacit and explicit knowledge as conceptualized in the SECI model [10]. FF enables individuals to move across different "ba" spaces, thereby promoting KS, combination, and internalization which are essential processes for fostering innovative behavior [11]. Organizations must enhance employees' knowledge, skills, and multi-functional competencies to empower proactive creativity. FF serves as a foundation for innovation, enabling employees to go beyond the formal boundaries of their job roles. Studies conducted in manufacturing firms and SMEs in Pakistan have confirmed the positive impact of FF on IWB [12, 13]. In the context of software sector in HCMC where technology evolves rapidly and customer demands are diverse, FF enables employees to adapt, innovate, and continuously improve.

H4: Functional flexibility has a positive impact on IWB in the software sector in HCMC.

2.6.4 KS and FF

According to the theory of organizational knowledge creation, knowledge is formed through the dynamic interaction between tacit and explicit knowledge, and its diffusion enhances both understanding and FF [8]. The sharing of knowledge, particularly tacit experiences, enables individuals to internalize insights and collaborate more effectively. Research has consistently demonstrated that KS is a foundational factor contributing to FF [10]. An empirical study conducted in Pakistan involving SME employees revealed that KS positively influences FF. Similarly, another study with SME employees confirmed that KS significantly enhanced FF. In the software sector where agile methodologies are prevalent, KS frequently occurs through daily stand-up meetings, code reviews, and technical workshops.

H6: Knowledge sharing has a positive effect on FF in the software sector in HCMC.

2.6.5 KMIC and FF

According to the theory of organizational knowledge creation [8], knowledge is developed through continuous interactions among individuals, groups, and organizational systems, and becomes valuable when supported by an appropriate infrastructure for its creation, sharing, and application. When the KMIC is effectively designed, employees can access to information more easily, develop skills, enhance motivation, and improve adaptability, thereby contributing to the overall effectiveness of human resources. According to [10], such infrastructure strengthens learning processes, which in turn fosters FF. In the software sector, which is characterized by a knowledge-intensive and innovation-driven environment, knowledge management infrastructure promotes continuous learning, supports multi-role capabilities, and enhances adaptability to new technologies.

H2: The capability of knowledge management infrastructure is positively associated with FF in the software sector in HCMC.

2.6.6 TL and FF

Social Exchange Theory suggests that employees reciprocate leaders' trust and support through positive behaviors, including greater adaptability beyond formal job boundaries [6]. Multiple research highlight the mediating role of employee adaptiveness in the relationship between paradoxical leadership and employee creativity, this thesis shifts the focus to TL. FF is conceptualized as a tangible expression of adaptability, reflecting employees' capacity to adjust their roles and competencies in response to changing demands. TL cultivates a proactive disposition among employees and enhances their adaptability, thereby establishing the foundation for FF. Within the dynamic and rapidly evolving software sector, TL plays a critical role in fostering multi-skilling and strengthening employees' FF by promoting inspiration, empowerment, and continuous learning.

H5: Transformational leadership positively influences FF among employees in the software sector in HCMC.

2.6.7 FF mediates the relationship between TL and IWB

According to SET, employees tend to reciprocate positive support from leaders by engaging in behaviors that contribute to the organization [6]. In the software sector in HCMC, TL promotes innovation by inspiring employees, fostering continuous learning, and empowering them in task execution. Such support enhances employees' perceptions of trust and developmental opportunities, thereby encouraging them to proactively broaden their roles, undertake multiple responsibilities, and adapt flexibly to changing project demands.

This FF enables employees to access to diverse knowledge sources, experiment with novel solutions, and propose effective improvements, which in turn facilitates IWB. Therefore, FF is expected to act as a mediating mechanism through which TL influences IWB.

H7a: FF mediates the relationship between TL and IWB in the software sector in HCMC.

2.6.8 FF mediates the relationship between KMIC and IWB

According to [8], through the SECI model, and [11], via the concept of “Ba,” the process of knowledge creation and diffusion depends on supportive infrastructure, including technology, organizational structure, culture, and interactive space. Knowledge management infrastructure facilitates employees' access sharing, and application of knowledge, thereby enhancing their skills, knowledge, motivation, and proactivity. Numerous studies have confirmed that such infrastructure increases FF, which is a critical driver of creativity and innovative behavior. Previous research [9] indicated that knowledge management infrastructure positively influences functional flexibility, subsequently impacting innovative behavior. In software sector where technological changes occur rapidly, knowledge management infrastructure enables employees to reskill, adapt to new projects, and foster innovation. Therefore, this study proposed the following hypothesis:

H7b: FF mediates the relationship between KMIC and innovative behavior among software employees in HCMC.

2.6.9 FF mediates the relationship between KS and IWB

Organizational knowledge is continuously shaped through interactions among individuals, groups, and organizations with KS as a core mechanism [8]. SECI model and “ba” concept highlight the role of dynamic context in facilitating knowledge creation and diffusion [11]. FF, which is the ability to apply knowledge across diverse tasks, is essential for converting knowledge into innovation. Studies have suggested that intrinsic motivation and risk-taking may mediate this process. KS expands understanding, boosts proactivity, and enhances flexibility, thus fostering innovation. Moreover, it supports knowledge processes that improve adaptability. Prior research [12] found that FF mediated the indirect effect of KS on IWB. In competitive software sectors, FF drives innovation while KS updates technical know-how. Thus, this study proposed:

H7c: FF mediates the relationship between KS and IWB in software firms in HCMC

Proposed Research model

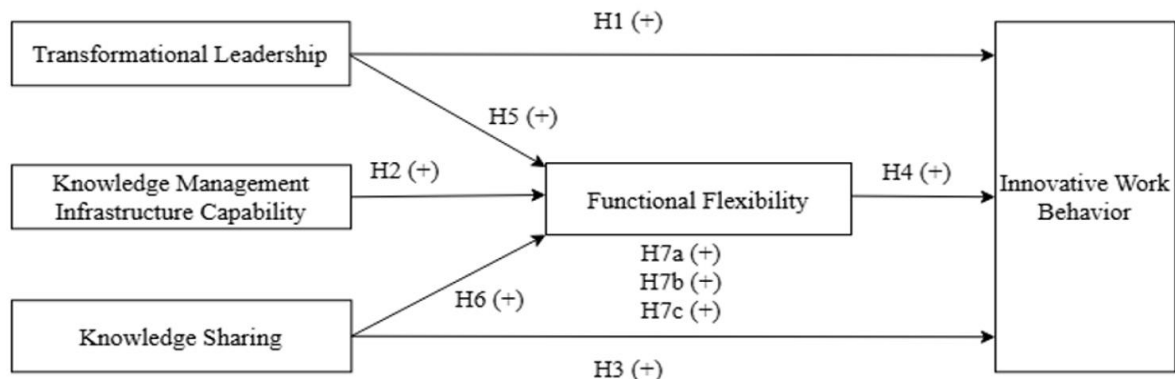


Figure 1 Proposed research model

Measurement scales

The questionnaire employed a five-point Likert scale. TL was measured using seven items developed by [13]. Respondents, who were subordinates, evaluated their supervisors based on four dimensions: (i) idealized influence, (ii) inspirational motivation, (iii) intellectual stimulation, and (iv) individualized consideration. KS was assessed through six items drawn from [14], capturing behaviors related to the sharing of information, experiences, and insights among individuals. KMIC of five items covering: (i) organizational culture (2 items), (ii) flexible structure (2 items), (iii) IT support (2 items) adapted from [2] respectively.

FF was measured with six items based on works by assessing employees' ability to adapt to change through modifications in roles, tasks, and skill sets. IWB included five items from, evaluating the frequency with which individuals engage in innovative actions within the workplace [15].

3 Research Methods

3.1 Data analysis

In this study, SmartPLS3 was employed to test the proposed hypotheses. The PLS procedure consisted of two steps: (1) assessing the measurement model to examine the reliability and validity of the scales, and (2) testing the hypotheses using the bootstrapping technique.

3.2 Sample and data collection

A non-probability sampling method, combining convenience and snowball sampling, yielded a total of 400 responses. After screening, 20 invalid responses were removed, leaving 380 valid responses collected between October 18 and December 18, 2025. The

sample consisted primarily of male participants, most of whom are from 26 to 30 years old and held a bachelor's degree. Participants mainly work in Testing/Quality Assurance, Software Engineering, or Data Analytics, with nearly half having less than five years of professional experience. Most respondents are employed at limited liability companies with at least 100 employees. The 26-30 age group represents the largest proportion of the sample.

4 Results

4.1 Measurement model analysis

The R^2 values exceeding 0.26 indicated a large effect, while values between 0.13 and 0.26 represented a medium effect. As shown in Table 1, the model accounts for 42.9% of the variance in FF, indicating a large explanatory effect. For IWB, the R^2 of 22.2% reflected a medium effect size. The close alignment between the R^2 and adjusted R^2 values suggests that the model is both stable and appropriately estimated.

Table 1 R square

	R Square	R Square Adjusted
FF	0.429	0.424
IWB	0.222	0.215

The study employed the Fornell-Larcker criterion and the HTMT ratio for discriminant validity. As shown in Table 2, the square roots of AVEs for all constructs exceeded their inter-construct correlations, thereby satisfying the Fornell-Larcker criterion. The HTMT values reported in Table 3 fell below the threshold of 0.90, indicating low conceptual overlap and clear construct distinctiveness. Collectively, both criteria

confirmed strong discriminant validity of the measurement model, providing a robust foundation for subsequent structural model analysis.

Table 2 Fornell-Larcker criterion

	FF	IWB	KMIC	KS	TL
FF	0.867				
IWB	0.397	0.852			
KMIC	0.573	0.385	0.854		
KS	0.278	0.281	0.260	0.846	
TL	0.568	0.390	0.549	0.249	0.851

Table 3 Heterotrait-Monotrait ratio (HTMT)

	FF	IWB	KMIC	KS
FF				
IWB	0.440			
KMIC	0.635	0.424		
KS	0.304	0.306	0.283	
TL	0.645	0.437	0.617	0.278

The measurement model provides strong evidence of convergent validity. Table 4 shows that all indicator loadings exceeded the recommended threshold, confirming indicator reliability. Regarding internal consistency, the Cronbach's alpha coefficients for all constructs were above the acceptable threshold, reinforcing the reliability of the measurement scales. Overall, the results indicate that the constructs demonstrate satisfactory reliability and convergent validity.

Table 4 Validity and Reliability results

Variables	Items	Outer Loadings	Cronbach's alpha	Composite Reliability	AVE
TL	TL1	0.854	0.873	0.913	0.725
	TL2	0.849			
	TL3	0.854			
	TL4	0.847			
KMIC	KMIC1	0.841	0.907	0.931	0.729
	KMIC2	0.853			
	KMIC3	0.849			
	KMIC4	0.856			
	KMIC5	0.869			

Variables	Items	Outer Loadings	Cronbach's alpha	Composite Reliability	AVE
KS	KS1	0.849	0.921	0.938	0.716
	KS2	0.852			
	KS3	0.843			
	KS4	0.848			
	KS5	0.862			
	KS6	0.823			
FF	FF1	0.863	0.890	0.923	0.751
	FF2	0.876			
	FF3	0.866			
	FF4	0.861			
IWB	IWB1	0.858	0.906	0.930	0.726
	IWB2	0.849			
	IWB3	0.856			
	IWB4	0.858			
	IWB5	0.839			

4.2 Hypothesis testing

The study employed PLS-SEM with 5,000 bootstrap resamples to evaluate the proposed hypotheses, in accordance with established methodological recommendations for examining mediation effects. The results presented in Table 6 indicated that all hypothesized relationships (H1– H6) were statistically supported.

The findings indicated that all six hypotheses were supported. Specifically, H1 was supported, showing that TL had a significant positive effect on IWB ($\beta = 0.220$, $t = 3.958$, $p = 0.000$). H2 was also supported, indicating that KMIC positively influenced FF ($\beta = 0.356$, $t = 8.406$, $p = 0.000$). Likewise, H3 confirmed that KS significantly and positively affected IWB ($\beta = 0.163$, $t = 3.114$, $p = 0.002$). H4 was supported as FF exerted a positive effect on IWB ($\beta = 0.227$, $t = 4.281$, $p = 0.000$). In addition, H5 demonstrated that TL significantly enhanced FF ($\beta = 0.348$, $t = 7.759$, $p = 0.000$). Finally, H6 was supported, revealing that KS had a positive and significant effect on FF ($\beta = 0.099$, $t = 2.435$, $p = 0.015$). Overall, the results confirm the proposed structural relationships in the model.

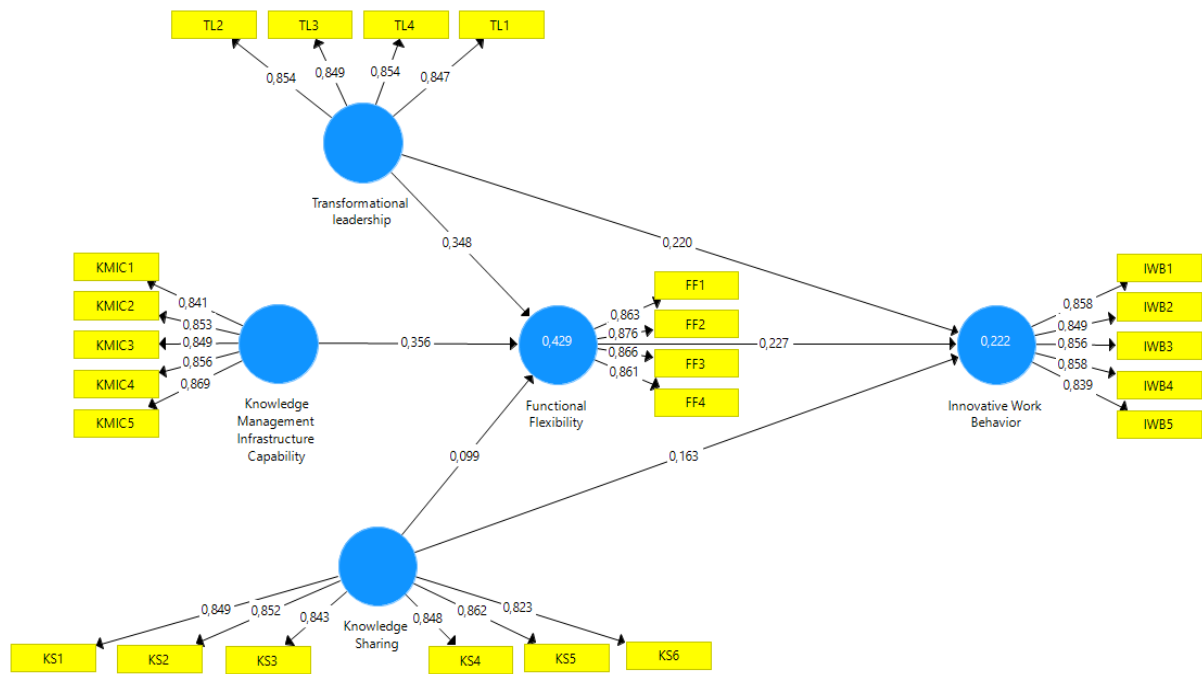


Figure 2 Structural equation model results

Table 6 Path coefficients

Hypothesis	Direct effects	β	T Statistics	P Values	Result
H1	TL $\rightarrow$ IWB	0.220	3.958	0.000	Supported
H2	KMIC $\rightarrow$ FF	0.356	8.406	0.000	Supported
H3	KS $\rightarrow$ IWB	0.163	3.114	0.002	Supported
H4	FF $\rightarrow$ IWB	0.227	4.281	0.000	Supported
H5	TL $\rightarrow$ FF	0.348	7.759	0.000	Supported
H6	KS $\rightarrow$ FF	0.099	2.435	0.015	Supported

Table 7 Specific indirect effects

Hypothesis	Specific Indirect Effects	β	P Values	Result
H7a	TL $\rightarrow$ FF $\rightarrow$ IWB	0.079	0.000	Supported
H7b	KMIC $\rightarrow$ FF $\rightarrow$ IWB	0.081	0.000	Supported
H7c	KS $\rightarrow$ FF $\rightarrow$ IWB	0.022	0.039	Supported

Table 8 Mediation analysis

Mediated Relationship	β (Indirect)	P Values	β (Direct)	P Values (Indirect)	Mediation (%)	Mediation Type
TL $\rightarrow$ FF $\rightarrow$ IWB (H7a)	0.079	0.000	0.220	0.000	26.4	Partial mediation
KMIC $\rightarrow$ FF $\rightarrow$ IWB (H7b)	0.081	0.000	-	-	100	Full mediation
KS $\rightarrow$ FF $\rightarrow$ IWB (H7c)	0.022	0.039	0.163	0.002	11.9	Partial mediation

5 Discussions and Implications

This study provides a more nuanced understanding of IWB in software sector in HCMC by showing that innovation is shaped not only by knowledge-related resources but also by leadership style and employees' adaptive capacity. This finding suggests that in

software project environments, where tasks are dynamic, deadlines are tight, and requirements frequently change, employees are more likely to generate, promote, and implement new ideas when they are inspired, empowered, and supported by leaders who encourage experimentation and learning. TL appears to

exert the strongest influence because it directly affects the motivational and psychological conditions under which innovative behavior occurs.

The findings also indicate that KMIC does not directly produce IWB, but works through FF. This implies that organizational culture, structure, and IT-based knowledge systems are important enabling conditions, yet they do not automatically lead to innovation unless employees are able to flexibly apply, transfer, and recombine knowledge across different tasks and project demands.

5.1 Theoretical implications

This study made several contributions to management theory. First, it extended Social Exchange Theory and the theory of organizational knowledge creation by integrating TL, KS, and KMIC into a unified framework for explaining IWB in the software sector. In doing so, the study showed that both relational mechanisms and knowledge-related organizational conditions were necessary for understanding how innovation emerged in knowledge-intensive workplaces.

Second, the study highlighted FF as a key mediating mechanism through which leadership and knowledge-related resources were translated into IWB. This finding enriched existing theory by showing that innovation was not merely the result of available support systems, but of the interaction between those systems and employees' adaptive capabilities.

Third, the study provided context-specific evidence from software firms in HCMC, where work was typically organized around projects, iterative development cycles, and cross-functional collaboration. By linking innovation outcomes to the project-based nature of software work, the study contributed to a more grounded understanding of how leadership, knowledge processes, and flexibility interacted in fast-changing professional environments.

5.2 Practical implications

The findings offered several practical implications. First, firms should invest in developing TL capabilities. In project-based environments, where uncertainty and

interdependence were high, such leadership was particularly important for sustaining creativity and adaptive problem-solving.

Second, firms should not assume that investments in knowledge management infrastructure alone will automatically improve innovation. Accordingly, software firms should strengthen FF through multi-skilling, cross-functional training, job rotation, and continuous skill development.

Third, organizations should foster a knowledge-sharing culture in which employees feel safe to discuss mistakes, exchange lessons learned, and collaboratively improve work processes. In software development, activities such as code reviews, retrospectives, technical sharing sessions, and peer learning forums can create opportunities for knowledge conversion and innovation. Building a trustworthy environment in which errors were treated as learning opportunities rather than personal failures can further strengthen both FF and IWB.

This study had several limitations. The use of convenience sampling might have reduced the representativeness of the findings, and the cross-sectional design limited causal inference.

Future research should consider using longitudinal designs and broader sampling strategies to improve robustness and generalizability. It would also be valuable to compare different types of software firms, such as outsourcing companies, product-based firms, startups, and large-scale technology enterprises, in order to examine whether the relationships identified in this study varied across organizational contexts. Such comparative research would provide a deeper understanding of how leadership, knowledge management infrastructure, and FF influence IWB under different business models and work structures.

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Giải pháp thúc đẩy hành vi đổi mới sáng tạo: trường hợp ngành phần mềm tại Thành phố Hồ Chí Minh

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Tóm tắt Trong bối cảnh ngành phần mềm tại Thành phố Hồ Chí Minh tăng trưởng nhanh và cạnh tranh ngày càng mạnh, hành vi đổi mới sáng tạo trong công việc là yếu tố quan trọng để duy trì lợi thế cạnh tranh. Nghiên cứu này phân tích các tiền tố của hành vi này và kiểm định vai trò trung gian của tính linh hoạt chức năng, dựa trên lý thuyết trao đổi xã hội và lý thuyết động về sáng tạo tri thức tổ chức. Dữ liệu định lượng được thu thập từ 380 phản hồi hợp lệ bằng chọn mẫu phi xác suất, kết hợp khảo sát trực tiếp thuận tiện và lấy mẫu dây chuyền trực tuyến. SmartPLS3 được sử dụng để kiểm định mô hình. Kết quả cho thấy lãnh đạo chuyển hóa, chia sẻ tri thức và năng lực hạ tầng quản trị tri thức góp phần gia tăng tính linh hoạt chức năng; đồng thời, tính linh hoạt chức năng giữ vai trò trung gian trong việc chuyển hóa các nguồn lực lãnh đạo và tri thức thành hành vi đổi mới của nhân viên.

Từ khóa Lãnh đạo chuyển đổi; chia sẻ tri thức; linh hoạt chức năng; năng lực hạ tầng quản trị tri thức; hành vi đổi mới sáng tạo trong công việc.



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Thể lệ gửi bài đăng
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Tạp chí Khoa học và Công nghệ - Đại học Nguyễn Tất Thành xuất bản 1 tháng/kì; tập hợp công bố kết quả nghiên cứu khoa học, thông tin chuyên đề, bài viết tổng luận của các nhà nghiên cứu, cán bộ, giảng viên trong và ngoài trường. Ngôn ngữ thể hiện: tiếng Việt và tiếng Anh. Chất lượng công bố và trình bày theo chuẩn của một Tạp chí khoa học.

1. Bài gửi đăng là công trình nghiên cứu có nội dung mới và chưa gửi đăng trên bất kì Tạp chí khoa học nào hoặc các dạng xuất bản phẩm khác. Sau khi tác giả thực hiện các yêu cầu chỉnh sửa (của ít nhất 2 phản biện nhận xét, đánh giá), bài đạt yêu cầu sẽ được đăng trong số báo gần nhất.

2. Bài viết không quá 08 trang khổ A4 (khoảng 4.000 - 6.000 từ) kể cả bảng biểu, đồ thị, tài liệu tham khảo với font Time New Roman, font size 11 normal; margin (top/bottom 2.54 cm; left/right 2.54 cm) gồm:

- Tên bài, họ và tên tác giả/nhóm tác giả, cơ quan công tác, email; tóm tắt nội dung và những kết quả chính (cỡ chữ 11, không quá 10 dòng), từ khóa;
- Nội dung, kết quả và thảo luận, định hướng mới và kết luận, lời cảm ơn. Các số liệu, kết quả thu được là tài liệu gốc, được in và chụp trực tiếp từ các thiết bị thí nghiệm, biện luận phải được trình bày rõ ràng và chính xác;
- Tài liệu tham khảo sắp xếp theo thứ tự trích dẫn trong bài;
- Ở cuối bài viết bằng tiếng Việt (hoặc Anh) phải có: tên bài, họ và tên tác giả/nhóm tác giả, cơ quan công tác, tóm tắt và từ khóa bằng tiếng Anh (hoặc Việt).

3. Bản thảo điện tử kèm theo Phiếu đăng kí nộp bài (word file và pdf file có chữ kí của các tác giả) gửi theo địa chỉ:

Tạp chí Khoa học và Công nghệ - Đại học Nguyễn Tất Thành,
300A Nguyễn Tất Thành, phường Xóm Chiếu, Tp. Hồ Chí Minh (P. A407).

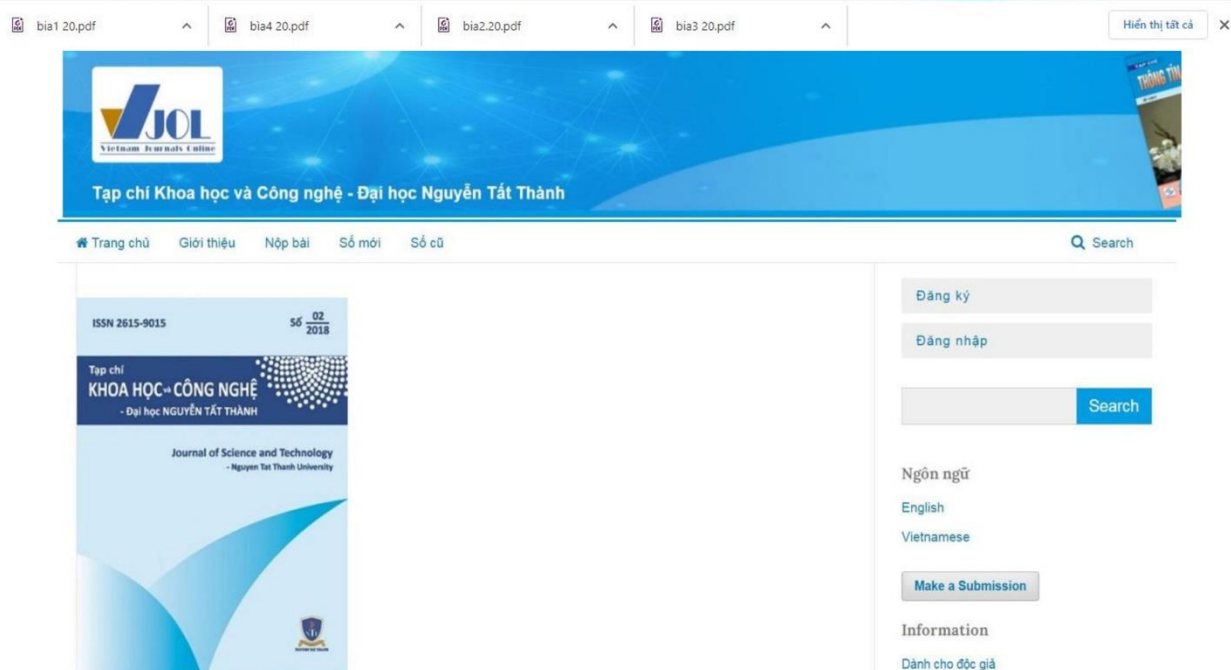
Email: tapchikhcn@ntt.edu.vn;
blgiang@ntt.edu.vn

Mọi chi tiết, xin liên hệ: mobile: 0918.218.434, 0902.858.828;
tel.: 028.7108.0889 (ext: 341)

Ban Biên tập



<http://jst.ntt.edu.vn>



Website Tạp chí Khoa học Việt Nam trực tuyến (Vietnam Journals Online- VJOL)

<https://vjol.info.vn>

Tạp chí khoa học Việt Nam trực tuyến (Vietnam Journals online- VJOL) là cơ sở dữ liệu cho phép độc giả truy cập các Tạp chí khoa học xuất bản tại Việt Nam.

Website VJOL hiện do Cục Thông tin Khoa học và Công nghệ Quốc gia (Bộ Khoa học và Công nghệ) quản lý, duy trì và phát triển nhằm tăng cường sự hiện diện và tầm ảnh hưởng của các tạp chí khoa học Việt Nam đối với bạn đọc trong nước và quốc tế, đồng thời đẩy mạnh quá trình chuyển đổi số của các tạp chí khoa học thông qua nền tảng VJOL